

2025 Final Annual Report

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EXECUTIVE SUMMARY

The primary goal of this study was to investigate piscivorous colonial waterbird predation on juvenile salmonids (smolts; *Oncorhynchus* spp.) in the Columbia River basin (CRB). Piscivorous colonial waterbird species investigated included Caspian terns (*Hydroprogne caspia;* hereafter "terns"), double-crested cormorants (*Nannopterum auritum;* hereafter "cormorants"), Brandt's cormorants (*Phalacrocorax penicillatus*), California gulls (*Larus californicus*) and ring-billed gulls (*L. delawarensis*; hereafter collectively as "gulls"), American white pelicans (*Pelecanus erythrorhynchos*; hereafter "pelicans"), and great blue herons (*Ardea herodias*) and great egrets (*A. alba*; hereafter collectively as "herons"). Managed predator species included terns and cormorants from select colonies in the Columbia Plateau region (CPR) and the Columbia River estuary (CRE). More specifically, the study sought to (1) estimate colony sizes for colonies within foraging range of smolts in the middle Columbia River, lower Snake River, lower Columbia River (LCR), and the CRE, (2) estimate colony-specific predation rates (percentage of fish consumed) on Endangered Species Act (ESA) listed smolts, (3) estimate the cumulative or system-wide effects of predation from all colonies combined on smolts, (4) estimate the additive effects of predation on smolt survival, and (5) evaluate the efficacy of ongoing management plans aimed at reducing predation on smolts.

A total of 36 active tern, cormorant, pelican, gull, and heron colonies were monitored in the CRB during the 2025 breeding season. Of those, cormorant (13) and gull (9) colonies were the most prevalent, followed by tern (6), heron (6), and pelican (2) colonies. Most of the colonies (27) were in the CPR, with 9 colonies located in the CRE. Of the colonies monitored, gulls were the most numerous (> 39,000 breeding individuals), followed by cormorants (7,950 breeding pairs), pelicans (3,984 individuals), terns (888 pairs), and herons (312 pairs). The number of active breeding colonies observed in 2025 was similar to previous years. The size and location of colonies, however, has changed over the course of the last ten years, and monitoring has been expanded to include new predator species and colonies.

Tern and cormorant colonies continue to be managed as part of three separate management plans in the CRB (*Estuary Tern Management Plan*, *Estuary Cormorant Management Plan*, and the *Inland Avian Predation Management Plan*). Results of this study indicated that the numbers of terns nesting in the CRE continue to be below management targets. Management in the CRE has also contributed to declines in the Pacific Flyway breeding population of terns, which raises concerns about their conservation status. The designated tern colony on East Sand Island (ESI) in the CRE, formerly the largest tern colony in the world, has rapidly declined and productivity (number of fledglings produced) has been low to non-existent, with complete nesting failure observed in six out of the last ten years (2016–2025). The number of terns on ESI (634 pairs in 2025) remains well below the target colony size identified in the management plan (3,125–4,375 pairs); a number, that if maintained, would help to sustain the Pacific Flyway breeding population of terns.

Predation rates on smolts by terns nesting on ESI have been greatly reduced since management actions were first implemented in 2008, with predation rates on ESA-listed steelhead (*O. mykiss*) populations reduced by upwards of 80%, reductions that exceed those specified in the management plan. Terns attempting to nest on Rice Island, located 26 river kilometers upstream of ESI, have partially offset the lower predation impacts by terns on ESI. For instance, in 2022, estimated predation rates by terns that nested on Rice Island were as high as 2.9% (95% credible interval = 1.6–5.1%) of Middle Columbia River steelhead. In 2023–2025, adaptive management was successful at preventing a sustained tern colony on Rice Island, although larger numbers of terns continued to attempt to nest on Rice Island during the smolt outmigration period. Collectively, results of tern management actions in the CRE indicate that adaptive management will be necessary to encourage nesting at ESI up to the target colony size and to prevent terns from relocating to nest at other sites in the CRE, especially at sites in the upper estuary where predation rates on smolts are significantly higher on a per capita (per bird) basis.

Management at the ESI double-crested cormorant colony has resulted in the dispersal of birds away from ESI, and 2025 marked the seventh consecutive year with little to no cormorant nesting on ESI during the smolt outmigration period. The failure of the ESI colony has resulted in an expansion of cormorant colonies on the Astoria-Megler Bridge (AMB), Longview Bridge, channel markers, and Troutdale Towers (TRT) located in the upper CRE and LCR. For instance, the double-crested cormorant colony on the AMB has increased from 333 breeding pairs in 2014 (prior to management actions on ESI) to 5,310 pairs in 2025, the second largest number observed to-date and similar to the 5,348 pairs observed in 2024. The AMB is located upstream of ESI in the freshwater mixing zone of the CRE, and the other colony locations (Longview Bridge, channel markers, and TRT) are in the freshwater zone of the estuary, aquatic environments where fewer alternative, non-salmonid prey fish are available. Consequently, smolts are consumed in greater proportion to their availability on a per capita basis by cormorants that nest at colonies in the mixing and freshwater zones of the CRE. Estimates of predation rates by cormorants nesting on the AMB in 2025 ranged from 5.5% (0.6–20.7%) on Snake River (SR) Sockeye Salmon (*O. nerka*) to 13.9% (5.8–28.7%) on SR subyearling Chinook Salmon (*O. tshawytscha).* Predation by AMB cormorants was also substantial on steelhead, with estimates of 9.8% (7.0–14.2%) and 12.7% (3.3–27.2%) on SR and Upper Columbia River (UCR) steelhead, respectively, in 2025. Estimates of predation by cormorants on TRT in 2025 ranged from 0.8% (1.0–5.7%) on SR subyearling Chinook to 5.9% (3.2–12.1%) on UCR steelhead. Taken together, results indicate that predation by cormorants in the CRE are now greater than those observed prior to implementation of management actions on ESI. As such, adaptive management to reduce double-crested cormorant predation in the upper CRE and LCR is necessary to achieve the goals of the *Estuary Cormorant Management Plan*. Management actions, such dissuading double-crested cormorants from the AMB and reattracting them to ESI, could reduce predation rates on steelhead and salmon by cormorants in the CRE by upwards of 70%, resulting in millions of additional smolts surviving outmigration to the Pacific Ocean each year.

Estimates of predation on smolts by Brandt's cormorants nesting on the AMB were significantly lower than those of double-crested cormorants and were amongst the lowest of all predator species and colonies evaluated in the CRE in 2025, with estimates of predation < 1.0% for most ESA-listed salmonid populations evaluated. Predation by double-crested cormorants in the CPR as high as 4.9% (3.3–8.1%) and 5.7% (3.5–9.6%) for SR sockeye and SR steelhead, respectively, were observed for birds breeding on Crescent Island in McNary Reservoir, a colony that has rapidly increased from less than 40 breeding pairs in 2021 to more than 500 pairs in 2024 and 2025, making it one of the largest cormorant colonies in the CPR. Numbers of cormorants breeding at other sites in the CPR were lower than those observed on Crescent Island, with estimates of predation < 2.0% per salmonid species, per colony, in 2025.

The ongoing management of tern colonies in the CPR as part of the *Inland Avian Predation Management Plan* has resulted in a shift in the number and location of tern colonies in the region. A total of 254 breeding pairs of terns nested at colonies in the CPR in 2025, a 71% reduction compared to the pre-management period (average of 875 pairs during 2007–2013). Efforts to dissuade terns from nesting on Goose Island and other islands in Potholes Reservoir were largely successful in 2025, with 30 breeding pairs observed on Goose Island, meeting the target colony size goal of less than 40 pairs. For the second consecutive year, adaptive management techniques (i.e., human hazing, fencing, and predator effigies) were able to successfully prevent nesting shortly after a tern colony site was identified by aerial and boat-based surveys in northern Potholes Reservoir in 2025. Management actions at Crescent Island in McNary Reservoir were initially successful at preventing colony formation during 2014–2021, but active management ceased in 2021. Terns subsequently re-established a colony in 2022 (149 breeding pairs) and terns again nested on Crescent Island in 2023 (88 pairs), 2024 (186 pairs) and 2025 (147 pairs), exceeding the colony size threshold identified in the Plan. A lack of active and passive tern dissuasion, coupled with increases in numbers of breeding gulls and pelicans on Crescent Island, have left the island largely unvegetated, resulting in additional habitat for ground nesting piscivorous waterbird use in the future. Adaptive management to raise the water elevation of John Day Reservoir was implemented to inundate tern nesting habitat in the Blalock Islands in John Day Reservoir during 2021–2025, and no nesting by terns was observed at that site during this period. Due to high fidelity to historical nesting sites, adaptive management will continue to be important, especially at Crescent Island and islands in Potholes Reservoir, to meet the stated goals of the management plan.

One of the primary goals of the *Inland Avian Predation Management Plan* is to reduce predation rates on smolts by terns to less than 2% per ESA-listed salmonid population, per colony. In 2025, that objective was achieved, except for a colony on Banks Lake, WA, where an estimated 2.2% (1.5–3.3%) of UCR steelhead were consumed. Predation rates were < 1.0% per salmonid population for terns nesting at the other five active colonies in the CPR, which included terns on an island in Lenore Lake, northern Potholes Reservoir, Goose Island in Potholes Reservoir, Finley Island in McNary Reservoir, and Crescent Island in McNary Reservoir. Since management commenced, average annual predation rates on UCR steelhead by terns nesting on Goose Island and elsewhere in Potholes Reservoir have been reduced from 15.7% (14.1–18.9%) prior to management (2007–2013) to 1.5% (1.0–2.7%) during the management

period (2014–2025). Cumulative predation (predation by all CPR tern colonies combined) was estimated to be 4.2% (3.2–5.7%) on UCR steelhead in 2025, reduced from an average of 19.6% (16.5–24.1%) prior to implementation of management actions in 2014. There was also evidence that the survival of UCR steelhead smolts has, on average, significantly increased in the river reach where terns from Goose Island and Crescent Island forage following implementation of the management plan. However, due to predation on smolts by terns that have re-nested on Crescent Island and terns that are now nesting in larger numbers at other sites (e.g., Banks Lake), adaptive management will continue to be necessary to achieve the goals of the Plan in 2026 and beyond.

An investigation of predation on smolts by other piscivorous waterbird species indicated that smolt consumption rates were highly variable depending on the predator species, colony location, colony size, and salmonid species. In 2025, predation by some gull colonies in the CPR were significantly higher than those of terns and were amongst the highest observed since system-wide studies of gull predation commenced in 2012. For instance, estimates of gull consumption rates as high as 11.3% (6.2–19.9%) for UCR steelhead and 11.0% (5.6–13.2%) for SR Sockeye Salmon were documented for gulls nesting at Miller Rocks in The Dalles Reservoir and Crescent Island in McNary Reservoir, respectively. Estimates of consumption at other large (several thousand individuals) gull colonies in the CPR region, however, were low, with consumption rates less than 1.0% per ESA-listed salmonid population, indicating that not all gull colonies posed a potential threat to smolt survival. Estimates of predation by American white pelicans nesting on Crescent Island in McNary Reservoir and by herons nesting at several small colonies in the CPR, were generally < 0.5% per salmonid population, per colony, indicating these colonies posed little threat to the ESA-listed smolts considered in this study in 2025. Pelicans, however, can consume both juvenile and adult salmonids, and other studies have documented substantial predation effects on adult sockeye and juvenile salmonids from specific stocks (e.g., subyearling Chinook from the Upriver Bright population and Chinook, steelhead, and coho in the Yakima River).

An investigation of the cumulative effects of avian predation/consumption (predation by all colonies combined) indicated that predation was a substantial source of smolt mortality during outmigration to the Pacific Ocean. Predation was consistently the highest on UCR and SR steelhead populations and the lowest on SR Chinook. For instance, in 2025, the cumulative effects of avian predation were estimated at 9.4% (7.1–12.5%) on SR subyearling Chinook but were 32.1% (27.2–37.7%) and 34.2% (28.6–40.9%) on SR and UCR steelhead, respectively. Similar to years past, predation on SR sockeye was lower than that of steelhead, but higher than that of Chinook, with an estimated 23.7% (18.6–29.5%) of SR sockeye consumed by birds in 2025. Comparisons of total smolt mortality (1 - survival) to mortality due to avian predation/consumption indicated that avian predation by the colonies included in the study accounted for between 7.6% (5.4–10.6%) and 47.5% (38.3–59.6%) of all smolt mortality sources during outmigration to Bonneville Dam, depending on the salmonid population. By river reach, predation/consumption rates were highest for smolts between Lower Monumental Dam (for SR smolts) and Rock Island Dam (for UCR smolts) to McNary Dam and from Bonneville Dam to the Pacific Ocean in 2025. Reach-specific predation/consumption rates in 2025 were similar to or

higher than those in previous years (2008–2024). Predation by gull colonies in the CPR and by cormorant colonies in the CRE, were above average in 2025. Collectively, results indicate that the cumulative effects of avian predation/consumption remain a substantial source of smolt mortality in the CRB, particularly for ESA-listed steelhead and sockeye populations.

Finally, previously published studies have demonstrated that predation by terns was largely an additive source of mortality for steelhead, with significantly more steelhead estimated to survive in the absence of tern predation across populations (UCR, SR), spatial scales, and life-stages (smolt, adult). In contrast, gulls are known to consume dead or moribund fish, including those they steal or kleptoparasite from other avian predators, like terns and, consequently, attempts to measure the relationship between gull consumption and smolt survival have been less conclusive. To address this uncertainty, we updated the time-series with eight years of additional gull data and refined the additive modelling approach to simultaneously assess both gull and tern additivity on UCR steelhead smolts during outmigration from Rock Island Dam to Bonneville Dam during 2008–2025. Updated results indicated that higher levels of gull consumption were associated with lower levels of steelhead survival, even after accounting for the additive effects of tern predation. The relationship was statistically significant across all years and was the most pronounced in years with the highest levels of gull predation, like those observed in 2025. Our results provide the most compelling evidence to date that a significant proportion of the steelhead smolts consumed by gull colonies in the CPR would have survived passage to Bonneville Dam in the absence of gull predation. By analyzing both tern and gull predation in the same model, we were able to estimate that annual steelhead survival probabilities from Rock Island Dam to Bonneville Dam would have been, on average, 0.26 (0.14–0.38) higher in the absence of all tern and gull predation upstream of Bonneville Dam during 2008–2025.

BACKGROUND

Predation by piscivorous colonial waterbirds on out-migrating juvenile salmonids (smolts; *Oncorhynchus* spp.) has been identified as a factor that can significantly limit the survival of some Endangered Species Act (ESA)-listed populations (referred to as Evolutionarily Significant Units [ESUs] or Distinct Population Segments [DPSs]) of anadromous salmonids in the Columbia River basin (CRB). Addressing the impact of avian predation on smolt survival is a component of Biological Opinions and Reasonable and Prudent Alternatives (RPAs) associated with management of the Columbia River Power System and is a critical component of the Columbia River Basin Fish and Wildlife Program. To address concerns about the impact of avian predation on the survival of smolts originating from the CRB, the U.S. Fish and Wildlife Service (USFWS), the U.S. Army Corps of Engineers (USACE), U.S. Bureau of Reclamation (BOR), and their management partners developed and implemented three separate management plans to reduce predation rates on smolts by piscivorous waterbirds nesting at four breeding colonies in the CRB (USFWS 2005, USACE 2014, USACE 2015): the tern and cormorant breeding colonies on East Sand Island (ESI) in the Columbia River estuary (CRE), formerly the largest known colonies for the respective species anywhere; and the two largest tern colonies in the Columbia Plateau

region (CPR), those on Crescent Island in McNary Reservoir and on Goose Island in Potholes Reservoir. The primary goal of these management initiatives was to reduce predation rates (percentage of available smolts consumed) on ESA-listed salmonid ESUs/DPSs by reducing the number of birds breeding at each of these four colony sites. Management initiatives implemented have involved primarily non-lethal strategies for terns (i.e., passive and active nest dissuasion) with limited egg collection under permit and a combination of lethal and non-lethal strategies for cormorants (i.e., culling and egg-oiling, plus reduction of nesting habitat). As part of the management plans for terns, the USACE created or enhanced alternative nesting habitat for terns outside the CRB (i.e., in northeastern California, southern Oregon, and south San Francisco Bay) to compensate for reductions in tern nesting habitat on ESI in the CRE and for elimination of tern nesting habitat on Crescent and Goose islands in the CPR.

Despite some success in managing avian predators to increase smolt survival in the basin, several critical uncertainties remain, and recent developments suggest that predation impacts from piscivorous waterbird colonies may be increasing for some predator species and colonies. First, terns and cormorants from managed colonies in the CRB are relocating to nest at other colony sites within the basin where their per capita (per bird) impacts on smolt survival are as high or higher than at managed colony sites. Second, sufficient management efforts at the Crescent Island tern colony are no longer being implemented, and the colony has been re-established. Third, management to reduce the size of the tern and cormorant colonies on ESI have led to complete or near complete colony failure/abandonment which is contributing to the movement of birds from managed to unmanaged sites within the CRB and may be negatively affecting the conservation status of these bird populations/species. Finally, recent research indicates that smolt consumption rates by several unmanaged gull colonies in the CRB are as great as, and in some cases greater than, those of managed tern and cormorant colonies. Taken together, these developments indicate that continued research, monitoring and evaluation (RM&E) of avian predation in the CRB is warranted if adaptive management is to be successfully implemented and the intended benefits to increase smolt survival from management of avian predators are to be realized.

PROJECT OBJECTIVES

The primary objectives of this study in 2025 were to assess the distribution and size of piscivorous waterbird colonies that consume smolts and to evaluate the effects of predation on smolt survival. Results from this project were also used to investigate the efficacy of ongoing management plans for piscivorous colonial waterbirds, plans that aim to reduce the size of managed tern and cormorant colonies to increase survival of juvenile salmonids in the CRB, and to provide recommendations for adaptive management. As part of these objectives, this study located and estimated peak colony size at all piscivorous waterbird colonies within foraging range of smolts in the middle Columbia River, lower Snake River, lower Columbia River, and Columbia River estuary, including information on any new or incipient colonies; estimated colony-specific predation/consumption rates by multiple piscivorous waterbird species; and measured the cumulative or system-wide effects of avian predation on salmonid smolt survival,

including an investigation of factors that influence smolt susceptibility to avian predation and the additive effects of predation on smolt survival. Avian predation RM&E in 2025 was a continuation of work that commenced in 2020.

METHODS & ANALYSES

This work is part of an ongoing, comprehensive program to implement, monitor, and evaluate avian predation and avian predation management plans, including adaptive management, to reduce the impacts of avian predators on the survival of ESA-listed juvenile salmonids in the CRB. Similar to years past, action effectiveness monitoring was conducted in the CRB at both managed and unmanaged sites where terns and cormorants might disperse to re-nest in response to management. The focus of this work was to evaluate the efficacy of ongoing management initiatives to reduce avian predation on smolts and to help identify new and emerging avian predation concerns that may warrant future management consideration. The primary methods used in 2025 (see *below*) were the same as those used in previous years, ensuring that results were comparable across years.

NEST DISSUASION ACTIVITIES

As part of the *Inland Avian Predation Management Plan* (IAPMP), nest dissuasion activities on Goose Island in Potholes Reservoir were conducted by the BOR and its contractor, U.S. Department of Agriculture-Animal and Plant Health Inspection Service-Wildlife Services (USDA-APHIS-WS). These activities were summarized in an annual report completed by the USDA (USDA-APHIS-WS 2025). The Fisheries Field Unit (FFU) at the USACE carried out nest dissuasion efforts associated with the *Caspian Tern Management Plan in the Columbia River Estuary* on ESI and as an adaptive management component on Rice Island. A description of those activities in 2025 was summarized in reports produced by USACE (Roberts et al. 2026, Strong et al. 2026). No active nest dissuasion activities on terns were conducted by USACE at Crescent Island in 2025, as those activities ended in 2021. Some passive nest dissuasion activities, however, were performed as an adaptive management component to the IAPMP at Crescent Island and other islands (e.g., Badger Island; B. Parker, Columbia River Inter-Tribal Fish Commission, pers. comm). As part of the IAPMP both passive and active dissuasion was implemented at an unnamed island in North Potholes Reservoir by the USDA-APHIS-WS (USDA-APHIS-WS 2025). Finally, a gull colony on Miller Rocks (a series of rocky islets in The Dalles Reservoir) was managed to prevent gull nesting and reduce the size of the colony over the course of the last four (2022–2025) breeding seasons using both passive and active dissuasion techniques (T. De Boer, Yakama Nation, pers. comm.; see *below* for more details).

NESTING DISTRIBUTION & COLONY SIZE

Monitoring of piscivorous waterbird colonies in the CRB (*Figure 1*) was conducted from the air (fixed-wing aircraft and unmanned aerial vehicles), the water (boat-based surveys), and/or from

shore, with precautions taken to minimize disturbance to actively nesting colonial waterbirds. Whenever possible, counts of piscivorous waterbirds at prospective nesting colony sites were differentiated by behavior (i.e., nesting vs. roosting), age (i.e., adult vs. juvenile), and location on the island. Each island was also closely monitored for the formation of new satellite colonies (i.e., away from the former colony site or in and around areas of passive nest dissuasion). Data collection methodologies followed established protocols such that the data collected in 2025 could be compared with analogous data collected in previous years and at other colonies (Antolos et al. 2004; Adkins et al. 2014; Roby et al. 2015; Collis et al. 2016, 2017, 2018, 2019, 2020, 2021a; Roby et al. 2021a; Evans et al. 2022b; 2023, 2024, 2025).

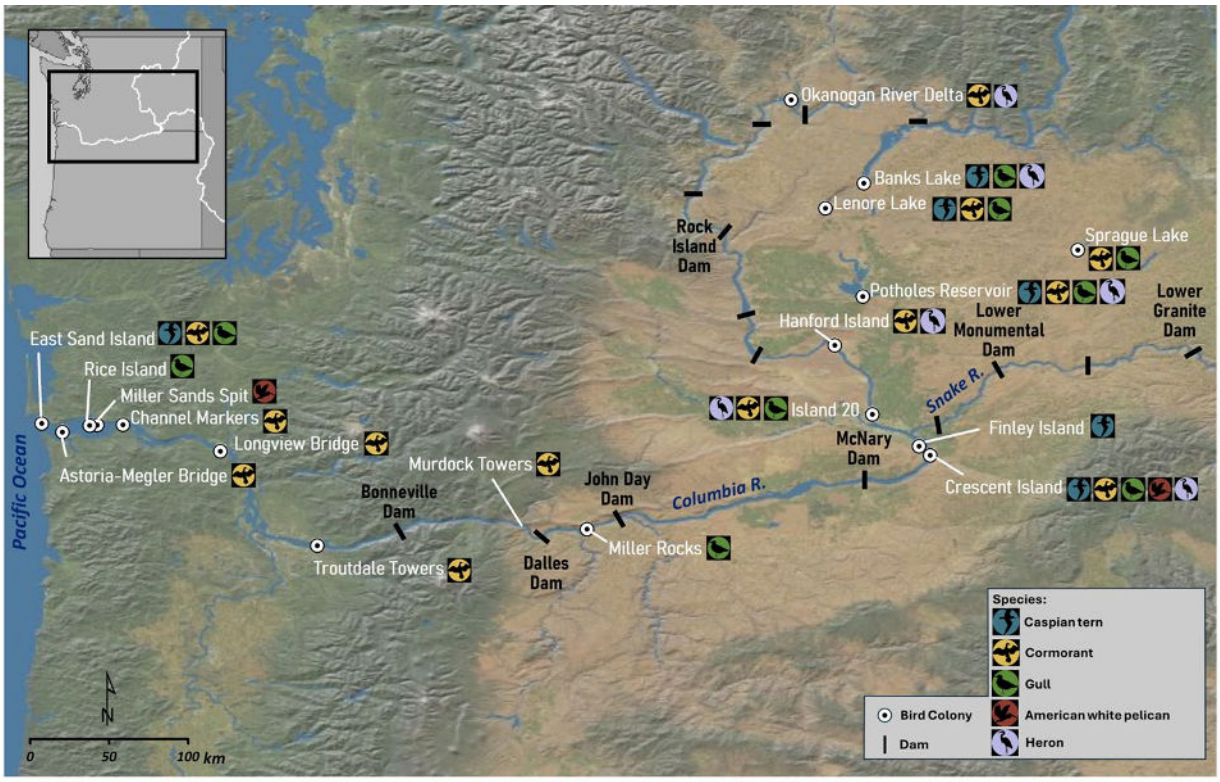


Figure 1. Study area in the Columbia River basin in 2025. Colonies include birds nesting on East Sand Island (ESI), Astoria-Megler Bridge (AMB), Rice Island, Miller Sands Spit (MSI), various channel markers, Longview Bridge, Troutdale Towers (TRT), Murdock Towers (MUT), Miller Rocks (MRI), Crescent Island (CSI), Finley Island (FIN), Island 20 (IS20), Handford Island (HAN), islands in Potholes Reservoir (PTI), Harper Island (HPI) in Sprague Lake, islands in Lenore Lake (LLI), islands in Banks Lake (BLI), and at the Okanagan River delta.

Periodic (bi-weekly to monthly) ground- and boat-based surveys were conducted throughout the breeding season to identify where piscivorous colonial waterbirds were nesting in the CRB in 2025. Three fixed-winged, basin-wide aerial surveys (18–19 May, 7–8 June, and 21–22 June) were conducted to help identify all active colonies of piscivorous waterbirds within the study area, to assess nesting chronology, and to estimate peak colony size. An early season flight to locate potential incipient tern colonies in the CPR was also conducted on 27 April. Colony size was estimated using high-resolution digital aerial imagery corresponding with the peak in

nesting activity at each colony and enumerating visible birds using ArcGIS (ArcGIS Pro 3.3.0, Redlands, CA). Colony size is reported as the number of birds on-colony, or, in the case of terns cormorants, and herons, the number of breeding pairs.

At Goose Island, and other suitable nesting colony sites in Potholes Reservoir, the activities of terns and gulls were monitored biweekly throughout the breeding season. The remaining piscivorous waterbird colonies (*Figure 1*) in the CPR were monitored one to two times per month throughout the breeding season. The cormorant colony on the AMB in the Columbia River estuary was monitored by boat on two occasions (21 May, 4 June) to determine nesting chronology and peak colony size. Finally, tern and cormorant colonies on ESI and the tern colony on Rice Island were monitored by the USACE-FFU in 2025 (Roberts et al. 2026, Strong et al. 2026).

AVIAN PREDATION RATES

Using previously established methods, smolt passive integrated transponder (PIT) tags recovered (detected) on bird colonies were analyzed to estimate colony-specific predation rates (percentage of available tagged fish consumed by birds) on ESA-listed salmonid ESUs/DPSs. The study also investigated the cumulative effects of avian predation (predation from all colonies combined) on smolts and compared mortality due to avian predation to total mortality (1 - survival). Results provide information on the system-wide effects of avian predation and identify which predator species (terns, cormorants, gulls, pelicans) and colonies posed the greatest potential threat to smolt survival in 2025.

PIT-tagging of Upper Columbia River Steelhead

To help ensure adequate numbers of ESA-listed UCR steelhead were available for predation analyses, smolts were intentionally captured, PIT-tagged, and released into the tailrace of Rock Island Dam (RIS) on the middle Columbia River as part of this study. Previous research has demonstrated that the UCR steelhead population is highly susceptible to predation by terns, cormorants, and gulls (Evans et al. 2012, Evans et al. 2019, Payton et al. 2020) and is therefore a suitable group to evaluate the efficacy of management actions aimed at reducing avian predation. Efforts to tag steelhead smolts at RIS as part of avian predation studies have also been ongoing since 2008, providing a long-term dataset in which to evaluate relative changes in predation associated with both managed and unmanaged piscivorous waterbird colonies (Evans et al. 2012, Evans et al. 2019). The tagging of steelhead smolts at RIS also provides a means to investigate factors that influence smolt susceptibility to avian predation (e.g., fish length, condition, rear-type {hatchery, wild}; Hostetter et al. 2023) and to determine to what degree avian predation is an additive versus compensatory source of mortality (Payton et al. 2020).

A detailed description of the sampling methods used to PIT tag steelhead at RIS are provided in Evans et al. (2014). In brief, steelhead were captured at the RIS juvenile fish trap, PIT-tagged (*Biomark* model APT12), measured (mm; fork-length), condition-scored (based on the presence/absence of injuries, descaling, and disease), and released into the tailrace of RIS

during the peak smolt out-migration period of April to June. Steelhead were randomly selected for tagging (i.e., tagged regardless of condition, rear-type, or size) and were tagged in-concert with, and in proportion to, the run-at-large to ensure that the tagged sample was representative of the steelhead population (tagged and untagged) passing RIS. All juvenile steelhead captured at RIS were part of the ESA-listed UCR steelhead DPS, as all hatchery and wild steelhead originating from tributaries upstream of RIS are part of the ESA-listed population (NOAA 2021). The sampling approach used at RIS as part of this study ensures that (1) steelhead from all naturally spawning stocks of the ESA-listed DPS are included in the sample (Entiat, Methow, Okanogan, and Wenatchee rivers), (2) that smolts of all sizes, conditions, and rear-types are included in the sample in proportion to their relative abundance, and (3) that smolts are tagged in-concert with the run at-large; criteria that allow us to make credible inference about the entire ESA-listed UCR steelhead population (see also Payton et al. 2020 and Evans et al. 2024a). The target sample size goal was to PIT-tag approximately 7,000 steelhead smolts for use in predation and survival analyses in 2025. This target sample size was selected because it was consistent with previous steelhead PIT-tagging efforts at RIS (Evans et al. 2014; Evans et al. 2019, Payton et al. 2020) and was estimated to generate predation rate estimates with a precision of approximately ± 2% for those colonies that forage on smolts in the middle Columbia River, such as the managed tern colony on Goose Island in Potholes Reservoir.

For several other ESA-listed salmonid ESUs/DPSs, adequate numbers of PIT-tagged smolts were available for inclusion in predation rate analyses based on other regional tagging studies that occurred on the lower Snake River or upstream of McNary and Bonneville dams on the Columbia River. However, since fish from these other studies were generally not randomly selected for tagging (e.g., fish were culled based on size, condition, and rear-type) and were not tagged in proportion to and in-concert with the run at-large (tagged and untagged), predation and survival results may be biased to an unknown degree relative to fish tagged at RIS as part of this study. Furthermore, in some cases, tagging of ESA-listed fish at collection sites at mainstem dams has been discontinued (e.g., spring Chinook at RIS), so adequate sample sizes for use in this study may not exist. The addition of a PIT tag detection system or array at Lower Granite Dam spillway 1 in 2020, however, has dramatically increased the number of tagged SR smolts available for predation analyses (Axel et al. 2024).

Predation Rate Estimates

The previously published methods of Evans et al. (2012) and Hostetter et al. (2015) were used to recover smolt PIT tags from piscivorous waterbird colonies and to estimate colony- and salmonid ESU/DPS-specific predation rates on tagged fish. Detailed analytical methods are provided on the Monitoring Resources website (https://www.monitoringresources.org/) and in the Avian Predation Synthesis Report (Roby et al. 2021a). In brief, to recover (electronic detection) fish PIT tags, PIT tag antennas were used to scan the entire area occupied by nesting birds following the breeding season, with a minimum of two complete sweeps or passes conducted at each colony. This study employed a hierarchical Bayesian model that integrated multiple factors of uncertainty in the tag recovery process, including imperfect detection of PIT tags on bird colonies, on-colony PIT-tag deposition probabilities, and temporal changes in smolt

availability to birds nesting at each colony. Predation rates were modeled independently for each salmonid ESU/DPS and bird colony. The probability of recovering a PIT tag from a smolt on each colony was modelled as the product of the probability that (1) the fish was consumed (θ), (2) the PIT tag was deposited on-colony (ϕ), and (3) the PIT tag was detected on-colony after the breeding season (ψ_i):

$$k_i \sim \text{Binomial}(n_i, \theta_i * \phi * \psi_i)$$

where k_i is the number of smolt PIT tags recovered from the number available (n_i) in week *i*. The detection probabilities (ψ_i) and predation probabilities (θ_i) were each modeled as a function of time. The probability, ψ_i, that a tag, consumed in week *i* and deposited on the colony is detected, is assumed to be a logistic function of week. That is:

$$\text{logit}(\psi_i) = \beta_0 + \beta_1 * i$$

where β_0 and β_1 are both derived from non-informative priors (normal [0, 1000]).
Predation rates nearer together in time are more similar than those further apart in time (Evans et al. 2016; Payton et al. 2019). To reflect this, variation in weekly predation probabilities, θ_i, was modeled as a random walk process with mean μ_θ and variance σ_θ^2, where:

$$\text{logit}(\theta_i) = \mu_\theta + \sum_{w \le i} \varepsilon_w$$

and $\varepsilon_w \sim normal(0, \sigma_\theta^2) \; \forall \; w$. Non-informative priors were placed on these two hyperparameters: $\text{logit}^{-1}(\mu_\theta) \sim uniform(0,1)$ and $\sigma_\theta^2 \sim uniform(0,20)$. This allows each week (*i*) to have a unique predation probability (θ_i), while still sharing information among weeks improving precision.

Informative Beta priors were used to model deposition probability(ϕ). The shape parameters (α, β) are dependent on the predator species (cormorant, tern, gull, pelican, or heron) and are assumed to be mutually independent from colony to colony. For terns, α = 16.20 and β = 6.55 was assumed, for cormorants α = 15.98 and β = 15.29 was assumed, for gulls α = 33.71 and β = 183.61 was assumed, for pelicans α = 6.70 and β = 7.37 was assumed, and for herons α = 19.75 and β = 20.22 was assumed.

Weekly predation estimates were defined as the estimated number of PIT-tagged smolts consumed divided by the total number available each week. Annual predation rates were derived as the sum of the estimated number of PIT-tagged smolts consumed each week divided by the total number of PIT-tagged smolts available:

$$\sum_{i \in breeding\ season} (\theta_i * n_i) \Big/ \sum_{i \in breeding\ season} (n_i)$$

Summation of weekly consumption estimates is necessary to accurately reflect weekly variation and autocorrelation of predation rates and thus to create unbiased annual rates with accurate assessments of precision (Hamilton 1994).

For some colonies and years included in the study, not all nesting areas or habitat used by birds during the breeding season were accessible to researchers (e.g., truss sections of the Astoria-Megler Bridge cormorant colony). In these few cases, per capita predation rates ($\dot{\theta}_p$) were calculated in areas or plots (p) that were scanned for PIT tags and where a known number of birds nested as:

$$\dot{\theta}_p = \sum_p {\sum_i (\theta_{ip} * n_i) / \sum_i (n_i)} \Big/ C_p$$

where C_p is colony size within plot p. Colony-wide estimates of predation (predation by all breeding pairs at that colony) were then calculated by multiplying the sum of the per capita predation rates by the peak colony-wide measure of colony size as:

$$\theta = \frac{\sum_i \sum_p (\theta_{ip} * n_i) \Big/ \sum_i n_i * C_{all}}{\sum_p C_p}$$

Models were analyzed using the software STAN (2022), accessed through R version 3.6.3 (RDCT 2014), and using the rstan package (version 2.30; SDT 2022). Reported predation rate estimates represent simulated posterior medians along with 95% highest (posterior) density intervals (95% Credible Interval [CRI]) calculated using the HDInterval package (version 0.2.0; Meredith and Kruschke 2016). Annual predation rates were calculated for salmonid ESUs/DPSs where ≥ 500 PIT-tagged individuals were available to birds at each colony to avoid imprecise results that may occur from small sample sizes of available PIT-tagged smolts (Evans et al. 2012).

Efficacy of Avian Predation Management Plans: Predation rate estimates were used to compare smolt losses prior to and during implementation of management actions at tern and cormorant colonies in the CRB, data critical to evaluate the effectiveness of management plans aimed at reducing predation rates at managed bird colonies. Data to address the efficacy of these management plans varied by plan, funding, and year and are detailed below for research conducted in 2025:

Inland Avian Predation Management Plan – A stated goal of the IAPMP is to reduce the impact of predation by terns on ESA-listed salmonids to less than 2.0% per salmonid ESU/DPS, per colony, per year (USACE 2014). To help evaluate the efficacy of the IAPMP at reducing predation impacts to those levels, predation rates were compared between the pre-management period (2007–2013) and the management period (2014–2025) at both managed and unmanaged tern colonies in the CPR. The IAPMP also uses the most recent 3-year average

during the management period (currently 2023–2025) to further evaluate if management goals are being met or whether additional adaptive management is necessary (USACE 2014).

Estuary Caspian Tern and Double-crested Cormorant Management Plans – A stated goal of estuary tern and cormorant management plans was to reduce the size of the tern and cormorant colonies on East Sand Island by about 60% and thereby reduce tern and cormorant predation rates on ESA-listed salmonids in the CRE by about 60% (USFWS 2005, USACE 2015). Since 2019, it has been the responsibility of the USACE and its contractors to recover smolt PIT tags and to estimate ESU/DPS-specific predation rates and per capita predation rates by terns and cormorants on East Sand Island. During 2021–2025, the USACE Fisheries Field Unit (FFU) recovered smolt PIT tags from the tern and cormorant colonies on East Sand Island (Roberts et al. 2026) but estimates of ESU/DPS-specific predation rates and per capita predation rates have not been generated since 2020 (see also Appendix A).

As part of this study, smolt PIT tags recovered by the USACE-FFU on East Sand Island were incorporated to evaluate the cumulative or system-wide effects of avian predation on select salmonid ESU/DPSs, those that must migrate through the forage range of multiple colonies, in 2025. Estimates of cumulative predation effects rely on a different analytical framework (see *Cumulative Predation and Survival* section) than those of colony-specific predation rates (see *Predation Rate Estimates* section), so the two estimates are not directly comparable to one another (Payton et al. 2019, Evans et al. 2021). Furthermore, several other ESA-listed salmonid ESUs/DPSs (e.g., those that originate from the middle and lower Columbia rivers) are susceptible to predation by East Sand Island terns and cormorants, but those impacts have not been documented since 2020.

Cumulative Predation and Survival: This study evaluated the cumulative and reach-specific effects of avian predation on UCR steelhead tagged at or detected (i.e., previously tagged recaptures) at RIS and on SR steelhead, yearling Chinook, subyearling Chinook, and sockeye that were tagged/detected at Lower Granite Dam (LGR) in 2025. Downstream river reaches were defined by locations where PIT-tagged smolts were redetected alive following passage at RIS and LGR, which include Little Goose Dam and Lower Monumental Dam (LMN) for SR migrants and McNary Dam (MCN), John Day Dam (JDJ), Bonneville Dam (BON), a net trawl detection system and at antennas located on pile dikes downstream of BON for both SR and UCR migrants. Predation rates were based on the proportion of available smolts consumed by birds within each river reach or from all reaches combined, and survival rates were based on the proportion that survive out-migration through each river reach or all reaches combined. In addition to avian predation, salmonid smolts are also subject to non-avian sources of mortality (e.g., hydroelectric dam passage, predation by piscivorous fish, disease, and other factors) and determining to what degree avian predation limits survival relative to these other sources of mortality is critical for prioritizing recovery actions for ESA-listed salmonids (Evans et al. 2016, Payton et al. 2019, Evans et al. 2022a). As such, estimates of avian predation were compared to estimates of total smolt mortality (1 - survival) to determine what proportion of all mortality sources were due to bird predation in 2025. To help describe recent trends in predation and

survival, results from 2025 were compared with those from years past, including the results of previously published studies (Evans et al. 2022a, Evans et al. 2025), which date back to 2008.

Tagged smolts included in these analyses were both hatchery and wild (natural origin) fish. Not all hatcheries are included in the ESA-defined ESUs/DPSs, and in the case of Chinook, yearlings may be a mixture of two distinct ESA-listed populations (spring- and fall-run; NOAA 2021); thus, populations reflected the same species and age-class but some unknown proportion of the fish were likely not part of the ESA-listed ESU or were a mixture of two ESA-listed ESUs. Only naturally emigrating, in-river smolts within each river reach were included in analyses, with all transported smolts excluded following, but not prior to, their removal from the river in fish barges or trucks. Adults returning to the Columbia River following ocean residency were also detected at PIT tag arrays located in fishways at BON, the first dam encountered by UCR and SR adults following ocean residency (see Evans et al. 2022a for additional details).

The joint mortality and survival (JMS) estimation methods of Payton et al. (2019) were used to independently estimate reach-specific and cumulative predation and survival probabilities for each salmonid species, population, and age-class evaluated. As described above, this hierarchical state-space Bayesian model incorporated both live and dead detections of PIT-tagged fish in space and time to simultaneously estimate consumption and survival through up to nine sequential river reaches (or segments), defined by passive recapture opportunities in which smolts were assumed to only travel downstream. In brief, the model used two vectors, $\boldsymbol{y}$ and $\mathbf{r}$, to describe each fish's recapture and recovery history throughout each downstream river reach and each of the bird colony recovery sites under consideration. Each vector $\mathbf{y}$ was a J-length vector –$J = 9$ for SR fish and $J = 5$ for UCR fish– where y_j was an indicator variable of a fish's recapture at recapture opportunity j for $j \in \{1,2, \dots, J-1\}$ and $y_J = 0$ as there was no live recapture site downstream of the net detector in the Columbia River estuary. Recoveries were indicated by $\mathbf{r}$, a D-length vector, where D represents the number of recovery areas each year, with a single element equal to one and the rest of the elements are zero, where $r_d = 1$ indicated recovery on colony d for $d \in \{1,2, \dots, D-1 = 14\}$, and $r_D = 1$ indicated a fish was unrecovered. Parameters used in the model included:

> **$\boldsymbol{\Theta}$,** a DxJ matrix where $\Theta_{d,j}$ represented the probability (from release) that a fish survived to recapture opportunity $j-1$ –where $j = 0$ represents release from RIS/LGR– and then subsequently succumbed to depredation by colony d for $d \in \{1,2, \dots, D-1\}$ or some other cause of mortality for $d = D$, prior to arrival at recapture opportunity $j+1$. Implicit from this parameterization is that survival from release through segment k is equal to $1 - \sum_{j \le k} \sum_d \Theta_{j,d}$.
>
> **p,** a J-length vector where p_j represented the probability that a fish alive at recapture opportunity *j* was successfully recaptured. This study defines $p_J = 0$, as there is no recapture opportunity downstream of the Net Detector.

$\boldsymbol{\gamma}$, a D-length vector where γ_d represented the probability of recovering a fish which died due to depredation by colony d for $d \in \{1,2,\dots,D-1\}$, and γ_{15} = 0 represented the lack of recovery opportunity for fish which died from all other unspecified causes.

The model employed can be expressed by incorporating these parameters into recursive functions, $\chi_{j,d}$, defined to represent the probability a fish entering segment j is not subsequently recaptured and is recovered on colony d (i.e. r_d=1), such that

$$\chi_{j,d} = \theta_{j,d} * \gamma_d + \left(1 - p_{j+1}\right) * \chi_{j+1,d} \; for \; d \in 1,\dots,D-1,$$

or not recovered at all (i.e. r_{15}=1), such that

$$\chi_{j,D} = \sum_d \theta_{j,d} * (1 - \gamma_d) + \left(1 - p_{j+1}\right) * \chi_{j+1,D}.$$

Then, if m is defined as the final recapture opportunity at which the fish was seen, with $m = 0$ representing a fish never reseen following release, the portion of the aggregate likelihood associated with each fish's recapture/recovery history can be expressed as

$$L = \prod_{j \le m} \left(p_j^{\,y_j} * \left(1 - p_j\right)^{(1-y_j)} \right) * \prod_d {\chi_{m+1,d}}^{r_d},$$

where the former product describes a fish's recapture history prior to its final recapture and the latter product describes the fish's subsequent recovery or lack thereof following its final recapture.

Each year, a subset of tagged smolts were collected and removed from the river in fish barges or trucks at one of the first three capture/recapture sites on the lower Snake River: Lower Granite Dam, Little Goose Dam, or Lower Monumental Dam. Once collected for transportation, these fish were no longer available in-river and, as such, the capture-recapture-recovery history for these fish was truncated following their removal at each dam. The likelihood associated with the truncated capture-recapture history of each of these fish can be expressed as:

$$L = \prod_{j<m} \left(p_j^{\,y_j} * \left(1 - p_j\right)^{(1-y_j)} \right) * \left(1 - \sum_d \sum_{j<m} \theta_{j,d} \right)$$

Two further modelling considerations beyond those of Payton et al. (2019) were included to better inform the spatially explicit estimates of predation effects. First, the informed partitioning methods of Evans et al. (2022a) were also used to allow for a sharing of information across years to increase the precision of segment-specific estimates. In brief, a vector of aggregate life-path possibilities is constructed including the probability of survival to return as an adult, the cumulative probability (across all segments) of depredation by each colony, and segment specific probabilities of death from unspecified sources to be the basis for

modelling variations across days. The cumulative probability of depredation by each colony is subsequently partitioned across river segments with proportionate impacts among reaches assumed to be similar among years. Second, pelicans and gulls occurred in co-nesting areas on Badger Island during 2015–2024 and on Crescent Island in 2025, with portions of each genus' colonies overlapping spatially, creating “mixed” genus areas. This study employed the methods of Payton et al. (2023) to incorporate supplemental data (i.e., aerial nest count surveys) to inform what proportion of each genus was nesting in the “only” areas versus the “mixed” areas. Then, by assuming the odds of a tag consumed by a given genus was deposited in the single genus portions of each colony versus in the “mixed” area was similar to the odds of a bird of that genus nesting in the single genus portion, it was possible to estimate the portion of tags recovered from the “mixed” portion of the colony were attributable to each predator genera.

To measure inter-annual temporal variation in probabilities, fish were partitioned into weekly release groups with the assumption that fish released within the same week experienced similar rates of mortality/survival, recapture, and recovery (Payton et al. 2019). While all rates were assumed to be independent among years, weekly cohorts closer in time were assumed to be more alike than those further apart. The serial correlation in probabilities was assumed and accounted for as described by Payton et al. (2019). The prior distribution for the initial week’s detection probability in each year was defined to be $\mathrm{uniform}(0,1)$. Analogously, the prior distribution assigned for the life paths simplexes in the initial week of each year was assumed to be $\mathrm{D}(\mathbf{1})$, where $\mathbf{1}$ was an appropriately sized vector of ones. Weakly-informative priors of $\mathrm{half}-\mathrm{normal}(0, 1.5)$ were implemented for the variance parameters describing inter-weekly variation.

The recovery parameters, γ_d, represent the combined probability that a consumed tag was deposited on-colony, d, and the probability that the tag is subsequently detected (recovered) by researchers following the breeding season given tag deposition on a colony. The simulated posterior distributions of deposition probabilities and colony-specific detection probabilities which were derived, summarized, and presented in previous studies were employed here as informative prior distributions in the derivation of predation probability estimates.

Models were analyzed using the software STAN (SDT 2022), accessed through R version 3.6.2, and using the rstan package (version 2.19.3). To simulate random draws from the joint posterior distribution, four Hamiltonian Monte Carlo (HMC) Markov Chain processes were run. Each chain contained 4,000 warm-up iterations followed by 4,000 posterior iterations thinned by a factor of 4. Chain convergence was visually evaluated and verified using the Gelman-Rubin statistic (Gelman et al. 2013); only chains with zero reported divergent transitions were accepted. Posterior predictive checks compared simulated and observed annual aggregate raw recapture and recovery numbers to ensure model estimates reflected the observed data. Reported estimates represent simulated posterior medians along with 95% highest (posterior) density intervals (95% Credible Interval [CRI]) calculated using the HDInterval package (version 0.2.0).

Additive Effects of Predation: Understanding the degree to which avian predation limits fish survival is paramount to evaluating the efficacy of management actions aimed at increasing fish survival. Specifically, if reductions in predation rates are associated with higher rates of fish survival (i.e., avian predation adds to total mortality) or if most fish consumed by birds were destined to die regardless of avian predation (i.e., avian predation is compensated for by other mortality factors). The additive mortality hypothesis predicts that predation is directly related to survival in space and time. The compensatory mortality hypothesis predicts that predation and survival are unrelated in space and time, at least up to the point where the level of predation exceeds natural mortality, whereby the proportion of fish consumed by predators cannot be greater than the proportion that was alive at any given time (Payton et al. 2020).

Previously published research has consistently suggested that predation by terns was an additive source of smolt mortality during the smolt life-stage and a partially additive source of mortality to adulthood, with significantly more smolts estimated to survive outmigration and to return to Bonneville Dam as an adult in the absence of tern predation (Payton et al. 2020, Payton et al. 2021, Evans et al. 2022a, Evans et al. 2025). Assessments of the relationship between gull predation and fish survival, however, were not as conclusive as those observed in terns (Payton et al. 2021, Evans et al. 2022a). This was due to a shorter time-series of available data and that gulls co-nesting with terns may kleptoparasite (steal) fish from terns which, if unaccounted for, could bias estimates of gull additivity (Payton et al. 2021, Evans et al. 2022a). In 2025, to further investigate the relationship between gull predation and smolt survival, we updated the time-series to include data from all years where gull predation was measured (2008–2025) and we developed a model to simultaneously estimate the additivity in gulls and terns, ensuring estimates of gull additivity were not biased by terns (i.e., estimates of gull additivity account for tern additivity and vice versa).

Weekly and annual estimates of predation and survival probabilities on PIT-tagged UCR steelhead smolts, a population exposed to predation from multiple gull colonies, that were initially tagged/recapture/released at RIS (see *above*) were used to investigate to what degree gull predation was an additive source of mortality during outmigration from RIS to Bonneville Dam (BON). We further developed the mark-recapture-recovery model of Payton et al. (2020) to assess the strength, magnitude, and direction of the relationship between gull predation on steelhead smolt survival between RIS and BON, while simultaneously estimating the analogous relationship associated with predation by terns. To accomplish this, we aggregated cumulative survival and predation across all segments prior to this recapture (detection) point (denoted as the set [REACH]) and across mortality sources associated with each genus of avian predator (referred to as the set [GENUS]). We therefore sharpen our focus on the survival rate to the specified downstream dam, aggregated total mortality prior to this dam for each predator genus, and a single rate of other mortality due to all other mortality factors (referred to as the set [OTHER]). That is, we let

$$\theta_w^{\{LAXX\}} = \sum_{j\in\{REACH\}} \sum_{c\in\{LAXX\}} \theta_{j,c,w},$$
$$\theta_w^{\{CATE\}} = \sum_{j\in\{REACH\}} \sum_{c\in\{CATE\}} \theta_{j,c,w},$$
$$\theta_w^{\{OTHER\}} = \sum_{j\in\{REACH\}} \sum_{c\notin\{GENUS\}} \theta_{j,c,w},$$

and

$$s_w^* = 1 - \theta_w^{\{LAXX\}} - \theta_w^{\{CATE\}} - \theta_w^{\{OTHER\}}$$

where $\theta_{j,c,w}$ refers to predation in segment j, by birds associated with colony c, in week w, and s_w^* refers to survival through the segment. Therefore, within each year, we can construct simplex weekly rates of survival and aggregated predation as,

$$\left[s_w^* \quad \theta_w^{\{LAXX\}} \quad \theta_w^{\{CATE\}} \quad \theta_w^{\{OTHER\}}\right]^T.$$

Following Sandercock et al. (2011), additive mortality rests on an assumption of annual "baseline survival" rate, s^0. This parameter s^0 can be interpreted as the hypothetical survival rate in the absence of bird predation. Therefore, in the absence of tern and gull predation, we assume

$$s^0 + \epsilon_w = 1 - \Theta_w^{\{OTHER\}}$$

where s^0 is the yearly "baseline survival" rate across the segments of {REACH}, around which weekly survival rates were assumed to vary completely at random with the variation, denoted by ϵ_w.

Independent of the variation described above, any further decrease/increase observed in survival is assumed to be proportional to the level of genus-specific predation associated with that reach. The magnitude of this proportional relationship is measured with the parameter b.

$$s_w^* = (s^0 + \epsilon_w)\left(1 - b^{\{\text{LAXX}\}}\Theta_w^{\{\text{LAXX}\}}\right)\left(1 - b^{\{\text{CATE}\}}\Theta_w^{\{\text{CATE}\}}\right).$$

However, in cases in which $\theta_w^{\{LAXX\}}$ + $\theta_w^{\{CATE\}}$ accounts for all unmeasured mortality, the relationship between survival and predation necessarily becomes directly inverse. In cases in which survival is zero, there is necessarily no relationship between predation and survival.

The parameterization outlined above is in line with that of the conventional Sandercock et al. (2011) model but a departure from some of our previous analyses, like those of Payton et al. (2020). In Payton et al. (2020) the additivity model was reparametrized to develop a statistic which would directly measure the "portion of depredated fish that would not have died if not for avian predation", defined as the parameter a. However, a consequence of this approach is that the parameter a encompasses both the portion of fish "destined to die" and the extraneous super-additive/compensatory factors which may be impacting the system (e.g., latent mortality, kleptoparasitism; see Payton et al. 2020). For this analysis we disentangle these processes by defining $a = s^0 * b$, where b measures these latter effects; effectively reverting the parameterization to the original Sandercock et al. (2011) model. We then assume smolt survival among most reaches in the CRB is relatively similar among years rather than

uniformly distributed between zero and one each year. We therefore model this similarity $as\ s_y^0 \sim normal(\mu_{s^0}, \sigma_{s^0})$ for each year y. We also assume b is also similar among years; $b_y \sim normal(\mu_b, \sigma_b)$. Results herein are still presented with respect to a for the sake of comparability with previously published studies.

In this modelling effort there were four competing models to compare: (1) a null model with no additive predation effects (i.e., both $\{LAXX\}$ and $\{CATE\}$ included in $\{OTHER\}$), (2) a model with only LAXX additivity (i.e., $b^{\{\mathrm{LAXX}\}}$ estimated while $b^{\{\mathrm{CATE}\}}$ = 0), (3) a model with only CATE additivity (i.e., $b^{\{\mathrm{CATE}\}}$ estimated while $b^{\{\mathrm{LAXX}\}}$ = 0), and (4) a full model with both CATE and LAXX additivity. To determine which configuration of predator additivity best represented the observed data, we employed leave-one-out cross-validation (LOO-CV; Vehtari et al. 2017). LOO-CV provides a measure of out-of-sample predictive performance while accounting for model complexity, with lower expected log pointwise predictive density (ELPD) values indicating better model fit. LOO-CV, as evaluated with expected log pointwise predictive density (ELPD), is a Bayesian approach that estimates how well a model will predict new, unseen data. LOO-CV is preferred over DIC as a model selection tool because, unlike DIC, LOO-CV does not produce negative effective parameter counts, is invariant to reparameterization, and fully integrates over the posterior distribution rather than relying on a single-point summary (Vehtari et al. 2017). AIC and BIC are frequentist methodology (rather than Bayesian) and therefore inapplicable to this analysis. We used the difference in ELPD (ΔELPD) and associated standard errors to assess the relative support for each model. When expected log pointwise predictive density (ELPD) differences are normal they follow an approximately normal distribution, and the standardized differences (Diff_z) can therefore be interpreted approximately as normal distribution z-scores for statistical comparison.

RESULTS & DISCUSSION

NEST DISSUASION ACTIVITIES

Columbia Plateau Region

For the fifth consecutive year, the BOR and its contractor (USDA-APHIS-WS) installed passive nest dissuasion arrays (i.e., stakes, cable, and flagging) on upland habitat that was suitable for tern nesting on Goose Island in Potholes Reservoir prior to the onset of the breeding season. During regular visits to Goose Island to conduct active hazing and to collect any tern eggs (under permit) that were discovered, the contractor made necessary repairs to the passive nest dissuasion array as needed. USDA-APHIS-WS also used boats to patrol islands in northern Potholes Reservoir for terns and to use passive and active nest dissuasion techniques to prevent tern nesting in these areas, as warranted. For the second consecutive year, monitoring efforts associated with this study discovered a tern and gull colony on an unnamed island in northern Potholes Reservoir. In 2025, our research team first identified the gull colony during a reconnaissance aerial survey on April 27. The tern colony was then located by our research team during a boat survey on May 7, and we immediately notified USDA-APHIS-WS of its

existence. Within 48 hours of notification, USDA-APHIS-WS deployed their team to remove tern eggs (under permit) and install passive dissuasion (i.e., flagging, fencing, and a coyote effigy) on the island, with no (zero) breeding terns observed at the site for the remainder of the 2025 breeding season. The gull colony, which surrounded the tern colony, continued to nest on the island following installation of tern dissuasion arrays. Further details on the nest dissuasion efforts performed by USDA-APHIS-WS on Goose Island and elsewhere in Potholes Reservoir can be found in annual reports prepared by the contractor (USDA-APHIS-WS 2022–2025).

No active hazing of terns during the breeding season has been conducted on Crescent Island since 2015, the first year of management actions on Crescent Island. The growth of willows planted on Crescent Island in 2016 had mostly eliminated upland habitat suitable for tern nesting on Crescent Island by 2019 (*Figure 2*). Consequently, the other passive nest dissuasion materials (i.e., fencing, stakes, and rope) that were previously installed on Crescent Island were removed prior to the 2020 breeding season. In 2020, however, gulls reestablished a breeding colony in semi-vegetated habitat on Crescent Island (see *below*) and have nested there every year since. Gulls were observed nesting in open areas where willows had died and/or where beaver herbivory on the willows had created patches of ground with minimal vegetative cover. The gull colony on Crescent Island rapidly grew from roughly 400 individuals in 2020 to 4,500 individuals by 2022. In addition to gulls, pelicans also started to nest on Crescent Island in 2024 (50 individuals) and the colony rapidly expanded to roughly 2,800 individuals in 2025 (*Table 2*; see also below). The growth of gull and pelican colonies on Crescent Island following the removal of passive dissuasion materials (i.e., silt fencing and ropes/flagging) and active hazing efforts, as well as beaver herbivory, has led to a substantial dieback of willows and other native vegetation (*Figure 2*). Prior to the 2023, 2024, and 2025 breeding seasons, agency personnel placed large woody debris in areas of unvegetated habitat on Crescent Island (B. Parker, Columbia River Inter-Tribal Fish Commission, pers. comm.) to try to prevent tern nesting. In all three years, however, terns successfully nested in areas adjacent to woody debris or on other open areas on Crescent Island. Successful management of terns on Crescent Island may thus require further planting of vegetation, beaver management, and the use of additional passive and active tern dissuasion in the future.



Figure 2. Aerial images of Crescent Island in 2016 (left panel) following plantings of native vegetation and passive dissuasion materials associated with IAPMP and in 2025 (right panel) depicting vegetation loss due to the subsequent removal of passive dissuasion materials, beaver herbivory, the reestablishment of gull and tern colonies, and the establishment of a large pelican colony.

As part of the IAPMP, provisions were made for adaptive management at tern colonies in the CPR that might grow as a direct result of management or for other reasons (USACE 2014). Both the Blalock Islands in John Day Reservoir and Badger Island in McNary Reservoir met the criteria for adaptive tern management outlined in the IAPMP, beginning in 2021 and 2022, respectively. In 2021–2025, the John Day pool level was raised during the tern breeding season to eliminate all upland habitat previously used by nesting terns at the Blalock Islands (USACE 2023). Prior to the 2022–2025 breeding season, agency personnel placed large woody debris on nesting areas used by terns on Badger Island (B. Parker, Columbia River Inter-Tribal Fish Commission, pers. comm.). There was some evidence that woody debris dissuaded terns from Badger Island in some, but not all, years (Evans et al. 2025). In 2025, no terns or other piscivorous colonial waterbirds, like gulls and pelicans, nested on Badger Island. Prior to 2025, pelicans had been observed nesting on Badger Island since at least 2004 and gulls were first observed in 2015, following management actions on nearby Crescent Island. Researchers from the Yakama Nation Fisheries, which installed trail cameras on Badger Island in 2025, observed coyotes (*Canis latrans*) on the island during the early portion of the breeding season; the presumed cause of colony abandonment (T. De Boer, Yakama Nation Fisheries, pers. comm.).

Over concerns of its impact to a cultural site of the Yakama Nation and the survival of juvenile salmonids, the gull colony on Miller Rocks was managed to prevent gull nesting and reduce the size of the colony in 2022–2025. A variety of nest dissuasion methods were used by the Yakama Nation and its contractors, including active hazing, air cannons, pyrotechnics, green lasers, and other dissuasion techniques. Limited lethal take (under permit) was also implemented to

reinforce non-lethal techniques (T. De Boer, Yakama Nation Fisheries, pers. comm). Results indicate that efforts to reduce the number of gulls while hazing occurred were initially successful, but a colony eventually formed once dissuasion activities ended in early May when permitted activities were no longer allowed. After colony reformation in May, the gull colony on Miller Rocks was estimated at 4,915 individuals in 2025, above the 10-year average annual count of 4,178 individuals (see also *Nesting Distribution and Colony Size below*).

Columbia River Estuary

Since 2020, the USACE-FFU has conducted tern monitoring and nest dissuasion efforts on ESI outside the 1-acre designated colony area and on Rice Island (Roberts et al. 2026 and Strong et al. 2026). These efforts included the installation of passive nest dissuasion arrays in areas where terns were observed to be prospecting for nest sites, active human hazing using colony walkthroughs, a green laser, the use of kinetic predatory decoys and effigies, audio playback systems, the collection of tern eggs (under permit), and other techniques (see Strong et al. 2026). Ring-billed gulls were also dissuaded using passive and lethal techniques on Rice Island in 2025, in an attempt to further discourage co-nesting of terns amongst ring-billed gulls (Strong et al. 2026). In 2025, efforts to prevent terns from nesting outside of 1-acre designated colony area on ESI and throughout Rice Island were successful at preventing sustained colonies (Roberts et al. 2026, Strong et al. 2026). Terns, however, were observed on Rice Island throughout April and May, with upwards of 800 individuals counted in early May, but no established, prolonged colony formed due to active hazing (Strong et al. 2026). No active nest dissuasion activities have been performed at the cormorant colony on ESI since 2018 (Blair and Tidwell 2024). No cormorant nesting attempts were observed on Rice Island in 2025 (Strong et al. 2026). For additional details on avian predation monitoring and management efforts on ESI and Rice Island in 2025, see Roberts et al. (2026) and Strong et al. (2026).

There was no hazing of cormorants on the AMB in 2025. Limited active hazing of cormorants was conducted by USDA-APHIS-WS on two spans (166–167) of the AMB for maintenance and painting in 2023 (M. Alex, USDA-APHIS-WS, pers. comm.). Cormorants reestablished nesting sites on locations where hazing occurred in 2023 during both the 2024 and 2025 breeding seasons. Nesting habitat for cormorants on the bridge is currently not limiting, as there continues to be more available habitat that the birds are not using.

NESTING DISTRIBUTION & COLONY SIZE

A total of 36 active tern, cormorant, pelican, gull, and heron colonies were monitored as part of this project during the 2025 breeding season (*Map 1*). Of those, cormorant and gull colonies were the most prevalent (13 and 9 colonies, respectively), followed by tern, heron, and pelican colonies (6, 6, and 2 colonies, respectively; *Table 1* and *Table 2*). Most of the colonies (27) were in the CPR, with 9 colonies located in the LCR and CRE (*Map 1*). As has been the case in the past, of the colonies monitored, gulls were the most numerous (> 39,000 breeding individuals) of all the piscivorous colonial waterbirds in the CRB, followed by cormorants (7,950 breeding pairs), pelicans (3,984 individuals), terns (901 pairs), and herons (312 pairs). Aerial imagery and counts

from this study have been published (Banet et al. 2026) and are available from Dryad at https://doi.org/10.5061/dryad.2z34tmq15.

Colony size estimates suggest that while management actions have achieved their objective of reducing double-crested cormorant and, especially, Caspian tern nesting populations in the Columbia River Basin (*Table 1*), these reductions appear to have contributed to broader declines in breeding populations of both species throughout the Pacific Flyway. For instance, based on a recently completed survey of the Pacific Flyway, the tern population has declined by more than 70% since the population peaked in 2009, one year after management on ESI began in 2008 (McGuire et al. 2025). This raises concerns about the conservation status of these populations, especially for the rapidly declining Pacific Flyway population of terns (see also Management Recommendations section). Whether terns and cormorants from the CRB have dispersed to locations outside of the CRB cannot be fully assessed within the scope of this study.

Table 1. Annual colony size (peak number of breeding pairs) for Caspian terns in the Columbia River (CR) basin during 2008–2025. "Active" denotes a colony that was active but not counted to determine peak number of breeding pairs. Estimates of colony size for Caspian terns on East Sand Island and Rice Island in 2020–2024 were provided by the U.S. Army Corps of Engineers (see also Roberts et al. 2025; Blair et al. 2025 for more details).

Colony	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Columbia River Estuary (CRE)																		
East Sand Is. (CR) [1, 2]	10,668	9,854	8,283	6,969	6,416	7,387	6,269	6,240	5,915	3,500	4,959	3,861	2,387	2,050	1,725	524	1,524	634
Rice Is. (CR) [3]	0	0	0	0	0	0	0	0	0	0	0	0	0	Active	Active	Active	Active	Active
Columbia Plateau Region (CPR)																		
Goose Is. (Potholes Res.)	293	487	416	422	463	340	159	2	0	0	0	0	6	22	16	12	20	30
Unnamed Is. (N. Potholes Res.)	0	0	0	0	0	0	0	0	144	0	0	0	0	0	0	0	126	15
Finley Is. (CR)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8
Crescent Is. (CR)	388	349	375	419	422	393	474	0	0	0	0	0	0	1	149	88	186	147
Burbank Slough (CR)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	23	0
Blalock Islands (CR)	104	79	136	20	6	26	45	677	483	449	313	379	150	0	0	0	0	0
Badger Is. (CR)	0	0	0	33	60	0	0	0	0	41	8	0	0	231	267	274	0	0
Twinning & Goose Is. (Banks Lk)	27	61	34	19	22	13	67	64	6	0	0	0	0	0	0	7	15	47
Harper Is. (Sprague Lk)	11	4	4	4	30	1	8	10	3	92	79	18	0	85	2	10	0	0
North Rock/Shoal Is. (Lenore Lk)	0	0	0	0	0	0	0	0	0	123	91	48	53	61	76	81	69	7
Total (CRE)	**10,668**	**9,854**	**8,283**	**6,969**	**6,416**	**7,387**	**6,269**	**6,240**	**5,915**	**3,500**	**4,959**	**3,861**	**2,387**	**2,050**	**1,725**	**524**	**1,524**	**634**
Total (CPR)	**823**	**980**	**965**	**917**	**1,003**	**773**	**753**	**753**	**636**	**705**	**491**	**445**	**209**	**400**	**510**	**472**	**439**	**254**
Grand Total	**11,491**	**10,834**	**9,248**	**7,886**	**7,419**	**8,160**	**7,022**	**6,993**	**6,551**	**4,205**	**5,450**	**4,306**	**2,596**	**2,450**	**2,235**	**996**	**1,963**	**888**

[1] Estimates during 2021 – 2024 were those of USACE – Fisheries Field Unit.

[2] Estimated based on aerial imagery from June 21, 2025. Larger numbers of terns were reported on ESI by USACE – FFU in July and August, after the traditional peak breeding season, but counts of nesting pairs from July and August were not available.

[3]Although terns were successfully dissuaded from establishing a sustained colony on Rice Island in some years, roosting, loafing, and attempted nesting was observed during 2021–2025, but accurate estimates of the number of breeding pairs in those years were not available due to active dissuasion.

Table 2. Size of California/ring-billed gull (LAXX), double-crested cormorant (DCCO), Brandt's cormorant (BRAC), American white pelican (AWPE), Great Blue Heron and Great Egret (ARXX) colonies in the Columbia River basin in 2025. "Active" denotes a colony that was active but not counted.

Colony	LAXX [1]	DCCO [2]	BRAC [2]	AWPE [1]	ARXX [2]
Columbia River Estuary					
East Sand Is.	Active [3]	Active [4]	Active [4]		
Astoria-Megler Bridge		5,310	1,350		
Rice Is.	Active [3]				
Channel Markers [5]		271			
Miller Sands Spit [6]				1,168	
Lower Columbia River					
Channel Markers [7]		128			
Longview Bridge		408			
Troutdale Towers		411			
Columbia Plateau Region					
Murdock Towers (Columbia River)		66			
Miller Rocks (Columbia River)	4,915				
Crescent Is. (Columbia River)	3,215	534		2,816	46
Island 20 (Columbia River)	9,189	19			114
Hanford Reach (Columbia River)		87			21
Okanagan River Delta (Columbia River) [8]		34			37
Goose Is. (Potholes Reservoir)	11,150	91			54
North Potholes Reservoir (Potholes Reservoir)	163				20
North Rocks/Shoal Is./other (Lenore Lake)	936	183			
Harper Is. (Sprague Lake)	3,312	408			
Twinning Is. and Goose Is. (Banks Lake)	6,249				20

[1] *Number of individuals.*
[2] *Number of breeding pairs.*
[3] *Includes glaucous-winged/western and ring-billed gulls. Data provided by the U.S. Army Corps of Engineers – Fisheries Field Unit (see Roberts et al. 2026 and Strong et al. 2026).*
[4] *Data provided by the U.S. Army Corps of Engineers – Fisheries Field Unit (see Roberts et al. 2026).*
[5] *Channel Markers from river kilometer 0 to river kilometer 53; see Appendix E for counts by channel marker.*
[6] *Data provided by Oregon State University (see Peck-Richardson and Orben 2026).*
[7] *Channel Markers located from river kilometer 127 to river kilometer 218; see Appendix E for counts by channel marker.*
[8] *Imagery for counts acquired by one unmanned aerial survey on May 6, 2025.*

Columbia Plateau Region

Caspian Tern Colonies: In 2025, terns nested at six locations in the CPR, including Goose Island in Potholes Reservoir (30 breeding pairs), an unnamed island in northern Potholes Reservoir (15 pairs), Crescent Island in McNary Reservoir (147 pairs; all "managed colonies" as part of the IAPMP; USACE 2014) and at three other unmanaged colony sites (Goose Island in Banks Lake {47 pairs}, Shoal Island in Lenore Lake {7 pairs}, and Finley Island in McNary Reservoir {8 pairs}; *Table 1*). No (zero) terns were observed nesting at Badger Island in McNary Reservoir,

presumably due to the presence of mammalian predators (coyotes) during the breeding season (T. DeBoer, Yakama Nation Fisheries, pers. com.). No (zero) terns were observed nesting at the Blalock Islands in John Day Reservoir due to raising reservoir elevation to inundate all previously used nesting habitat (B. Parker, Columbia River Inter-Tribal Fish Commission, pers. comm. and J. Macdonald, USACE, pers. comm.).

Although terns have been observed loafing on Finley Island in McNary Reservoir in years past, 2025 was the first year a documented breeding colony was confirmed (Roby et al. 2021). The colony, however, failed in May due to increasing reservoir levels that inundated the low-lying island. As such, it is unknown if Finley Island will be a suitable site for tern nesting in 2026 and beyond, but the lack of successful tern nesting at the site to-date suggests it may not be a sustainable breeding site. A colony of 47 pairs of terns was documented on Goose Island in Banks Lake, the largest number of terns observed in Banks Lake since 2015 (*Table 1*). This number exceeded the IAPMP target goal of < 40 pairs, per tern colony in the CPR (USACE 2014). Islands in Banks Lake are owned and managed by the BOR, so management may be possible and warranted, especially if the colony continues to increase in size in the coming years.

A colony of terns nested on the upper portion of Goose Island in Potholes Reservoir, with a peak count of 15 pairs in 2025 (*Table 1*, *Figure 3*). Smaller numbers of terns also attempted to nest on other areas of Goose Island in 2025, although none were successful due to active and passive management. In total, an estimated 30 pairs of terns attempted to nest on all areas combined on Goose Island in 2025. Since the IAPMP commenced in 2014, tern colonies on islands in northern Potholes Reservoir have been observed three times, in 2016, 2024, and 2025 (*Figure 3*), with adaptive management successful at dissuading nesting terns once implemented (see also *Nest Dissuasion Activities* section and USDA-APHIS-WS 2025). Collectively, data from Potholes Reservoir indicates that efforts to dissuade terns have largely been successful at achieving the goals of the IAPMP. Despite these successes, terns continued to show strong fidelity to islands in Potholes Reservoir, seeking out nesting habitat without dissuasion. This suggests that, without ongoing adaptive management, terns could continue to reestablish colonies at sites throughout Potholes Reservoir in the future.

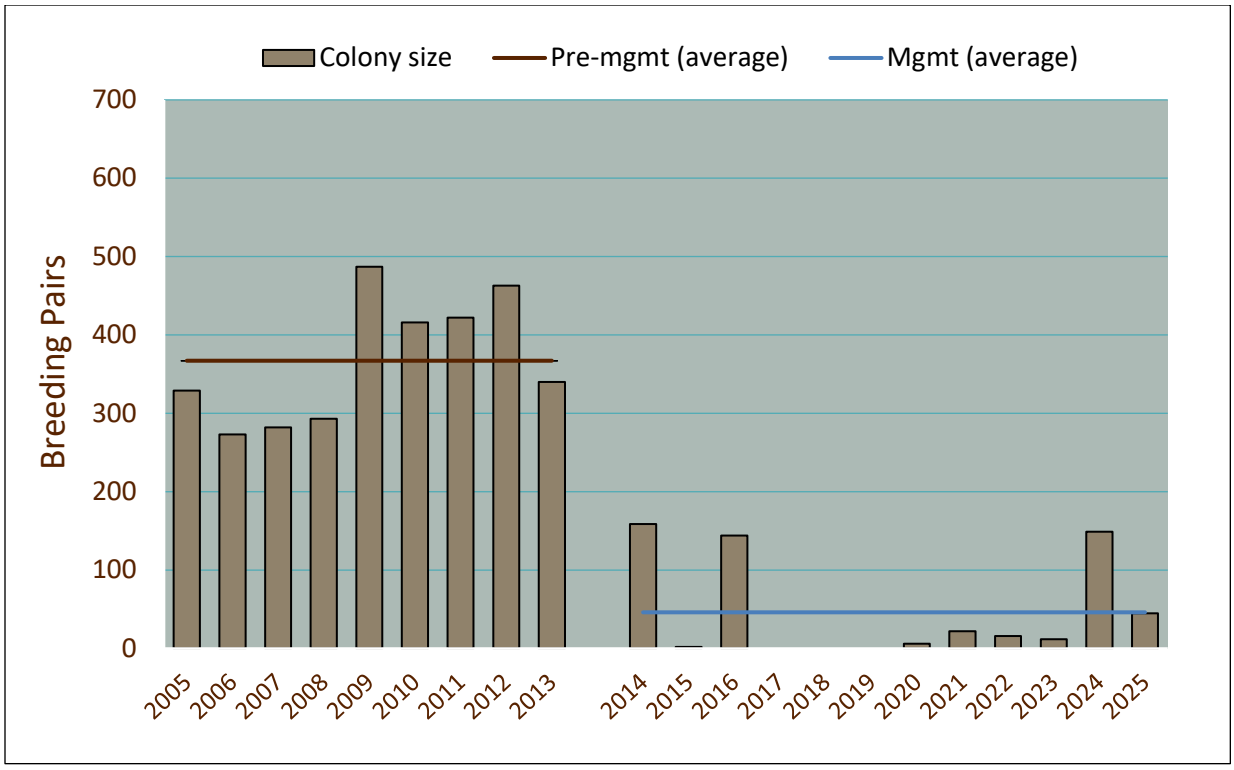


Figure 3. Annual colony size (number of breeding pairs) of Caspian terns at Goose Island and other islands in Potholes Reservoir before tern management (Mgmt; 2005–2013) and during tern management (2014–2025). In 2016, 2024, and 2025 terns nested on unnamed islands in northern Potholes Reservoir. Terns nested exclusively on Goose Island or surrounding islets in all other years.

For the fourth consecutive year, a colony of terns nested on Crescent Island in McNary Reservoir (*Table 1*, *Figure 4*). The re-establishment of a tern colony on Crescent Island was facilitated, in part, by the recent reduction in vegetative cover on the island (*Figure 2*). The vegetation loss is likely due to a combination of factors, including resumption of gulls nesting on the island, beaver herbivory, and the rapidly expanding pelican colony on Crescent Island. These factors have created patches of open unvegetated habitat that are suitable for tern nesting (see also Nest Dissuasion section *above*). Gulls likely also provide social attraction to nesting terns. In 2021, a single breeding pair of terns nested at the historical tern colony site on Crescent Island, the first successful tern nesting attempt on Crescent Island since 2014 (*Figure 4*). During 2022–2025, the Crescent Island tern colony has consistently been greater than the target threshold of 40 breeding pairs, with 149, 88, 186, and 147 pairs during 2022–2025, respectively (*Figure 4*). In the absence of active nest dissuasion and larger areas of non-vegetated habitat, the tern colony on Crescent Island is expected to continue to grow in the future, perhaps to levels observed prior to management (average of 461 breeding pairs; *Figure 4*).

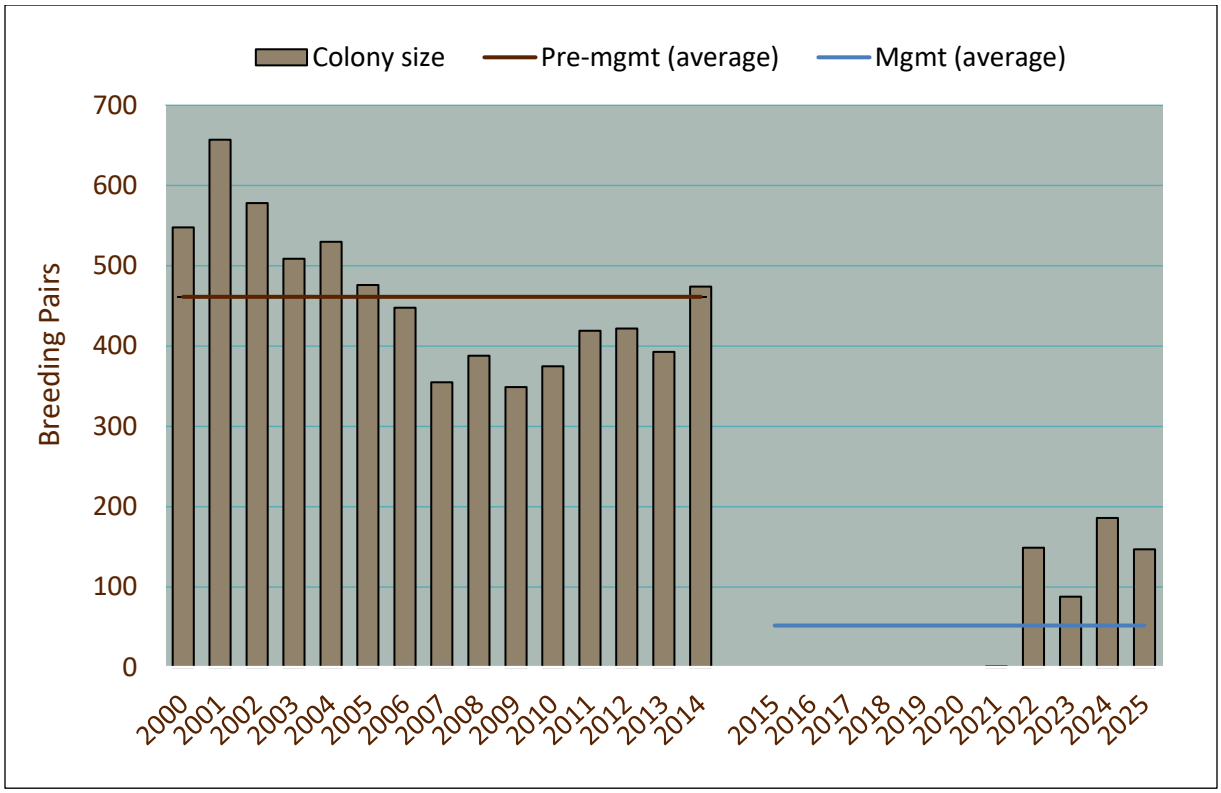


Figure 4. Annual colony size (number of breeding pairs) of Caspian terns at Crescent Island in the mid-Columbia River before tern management (2005–2014) and during tern management (2015–2025).

Results of this study indicate that management implemented at the two largest tern colonies in the CPR (Goose and Crescent islands) has resulted in a shift in the distribution of nesting terns in the region. In the first year of tern management on Crescent Island in 2015, most of the terns prevented from nesting at that site relocated to the nearby Blalock Islands. During the 2021–2025 breeding seasons, adaptive management (i.e., raising the John Day Reservoir elevation to inundate all nesting habitat formerly used by terns) was successful in preventing tern colony formation in the Blalock Islands. These actions resulted in a shift of terns away from the Blalock Islands and back to Crescent Island, where management actions have been largely scaled back, to Badger Island in 2021–2024, Goose Island in Banks Lake, islands in Lenore Lake, and most recently to Burbank Slough (in 2024) and Finley Island (in 2025) in McNary Reservoir. Despite these shifts in colony location and abundance, however, the over-all number of tern breeding pairs in CPR has decreased since management commenced in 2014 (*Figure 5*). In 2025, a total of 254 breeding pairs nested in the CPR from all colonies combined (*Table 1*, *Figure 5*), amongst the lowest number since management commenced in 2013, but still slightly greater than the maximum population size stipulated in the IAPMP of 200 breeding pairs (USACE 2014). The regional average annual post-management tern population size of 511 pairs during 2014–2025 represented a 42% reduction compared to the pre-management average annual size of 875 breeding pairs during 2007–2013 (*Figure 5*). While there is evidence that some terns displaced from the CPR have relocated to colonies outside of the basin (Lawes et al. 2021a, Banet et al. 2024), reductions in the number of terns nesting in the CPR have generally not resulted in

increased numbers of nesting terns in other regions of the Pacific Flyway (Peck-Richardson et al. 2019, Lawes et al. 2022, McGuire et al. 2025).

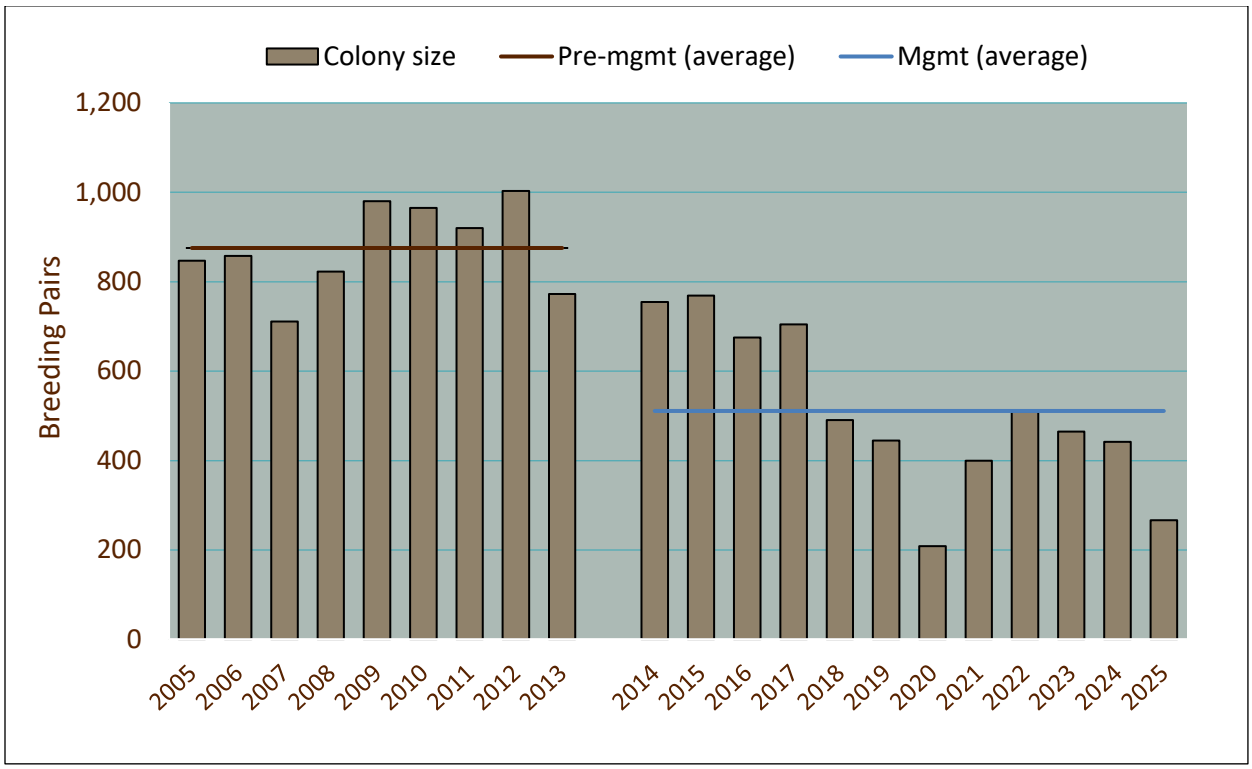


Figure 5. Annual and average number of Caspian tern breeding pairs nesting at all known colonies before tern management (2005–2013) and during tern management (2014–2025) in the Columbia Plateau region.

Other Piscivorous Waterbird Colonies: A total of eight active California and ring-billed gull colonies were monitored in the CPR in 2025, ranging in size from 163 individuals (North Potholes Reservoir) to over 11,000 individuals (Goose Island in Potholes Reservoir; *Table 2*). Nesting double-crested cormorants were confirmed at nine colonies in the CPR, with colony sizes ranging from 66 breeding pairs (Murdock Island transmission towers in CPR) to 534 pairs (Crescent Island; *Table 2*). A total of 7 mixed-species heron colonies (i.e., Great Blue Herons and Great Egrets) were monitored in the CPR, ranging in size from 20 pairs (Goose Island in Banks Lake) to 114 pairs (Island 20 in the middle Columbia River). One American white pelican colony was monitored in the CPR on Crescent Island in McNary Reservoir, with an estimated 2,816 individuals in 2025 (*Table 2*).

Lower Columbia River

In 2025, several active double-crested cormorant nesting colonies were identified downstream of Bonneville Dam, including the Longview Bridge (also referred to as the Lewis and Clark Bridge; 408 breeding pairs), transmission towers near Troutdale, Oregon (411 pairs), and numerous channel markers located between river kilometer 127–218 (128 pairs; *Table 2* and

Appendix E). Sites in the LCR continue to provide nesting habitat for cormorants dispersing from ESI in the CRE, including additional nesting habitat on the Longview Bridge, Troutdale Towers, and channel markers in the freshwater zone of the LCR.

Columbia River Estuary

Caspian Tern Colonies: An estimated 634 breeding pairs of terns nested on the prepared 1-acre colony site on ESI in 2025 (*Table 1*). This estimate was based on aerial imagery acquired in June, near the traditional peak breeding season (Roby et al. 2021). Larger numbers of adult terns were observed on ESI in July and August (Roberts et al. 2026), but no estimates of the number of breeding pairs was available. Efforts to prevent terns from nesting outside the designated 1-acre colony area on ESI were successful in 2025 (Robert et al. 2026). However, for the fifth consecutive year the tern colony on ESI was well below the target colony size identified in the *Estuary Tern Management Plan* of 3,125–4,275 pairs (USFWS 2008), raising concern regarding the status and sustainability of the Pacific Flyway breeding population of terns. Nesting phenology has also been delayed in recent years by approximately 1–2 months, with the largest number of terns observed in July as opposed to the traditional peak in late-May to early-June (Roby et al. 2021). Complete nesting failure has been observed at the ESI colony in at least six out of the last ten years (2016 –2025), with limited or unknown productivity in all other years (Roby et al. 2021, Roberts et al. 2026). The factor(s) causing colony failure in recent years are not clearly understood because close monitoring of the colony ceased after the 2019 breeding season. It is likely, however, that disturbance of the tern colony by bald eagles and concurrent predation on tern eggs and chicks by glaucous-winged/western gulls was the proximate cause of colony failure. During the same period, terns were counted on Rice Island, which is located in the freshwater zone of the estuary (Blair et al. 2026). When terns attempted to nest on Rice Island, active and passive dissuasion of the incipient tern colony was used to prevent colony formation. This adaptive management was successful in dispersing the terns from the incipient colony site on Rice Island in recent years, but large numbers (hundreds to thousands) of terns continue to prospect for nesting sites on Rice Island each year (see Roberts et al. 2024, Blair et al. 2024, and Roberts et al. 2026 for further details). Terns were also observed at Port of Astoria Piers and at piers at Tongue Point in 2025, although no nesting was confirmed at these locations. A warehouse rooftop on Port of Astoria Pier 2 was where the highest number of prospecting terns (hundreds of individuals) were observed in 2025.

Double-crested Cormorant Colonies: Double-crested cormorant colonies in the CRE in 2025 included the AMB (n=5,310 breeding pairs) and birds breeding on numerous channel markers (n=271 pairs) located between river kilometers 0 and 53 (*Table 2 and Appendix E*). Relatively large numbers of double-crested cormorants (upwards of 3,000 individuals) were also observed on the historical colony site on ESI in August and September (Roberts et al. 2026), after the traditional peak breeding season of June (Roby et al. 2021). It is unknown how many cormorants may have attempted to nest on ESI in 2025, but many of these birds may have been post-breeding, failed breeders, or fledglings from other colonies in Pacific Flyway. Assuming most birds were non-breeders, 2025 would mark the seventh year with little or no cormorant nesting on ESI (USACE-FFU 2020, 2021, 2022a; Blair et al. 2024; Roberts et al. 2025, Roberts et

al. 2026). Management and other factors (e.g., colony disturbance and nest predation) at the ESI cormorant colony have resulted in the dispersal of cormorants away from ESI, with most nesting cormorants in the CRE now located in the mixing zone of the CRE on the AMB (*Table 2, Figure 6*). The cormorant colony on the AMB has increased dramatically from 333 breeding pairs in 2014 (year prior to management at the ESI cormorant colony) to 5,310 breeding pairs in 2025 (*Table 2*), the second largest number observed to date and very similar to the 5,348 pairs observed in 2024 (*Figure 6*). Without efforts to dissuade cormorants from nesting on the bridge and to attract them to a alternative locations, like ESI, it is expected that the AMB colony will persist, and perhaps grow, as there is additional available nesting habitat on the bridge. This is a concern to fisheries managers because the per capita smolt impacts of cormorants nesting on the AMB is 2–5 times higher than those of cormorants nesting on ESI (Cramer et al. 2021b; Evans et al. 2022b; Evans et al. 2024a, Evans et al. 2025, and Avian Predation Rates section *below*).

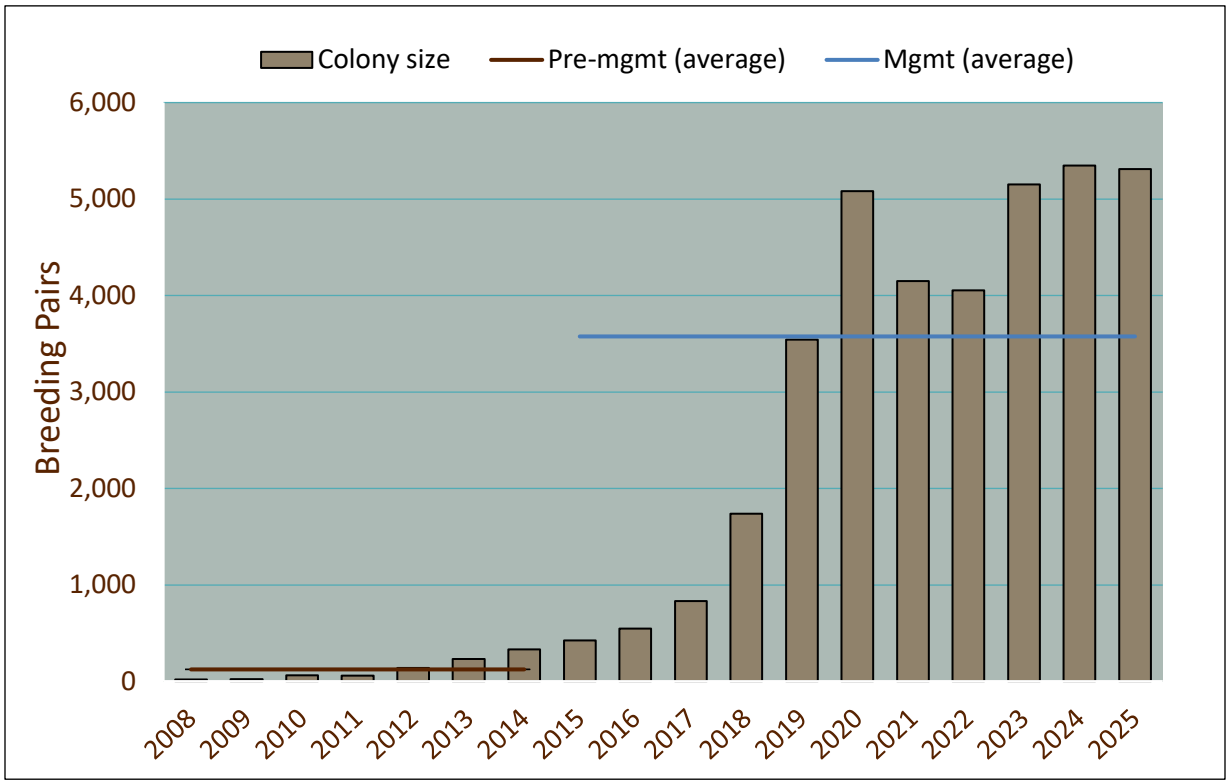


Figure 6. Annual and average number of double-crested cormorant breeding pairs nesting on the Astoria-Megler Bridge prior to management actions on East Sand Island to reduce the number of cormorants (2008–2014) and following management actions on East Sand Island (2015–2025) in the Columbia River Estuary.

Other Piscivorous Waterbird Colonies: Both glaucous-winged/western gulls and ring-billed gulls nested in the Columbia River estuary in 2025. The first confirmed nesting by ring-billed gulls in the Columbia River estuary was on ESI in 2004 (Roby et al. 2021a), where a colony persisted in each year up until this year. That colony was abandoned in April–May of 2023 and those birds

presumably relocated to nest on Rice Island where a sizable colony (>3,000 individuals; Blair et al. 2024) was counted in 2023 and 2024. In an effort to further discourage tern nesting, the ring-billed gull colony on Rice Island was dissuaded by USACE in 2025, and the colony was re-established on ESI (see also Strong et al. 2026 and Roberts et al. 2026). As in previous years, glaucous-winged/western gulls nested at dispersed colonies on ESI and Rice Island. Although estimates of colony size in recent years are unavailable, in past years these colonies numbered in the 1,000s (Roby et al. 2021a).

In addition to double-crested cormorants (DCCO) nesting on the AMB, two other cormorant species have nested on the AMB: Brandt's cormorants (BRAC), documented since at least 2018, and pelagic cormorants (*Phalacrocorax pelagicus*; PECO), documented since at least 1999 (Naughton et al. 2007, Roby et al. 2021). Both species continued nesting on the bridge in 2025. Peak counts for BRAC were 1,350 breeding pairs and for PECO were 133 breeding pairs, similar counts for both species compared with 2023 and 2024 (Evans et al. 2024a, Evans et al. 2025). The BRAC colony in the AMB continues to be the largest in the CRE and amongst the largest in the Pacific Northwest. Brandt's cormorants also attempted to nest on ESI in 2024 and large numbers were observed in July through September of 2025 (upwards of 4,000 adults; Roberts et al. 2026). ESI was formerly the largest colony site for BRAC in the CRE. Estimates of the number of BRAC pairs on ESI in 2024 and 2025, however, were not available due to the late formation of the colony (after aerial surveys associated with this study were completed) and the inability to distinguish BRAC from DCCO in data analyzed by the USACE-FFU (Robert et al. 2025, 2026).

There was one active American white pelican colony in the CRE in 2025 on Miller Sands Spit (*Table 2*). The Miller Sands Spit pelican colony was one of only two pelican colonies in the CRB in 2025; the other being on Crescent Island in McNary Reservoir in the CPR (*Table 2*). The Miller Sands Spit pelican colony was established in 2010, when 42 individuals attempted to nest there, although it has relocated to nearby Rice Island in some years (Cramer et al. 2021a). In 2025, the colony on Miller Sands Spit was estimated at 1,168 individuals (*Table 2*), a slight increase compared with 2024 (997 individuals) and similar to counts in 2023 (1,272 individuals; Evans et al. 2024a, Evans et al. 2025).

AVIAN PREDATION RATES

PIT-tagging of Upper Columbia River Steelhead

A total of 7,671 PIT-tagged UCR steelhead were sampled at RIS and available for predation and survival analyses in 2025, which includes 6,984 newly tagged smolts and 687 recaptured smolts (i.e., previously tagged and sampled at RIS). Of the tagged smolts, 6,201 and 1,470 were classified as hatchery and wild, respectively (see also *below* and *Appendix B*). Steelhead were tagged or recaptured during 6 April to 16 June, a period which accounted for >98% of all steelhead encountered in the trap in 2025. Mean fork length was 186 mm (standard deviation [SD]=23 mm; range=91–299 mm). An evaluation of external fish condition indicated that most steelhead were in good overall external condition in 2025, with 8.5% of steelhead observed

with disease (bacterial, fungal, or viral infections), severe body injuries (subcutaneous wounds/scars), or severe descaling (>20% of scales missing). For comparison, on average, 9.1% of steelhead tagged at RIS in previous years were classified as being in compromised conditions (Evans et al. 2014, Evans et al. 2025). The most common type of anomaly in 2025 was descaling, followed by body injuries and disease. Previously published studies have investigated whether steelhead smolts with signs of external injuries/damage/disease were more susceptibility to avian predation than seemly healthy or uncompromised smolts. Results indicated that compromised smolts were, on average, more likely to be consumed by terns, cormorants, and gulls compared with uncompromised smolts. Differences, however, were often small, with compromised smolts only slightly more likely to be consumed (e.g., 1.1–1.3 times) than uncompromised smolts (Hostetter et al. 2012, Evans et al. 2019, and Hostetter et al. 2023). Given small differences in the relative susceptibility of fish in different conditions and the finding that the majority of smolts were in good external condition, the effects of condition-dependent predation on steelhead smolt survival was estimated be low in most years (Evans et al. 2014, Evans et al. 2019, and Hostetter et al. 2023).

In an attempt to more accurately classify the rear-type (hatchery, wild) of steelhead sampled at RIS, all non-adipose clipped smolts with no signs of fin erosion or other marks/tags indicative of a hatchery environment – fish that met the visual classification criteria of a wild smolt (Evans et al. 2014) – were scanned for the presence of a coded wire tag (CWT), an indication that the fish was actually hatchery. In total, 8 or 0.5% of the 1,478 presumed wild steelhead sampled at RIS in 2025 had a CWT, fish that were reclassified as hatchery. In 2023 and 2024, 0.1% and 0.5%, respectively, of the presumed wild steelhead sampled at RIS had a CWT. To ensure the CWT detector/antenna was functioning, a subsample of 408 hatchery steelhead were also scanned for the presence of a CWT. In total, 68 or 16.7% of the hatchery steelhead had a detectible CWT, indicating the antenna was functioning properly. To test the efficacy of different CWT detectors at RIS, two different types of antennas, V-block and W style, were used to scan 596 steelhead (408 hatchery, 188 presumed wild), with each individual fish scanned by each antenna type. The two antenna types agreed with one another 99.8% (595/596) of the time, indicating both antenna types were likely to detect a fish with a CWT, if CWT was present. Finally, of the 6,201 fish classified as hatchery in 2025, 78.6% had their adipose fin clipped and 21.4% had fin erosion or other signs indicative of a hatchery environment. Collectively, results provide evidence that steelhead classified as wild in the RIS trap were likely wild, but the possibility remains that some hatchery fish could have been misclassified as wild (see also *Appendix B*).

PIT Tag Recovery

A total of 31 avian breeding colonies and several avian loafing sites were scanned for fish PIT tags following the 2025 breeding season. Breeding colonies used in predation rate analyses included 9 cormorant colonies, 7 tern colonies, 4 heron colonies, 3 gull colonies, 2 pelican colonies and 1 Brandt's cormorant colony (*Table 3*). Scanning was also conducted at other avian breeding sites where predation rates were not estimated (e.g., Channel Markers and Longview Bridge) and select avian loafing/roosting sites, areas where large numbers of piscivorous

colonial waterbirds were observed during the smolt out-migration period (*Table 3*). A total of 31,532 PIT tags from 2025 migration year smolts (Chinook, Coho Salmon {*O. kisutch*}, sockeye, and steelhead combined) were recovered on avian breeding colonies and avian loafing/roosting sites in 2025 (*Table 3*). Of these, more than 95% were tags recovered from colonies, with the largest number recovered on a mixed gull and pelican colony on Crescent Island (n=8,415), followed by cormorant colonies on Crescent Island (n=6,894), gulls on Miller Rocks (n=3,378) and Island 20 (n=2,224), and double-crested cormorants on the Astoria-Megler Bridge (n=1,607), Murdock Towers (n=1,462), and Troutdale Towers (n=1,365). Scanning at AMB was limited to areas accessible to researchers following the breeding season (see Methods & Analysis), resulting in a subsample of all available nests at that site (*Table 3*).

A total of 118 tags from other fish species and salmonid age-classes (adult salmonids) were also recovered on avian colonies in 2025. This included PIT tags from 40 adult sockeye, 12 adult steelhead, 4 adult Chinook, 8 Northern Pikeminnow (*Ptychocheilus oregonensis*), 5 Pacific Lamprey (*Entosphenus tridentatus*), 6 Smallmouth Bass (*Micropterus dolomieu*), 4 Walleye (*Sander vitreus*), and 7 Bull Trout (*Salvelinus confluentus)* from colonies in the CPR (*Table 3*). A total of 32 tags from Sea-run Cutthroat Trout (*O. clarkii clarkii*) were recovered on avian colonies in the CRE (*Table 3*). Tags from 54 hatchery released White Sturgeon (*Acipenser* transmontanus) were also recovered on cormorant colonies in the middle Columbia River from fish released during 2015–2019, but the year of consumption (2025 or years past) was unknown.

Finally, for the third consecutive year, PIT tags were recovered from select heron colonies or rookeries in the CRB. The primary impetus for scanning these sites was to learn if appreciable numbers of smolts were being consumed by herons, a less studied avian predator species (Cramer et al. 2021a). Herons are wading birds that forage in shallow water habitat, habitat that exists throughout the lower Yakima River. A relatively large number of tags (n=832) were recovered from a heron colony (114 breeding pairs) on Island 20 in the middle Columbia River (*Table 3*), the second time the site has been scanned and yielded large numbers of smolt PIT tags (Evans et al. 2025). Interestingly, similar to 2024, a high percentage (49%) of the tags recovered on the Island 20 heron colony were from fish that originated from the Yakima River, which enters the middle Columbia River downstream of Island 20. By comparison, of the tags recovered on the gull colony on Island 20 (n=2,224), just 5% were from Yakima River origin smolts. Results suggest that herons were disproportionately foraging on smolts from the Yakima River relative to gulls that also nested on Island 20.

Table 3. Numbers of 2025 Migration Year smolt PIT tags (Chinook Salmon, Sockeye Salmon, Coho Salmon, and steelhead combined) and tags from other fish species or life stages recovered at avian breeding/loafing sites in the Columbia River basin. Piscivorous colonial waterbird species included Caspian terns (CATE), California and ring-billed gulls (LAXX), double-crested cormorants (DCCO), great egrets and great blue herons (ARXX), Brandt's cormorants (BRAC), and American white pelicans (AWPE). The number of PIT tags recovered was not adjusted to account for tag loss due to on-colony PIT detection and deposition probabilities (see Table 4) and thus represents the minimum number of tagged fish consumed by birds at each location.

Location	Rkm	Bird Species	Area Use	Smolt PIT Tags	Other Fish PIT Tags[1]
Harper Island (Sprague L.)	Off-river	DCCO	Breeding	33	
Goose Island (Banks L.)	Off-river	CATE	Breeding	437	
Okanogan River Mouth	859	DCCO/ARXX	Breeding	396	
Goose Island (Potholes R.)	Off-river	CATE	Breeding	145	
		DCCO/ARXX	Breeding	24	
Unnamed Island (Potholes R.)	Off-river	CATE	Breeding	28	
Lenore Lake	Off-river	CATE	Breeding	21	
Hanford Island	592	DCCO	Breeding	123	
		ARXX	Breeding	86	
Island 20	549	LAXX	Breeding	2,224	1
		DCCO	Breeding	136	
		ARXX	Breeding	832	2
Foundation Island	518	Mixed	Loafing	38	
Finley Island	517	CATE	Breeding	554	
Badger Island	512	Mixed	Loafing	83	1
Crescent Island	510	CATE	Breeding	437	
		LAXX/AWPE	Breeding	8,415	60
		DCCO	Breeding	6,894	7
Miller Rocks	331	LAXX	Breeding	3,378	4
Murdock Towers[2]	299	DCCO	Breeding	1,462	5
Troutdale Towers	189	DCCO	Breeding	1,365	4
		Mixed	Loafing	14	
Lewis & Clark Bridge [3]	106	DCCO	Breeding	182	3
Channel Markers [3]	33-218	DCCO	Breeding	67	
Miller Sands Spit	38	AWPE	Breeding	82	
Astoria-Megler Bridge [3]	23	DCCO	Breeding	1,607	3
		BRAC	Breeding	124	1
		DCCO/BRAC	Loafing	601	1
		Mixed	Breeding	212	
Tongue Pier 5	28	Mixed	Loafing	39	
East Sand Island [4]	8	CATE	Breeding	1,493	26
Total				31,532	118

[1] Includes adult Sockeye Salmon (n=40; Crescent Island, Troutdale Towers), steelhead (n=12; Crescent Island, Badger Island), and Chinook Salmon (n=4; Crescent Island); Bull Trout (n=7; Crescent Island); Northern Pikeminnow (n=8; Island 20, Crescent Island, Murdock Towers, Astoria-Megler Bridge), Pacific Lamprey (n=5; Miller Rocks, Troutdale Tower), Smallmouth Bass (n=6; Crescent Island), Walleye (n=4; Crescent Island), and sea-run Cutthroat Trout (n=32; Troutdale Towers, Longview Bridge, Astoria-Megler Bridge, East Sand Island). Newly detected tags from hatchery White Sturgeon (n=54) that were released during 2015–2019 were also recovered on the Okanagan River Mouth and Hanford Island cormorant colonies, but the consumption year was unknown, so numbers were not reported herein.

[2] Includes tags (n=67) recovered by BPA in tower nests.

[3] Tags recovered from a sub-sample of nests.

[4] Tags recovered by the USACE-FFU.

Smolt Survival to Bonneville Dam

In addition to providing information on predation, recoveries of smolt PIT tags on bird colonies can also be used to increase the precision and accuracy of smolt survival estimates by increasing the sample sizes of tagged fish used in mark-recapture-recovery survival models (Hostetter et al. 2018, Payton et al. 2019, Payton et al. 2023). To provide information for use in smolt survival models, we recovered smolt PIT tags from several additional avian breeding and loafing sites that were not included in the original scope of work for this study. In 2025, scanning associated with this additional effort resulted in the detection of 4,293 PIT tags from 2025 migration year smolts that survived outmigration to below Bonneville Dam prior to being consumed by a bird. This information, coupled with 2025 smolt tags recovered by the USACE-FFU on the ESI tern colony (n=1,493), tags in live fish detected at antennas on pile dikes in the CRE (n=8,082), and tags of live fish detected by the National Marine Fisheries Service's net trawl in the CRE (n=9,023), were used by this and other projects (e.g., NOAA Fisheries Survival Study) to generate more accurate and precise estimates of smolt survival to Bonneville Dam in 2025 and years past (see also Hostetter et al. 2018, Payton et al. 2019, Payton et al. 2023).

PIT Tag Detection & Deposition Probabilities

Data necessary to accurately estimate predation rates is provided in *Table 4;* this includes results of the probability that a consumed PIT tag by a breeding bird from a given colony was deposited on-colony (i.e., deposition probability) and the probability that a deposited tag was detected by researchers following the breeding season (i.e., detection probability; Hostetter et al. 2015). Deposition probabilities were based on previous studies that empirically measured deposition rates for terns, cormorants, gulls, and pelicans at their breeding colonies (Hostetter et al. 2015, Evans et al. 2022c). No previously published estimate of deposition probability, however, was available for heron colonies. In lieu of empirical data, the average annual deposition estimate from pelican and cormorant colonies, the two species that most closely resemble herons in terms of their size (mass) and ability to consume large-sized prey items, were used. The validity of this assumption, however, is untested, so predation rate estimates by heron colonies should be interpreted cautiously (Banet et al. 2024). Detection probabilities were directly measured in 2025 based on the proportion of tags intentionally sown by researchers on each colony that were subsequently detected after the breeding season (see Hostetter et al. 2015 for details). Like results in previous years, detection probabilities were highly variable depending on the colony and when tags were deposited during the breeding season (*Table 4*). Finally, detection probabilities were not available for cormorants breeding on channel markers and the Longview Bridge in the CRE and, as such, predation rate estimates were not generated for those colonies in 2025.

Table 4. Detection efficiency (range during breeding season) and deposition (95% credible interval) estimates (depicted as a proportion) for smolt PIT tags on bird colonies during 2025. Estimates were used to generate predation rates based on the number of smolt PIT tags recovered on each colony following the breeding season (Table 3). Piscivorous colonial waterbird species include Caspian terns (CATE), California and ring-billed gulls (LAXX), double-crested cormorants (DCCO), Brandt's cormorants (BRAC), American white pelicans (AWPE), and Greg Blue Herons and Great Egrets (ARXX).

Location	Rkm	Bird Species	Detection (Range)	Deposition (95% CRI)
Sprague Lake	Off-river	DCCO	0.96	0.51 (0.34–0.70)
Banks Lake	Off-river	CATE [1]	0.34–0.85	0.71 (0.51–0.89)
Lenore Lake	Off-river	CATE [1]	0.53–0.56	0.71 (0.51–0.89)
Potholes Reservoir	Off-river			
Goose Island		CATE [1]	0.31–0.96	0.71 (0.51–0.89)
		DCCO/ARXX	0.99	0.49 (0.32–0.67) [2]
Unnamed Island		CATE	0.78	0.71 (0.51–0.89)
Hanford Island	592	DCCO	0.36–0.44	0.51 (0.34–0.70)
		ARXX	0.36–0.44	0.49 (0.32–0.67) [2]
Island 20	549	LAXX	0.88–0.96	0.15 (0.11–0.21)
		DCCO	0.60–0.80	0.51 (0.34–0.70)
		ARXX	0.60–0.80	0.49 (0.32–0.67) [2]
Finley Island	517	CATE	0.88	0.71 (0.51–0.89)
Crescent Island	510	LAXX [1]	0.67–0.81	0.15 (0.11–0.21)
		DCCO	0.38–0.64	0.51 (0.34–0.70)
		AWPE	0.72–0.80	0.47 (0.24–0.73)
		CATE	0.16–0.52	0.71 (0.51–0.89)
Miller Rocks	331	LAXX	0.52–0.84	0.15 (0.11–0.21)
Murdock Towers	308	DCCO	0.84–0.88	0.51 (0.34–0.70)
Troutdale Towers	189	DCCO	0.20–0.60	0.51 (0.34–0.70)
Miller Sands	38	AWPE	0.80-0.99	0.47 (0.24–0.73)
Astoria-Megler Br.	23	DCCO	0.38–0.80	0.51 (0.34–0.70)
		BRAC [1]	0.40–0.83	0.51 (0.34–0.70) [3]
East Sand Island	8	CATE	0.82–0.98	0.71 (0.51–0.89)

[1] Variation in detection was partially inferred from other years (see also Payton et al. 2019).
[2] Heron deposition was assumed to be the average of DCCO and AWPE (see also Banet et al. 2024).
[3] BRAC deposition was assumed to be the same as DCCO (see also Cramer et al. 2021a).

Sample Sizes of PIT-tagged Smolts

Sample sizes of PIT-tagged smolts detected at juvenile bypass facilities located upstream of the avian colonies investigated, those used to generate ESU/DPS-specific predation rates based on detections at Lower Monumental Dam, McNary Dam, John Day Dam, and Bonneville Dam (see Methods, Predation Rate Estimates), were often too small (< 500) to generate reliable estimates of predation in 2025. However, estimates of predation that were based on PIT-tagged smolts initially detected passing Lower Grante Dam (for Snake River migrants) or Rock Island Dam (for Upper Columbia River migrants), estimates that account for smolt survival to each

downstream river reach and dam (see Methods, Predation Rate Estimates, Cumulative Predation and Survival), were substantially larger for most of the salmonids species investigated in 2025. For instance, 86,789, 51,403, and 11,003 PIT-tagged, in-river (i.e., not transported) Snake River (SR) steelhead, yearling Chinook, and sockeye, respectively, were detected passing Lower Granite Dam, compared with 3,853, 2,157, and 166 Snake River steelhead, yearling Chinook, and sockeye, respectively, at Lower Monumental Dam in 2025. Thus, to provide the most precise estimates of predation rates in 2025, we report predation estimates that were based on smolts initially detected passing Lower Granite Dam or Rock Island Dam, unless otherwise noted in tables and figures. Sample sizes and ESU/DPS-specific predation rate estimates based on detections at Lower Monumental Dam, McNary Dam, John Day Dam, and Bonneville Dam are provided in Appendix A.

Efficacy of Avian Predation Management Plans

Inland Avian Predation Management Plan (IAPMP): Estimates of tern predation rates on the ESA-listed salmonid populations were below the 2% per colony threshold identified in IAPMP for all colonies evaluated, with the exception of terns breeding on an island in Banks Lake, WA, where terns consumed an estimated 2.2% (1.5–3.3%) of UCR steelhead in 2025 (*Table 5*). Estimates of predation rates on all other salmonid populations evaluated were less than 1% per colony in 2025 (*Table 5*), amongst the lowest since management actions associated with the IAPMP commenced in 2014. Predation by all tern colonies in the CPR combined (aggregate estimates of predation) on SR and UCR steelhead, the two most impacted salmonid populations, were 1.3% (1.0–1.8%) and 4.2% (3.2–5.7%) in 2025, rates that were below the 5% aggregate tern predation rate threshold identified in the IAPMP.

Table 5. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by Caspian terns nesting on Goose Island and an unnamed island in northern Potholes Reservoir, Goose Island in Banks Lake, Shoal Island in Lenore Lake, Crescent Island in McNary Reservoir, and Finley Island in McNary Reservoir during 2025. NA indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable predation estimates (see Methods). See Appendix A for estimates of predation by these and other tern colonies in previous years.

Salmonid	Potholes Goose Is.	North Potholes Is.	Lenore Lake Is.	Banks Lake Is.	Crescent Is.	Finley Is.
SR steelhead	< 0.1%	< 0.1%	< 0.1%	< 0.1%	0.8% (0.3–1.7)	0.2% (0.1–0.3)
SR Yearling Chin	< 0.1%	< 0.1%	< 0.1%	< 0.1%	0.1% (0.1–0.2)	0.1% (<0.1–0.1)
SR Subyearling Chin	< 0.1%	< 0.1%	< 0.1%	< 0.1%	0.4% (0.2–0.9)	0.1% (<0.1–0.1)
SR Sockeye	< 0.1%	< 0.1%	< 0.1%	< 0.1%	0.5% (0.2–1.2)	0.1% (<0.1–0.2)
UCR Steelhead	0.9% (0.5–1.7)	0.1% (<0.1–0.1)	< 0.1%	2.2% (1.5–3.3)	0.4% (0.1–1.1)	0.2% (0.1–0.3)
UCR Yearling Chin	NA	NA	NA	NA	NA	NA

Predation rates on UCR steelhead by Goose Island terns in Potholes Reservoir prior to implementation of management actions in 2014 were the highest of any tern colony in the CPR, averaging 15.7% (14.1–18.9%) during 2007–2013 (Collis et al. 2024; see also *Appendix A*). Estimated average annual predation rates on UCR yearling Chinook were 2.5% (1.7–3.6%) during 2007–2013 (Evans et al. 2025; see also *Appendix A*). In 2025, estimated predation rates on UCR steelhead by Goose Island terns were just 0.9% (0.5–1.7%). This was significantly lower than those observed prior to management (*Table 5*) and similar to those observed in 2024 and in several, but not all, years following implementation of management actions on Goose Island (*Appendix A*). No estimates of predation on UCR yearling Chinook have been available since 2021, the last year sufficient sample sizes of PIT-tagged smolts were available. Reductions in Goose Island tern predation rates on steelhead were roughly commensurate with reductions in colony size, with an average annual colony size of 367 breeding pairs prior to management and 22 pairs following management, including 30 pairs observed on Goose Island in 2025 (*Table 1*). In 2016, a colony of 144 breeding pairs of terns formed on an unnamed island in northern Potholes Reservoir, where recoveries of smolt PIT tags indicated that terns consumed an estimated 4.1% (2.9–6.3%) of UCR steelhead (*Appendix A*). In 2024 and 2025, an incipient tern colony again formed on a different, unnamed island in northern Potholes Reservoir (126 pairs and 15 pairs, respectively), but adaptive management actions were initiated in-season and predation rates were significantly lower (1.7% {1.0–3.2}) in 2024 and, especially, in 2025 (0.1% <0.1–0.1%}). Collectively, results suggest that active and passive dissuasion techniques can successfully prevent or limit terns from nesting on islands in northern Potholes Reservoir, thereby reducing impacts from tern predation. Due to the relatively large number of terns that continue to prospect for nest sites in Potholes Reservoir, however, continued management will likely be necessary to achieve the stated goals of the IAPMP in the future.

Predation rates by Crescent Island terns in 2025 were < 1.0% per salmonid population, with the highest estimates observed on SR steelhead at 0.8% (0.3–1.7%; *Table 5*). During 2015–2020, no (zero) terns nested on Crescent Island, effectively reducing the impact of tern predation to zero in those years (Collis et al. 2021a, 2021b). Prior to management actions in 2015, predation rates on smolts by Crescent Island terns were highest on steelhead populations, with an estimated average annual predation rate of 2.5% (2.2–2.9%) and 4.5% (4.2–5.1%) on UCR and SR steelhead, respectively (Evans et al. 2025; see also *Appendix A*). Average annual predation rates on salmon ESUs were less than 2% prior to management (*Appendix A*). Comparisons of predation rates prior to and during management indicated that management actions at Crescent Island have been successful in reducing predation rates at this site, especially on steelhead (see also Collis et al. 2024). However, due to the re-establishment of the tern colony on Crescent Island in recent years (2021–2025), future management will likely be necessary to achieve the goals of the IAPMP. For instance, the adaptive management colony size threshold of 40 pairs has now been exceeded for four consecutive years at Crescent Island (2022–2025), with an average annual colony size of 143 pairs during this time. In addition, the amount of reduced vegetation on Crescent Island (see *Figure 2*) has substantially increased the amount of suitable tern nesting habitat, further increasing the likelihood of continued tern nesting in the future.

Estimates of predation rates on smolts at the unmanaged tern colony on Shoal Island in Lenore Lake were < 0.2% on all salmonid populations evaluated in 2025 (*Table 5*), the lowest predation estimates observed at the site since the colony formed in 2015. In 2025, only 7 breeding pairs were observed on the island (*Table 1*), although this could be due to higher water levels that limited available breeding habitat in 2025. In 2022, estimates of predation by terns nesting at Lenore Lake on UCR steelhead were above the target threshold at 2.1% (1.4–3.4%; *Appendix A*), a year where 76 breeding pairs were observed. At larger colony sizes, variation in UCR steelhead predation rates does not appear to be closely related to variation in colony size. For instance, in 2017, there was an estimated 123 pairs nested in Lenore Lake, yet predation rates on UCR steelhead were estimated to be 1.0% (0.6–2.0%; *Appendix A*), suggesting that factors other than colony size alone may be related to predation effects by terns nesting in Lenore Lake. Collectively, results suggest that future monitoring and possible management of the tern colony in Lenore Lake may be necessary to achieve the goals of IAPMP. In 2025, however, predation rates and colony size were the lowest observed since the colony formed in 2015.

In 2025, there was no tern colony on Harper Island in Sprague Lake, WA (*Table 1*). Harper Island has had a relatively small tern colony (< 40 breeding pairs) in some years past (Collis et al. 2021b), except for 2021 when an estimated 85 pairs were documented (Evans et al. 2022b). Scanning for PIT tags at the Harper Island tern colony has occurred only once, in 2012 (Collis et al. 2021b). Predation rate estimates in 2012 indicated that terns consumed less than 0.3% of available salmonid ESUs/DPSs, with the highest rate observed on SR steelhead at 0.2% (Collis et al. 2021b). Predation rates on smolts were low, in part, due to the small size of the colony in 2012 (30 breeding pairs; Collis et al. 2021b) and the distance from Harper Island to the Snake River (65 km). With landowner permission, the tern colonies from 2021–2023 were scanned in the spring of 2024, and recent tag recoveries from Sprague Lake have been uploaded to a regional PIT tag database (i.e., PTAGIS), but no estimates of predation were available due to lack of information on year-specific detection probabilities on Harper Island in those years. Future monitoring of predation rates may be warranted if a large tern colony becomes established on Harper Island in the future.

For the first time since colonial waterbird monitoring efforts commenced on Badger Island in 2007, no (zero) piscivorous colonial waterbirds were observed breeding on the island in 2025 (see also *Predation by Other Piscivorous Waterbirds).* Since 2011, terns have periodically nested on Badger Island, with the largest colonies observed following management actions on nearby Crescent Island in 2015 (*Table 1*). Average annual predation rates on SR steelhead during 2021–2023, years with the largest tern colonies on Badger Island (range = 231–274 pairs annually), were 2.1% (1.6–2.8%; *Appendix A*), indicating terns consumed an appreciable proportion of steelhead in some years. Estimates of predation on salmon smolts, however, were less than 1.0% during this same period (*Appendix A*). During 2022–2024, adaptive management efforts (e.g., placement of woody debris) were implemented on the Badger Island tern colony, but they did not fully dissuade or discourage tern nesting in those years (Evans et al. 2025). Badger Island is large (> 7 ha), with numerous interior and shoreline areas where terns could potentially nest, factors that could make it challenging to prevent tern nesting in any given year using passive methods.

In 2025, an incipient tern colony formed on Finley Island in McNary Reservoir, a low-lying island located upstream of Badger Island. Peak colony size was 8 pairs and terns occupied the site starting from early April to late May but then abandoned the site when higher river levels inundated (flooded) the colony area. Terns have periodically been observed loafing on Finley Island in years past, but 2025 was the first year that nests with eggs were observed. As such, future monitoring of the site may be warranted, especially if future dissuasion activities are successful at preventing tern nesting at the Badger and Cresent Island colony sites, which are both located less than 10 km away.

For the fifth consecutive year since adaptive management was implemented (2021–2025), there was no tern colony on the Blalock Islands in John Day Reservoir, the site of the largest tern colony in the CPR during 2015–2020 (*Table 1*). The managed increase in water levels in the John Day Reservoir inundated the former colony sites on low-lying islands in the Blalock Islands starting in 2021. Immediately following management actions on the Crescent Island tern colony in 2015, the tern colony on Blalock Islands increased from an average of 57 breeding pairs (range = 6 to 136) during 2007–2014 to an average of 409 breeding pairs (range = 150 to 677) during 2015–2020 (Collis et al. 2021a, 2021b). During this period, estimated average annual predation rates by terns nesting on the Blalock Islands were comparable to or higher than those of terns nesting on Crescent Island during the pre-management period, particularly for ESUs/DPSs originating from the SR. For example, predation rates on SR steelhead by terns nesting on the Blalock Islands increased from an average of 0.5% (0.4–0.9%) prior to management actions at Crescent Island to an average of 4.0% (3.3–4.9%) following management at Crescent Island during 2015–2020 (Collis et al. 2021b). As such, increases in predation rates on smolts by terns nesting on the Blalock Islands had initially offset the benefits achieved by the elimination of the tern colonies on Crescent and Goose islands due to management. Since implementation of adaptive management actions to inundate the Blalock Islands, however, predation rates have been greatly reduced (if not eliminated) during 2021–2025. Results indicated that the elimination of nesting habitat on the Blalock Islands successfully prevented the formation of a tern colony and therefore greatly reduced predation by terns at this site in recent years (see also Collis et al. 2024).

As it relates to management actions aimed at reducing tern predation in the CPR, reductions in predation rates by terns on UCR steelhead were associated with reductions in the size of the Goose Island tern colony in Potholes Reservoir and, to a lesser extent, the Crescent Island tern colony in McNary Reservoir; the two tern colonies managed as part of the IAPMP (USACE 2014). In 2020 and 2023, record low tern predation rates were commensurate with record high estimates of UCR steelhead survival in the river reach (RIS to MCN) where terns were managed (Figure 7). In 2021, 2022, and 2024, increases in predation rates coincided with lower estimates of UCR steelhead survival. Terns nesting at the colonies on Goose, Lenore Lake, Crescent, and Badger islands foraged on smolts in the Columbia River between RIS and MCN, and an increase in predation rates by terns nesting at these colonies coincided with lower UCR steelhead smolt survival. In 2025, near record low tern predation rates were not associated with near record high survival rates. On average, however, over the course of the entire *Inland Avian Predation Management Plan* period (2014–2025), tern predation rates on UCR steelhead smolts have

significantly declined following management actions at Goose and Crescent islands (*Figure 9*). For instance, on average, tern predation rates on UCR steelhead have decreased from 19.6% (16.5–24.1%) to 3.6% (3.0–4.5%) following management while survival has increased from 59.8% (56.6–63.3%) to 69.5% (62.9–75.1%; *Figure 7*). These results provide evidence that management efforts to reduce tern predation have demonstrably improved smolt survival in managed areas; however, if tern colonies reform or grow, smolt survival could decrease to pre-management levels. Estimates from 2025, and to a lesser degree in 2015 and 2021 (the two other years with below average annual tern predation and survival), the relationship between tern predation on steelhead and steelhead survival was less apparent on an annual basis. However, these years coincided with higher gull predation and higher rates of non-avian mortality factors (Evans et al. 2022a). These factors include below average flows, higher temperatures, and potentially other, unidentified biotic and abiotic factors that may have negatively impacted smolt survival in those years. Additional research to better understand these factors, which will require an investigation of weekly tern predation, gull predation (see *below*), total mortality, and survival within and across years may be warranted.

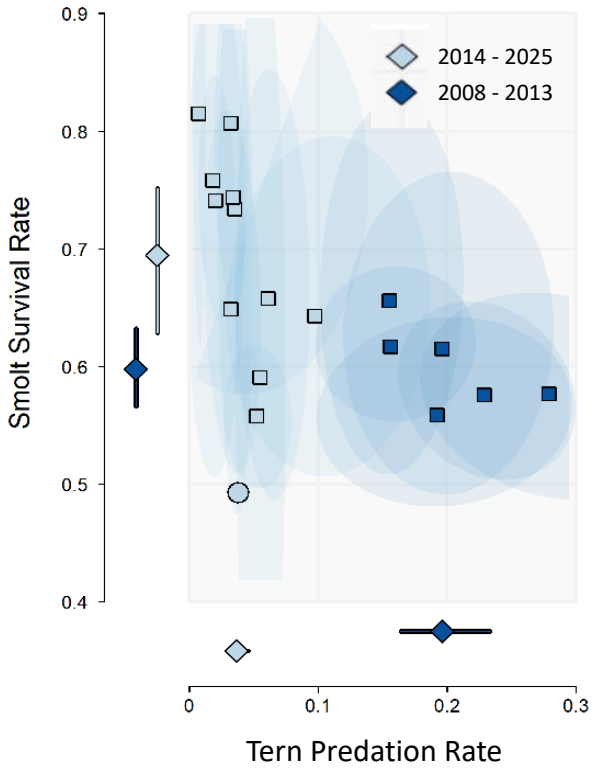


Figure 7. Caspian tern predation rates and survival rates of Upper Columbia River steelhead smolts during passage from Rock Island Dam to McNary Dam during (dark blue squares) and following (light blue squares) management actions that reduced the size of tern colonies at Goose Island and Crescent Island. Median annual rates and average annual rates for the entire study period (dark blue and light blue diamonds) are also shown. Error bars represent 95% credible intervals for annual averages and shaded ellipses represent 95% credible regions for the joint estimation of survival rate and predation rate (see also Payton et al. 2020). The light blue circle indicates the estimate from 2025.

In summary, predation rates by terns nesting at colonies in the CPR in 2025 were amongst the lowest observed since management actions associated with the IAPMP commenced in 2014, with the target goal of predation rates less than 2% per ESU/DPS, per tern colony, achieved at all colonies, with the exception of terns nesting on an island in Banks Lake, WA. Reductions in tern colony sizes at both Goose Island in Potholes Reservoir and, to a lesser extent, Crescent Island, have reduced predation by terns breeding at these sites since management was initiated in 2014. There is also evidence that reductions in tern predation at managed colonies have, on average, resulted in higher survival of UCR steelhead in the river reaches where management occurred. Adaptive management actions that used active and passive dissuasion on an unnamed island in northern Potholes Reservoir and that raised water levels in the John Day Reservoir, have also successfully prevented or limited tern nesting in recent years. Terns, however, continue to demonstrate strong fidelity to the CPR and adaptive management will likely be needed in 2026 and beyond to continue to achieve the goals of the IAPMP. This may especially be true at Crescent Island, where the tern colony has been re-established since management actions ceased and there are now large amounts of suitable nesting habitat expanded for future tern use.

Estuary Caspian Tern & Double-crested Cormorant Management Plans: Salmonid ESU/DPS-specific estimates of predation by terns nesting on ESI in 2025 were not available, estimates that were generated with funding from USACE in the past. Smolt PIT tags, however, were recovered on ESI by the USACE-FFU following the 2025 breeding season (*Table 3*), so the data necessary to generate ESU/DPS-specific predation estimates exists. The number of smolt PIT tags recovered on ESI tern colony in 2025 (n=1,493) was well below the annual average number recovered prior to tern management in 2008 (n=26,553 annually during 2000–2007; Roby et al. 2021). The peak or maximum numbers of terns on ESI in 2025 also occurred in July and August (Robert et al. 2026), after most salmonids had passed through the estuary. The estimated number of terns breeding on ESI in 2025 was also amongst the lowest observed to-date, with an estimated 634 pairs (*Table 1*) compared with an average of 9,079 pairs prior to management during 2000–2007 (Collis et al. 2024).

Estimates of cumulative predation and survival on salmonids – estimates that use a different modelling structure than those that generate ESU/DPS-specific estimates (see Methods, Avian Predation, Cumulative Predation and Survival section) – were generated for ESI terns in 2025 and indicated predation rates of 1.0% (0.8–1.4%) and 1.3% (0.6–2.6%) on SR and UCR steelhead, respectively, the two populations most susceptible to tern predation. Estimates were similar to those observed in 2024 and were amongst the lowest observed on steelhead since management was initiated to reduce the size of the colony in 2008. For instance, prior to management (2000–2007), average annual estimates from the ESU/DPS-specific model were 25.3% (22.7–28.3%) and 17.2% (15.2–19.5%) for SR and UCR steelhead, respectively, but again, relative comparisons presented herein should be interpreted cautiously due to differences in the modelling framework used in recent years relative to years past when the USACE was funding predation studies on ESI. See Collis et al. (2023) for relative comparisons of ESI tern predation using the same modeling framework, data that dates back to 2021, the last year updated estimates using the same modelling framework were available.

In 2024 and 2025, few terns were documented on Rice Island in the CRE due to active and passive dissuasion implemented by USACE as part of ongoing management at that site (Roberts et al. 2026). In 2022 and 2023, a tern colony on Rice Island successfully formed, and predation rates were as high as 2.9% (1.6–5.1%) on Middle Columbia River (MCR) steelhead (Evans et al. 2023). Results indicate that continued implementation of nest dissuasion activities and continued monitoring of tern nesting sites throughout the CRE will be necessary to ensure that smolt survival gains achieved as part of the *Estuary Tern Management Plan* are not offset by increased tern predation rates at other sites in the CRE, like Rice Island. Collis et al. (2023) provides a detailed assessment of the efficacy of the estuary tern management plan to reduce predation on ESA-listed smolts in the CRE, with data reported through the 2021 breeding season.

In 2025 there was evidence of late-season (July – September) breeding attempts by double-crested cormorants on ESI, but an estimate of the number of pairs was not available (Roberts et al. 2026). Furthermore, researchers from USACE-FFU, which were tasked with monitoring terns and cormorants on ESI, were not able to distinguish double-crested cormorants from Brandt's cormorants in aerial imagery (Roberts et al. 2026). The USACE-FFU did not recover smolt PIT tags on ESI cormorant colony following the 2025 breeding season, so predation rate estimates could not be generated, work that had previously been funded by USACE. The late arrival of cormorants on ESI, however, likely resulted in low predation effects in 2025, as most PIT-tagged salmonid smolts pass through the estuary in April – June. In 2022, 2023, and 2024, small numbers of cormorants attempted to nest on ESI. Just 99, 50, and 17 smolt PIT tags, respectively, were recovered by USACE following the breeding season, suggesting that predation rates by cormorants (both double-crested and Brandt's) on ESI have been low in recent years, but predation rate analyses and accurate species-specific (Brandt's versus double-crested cormorants) counts were lacking.

An unintended consequence of management actions at the double-crested cormorant colony on ESI during 2015–2019 was the abandonment of the colony site and the subsequent expansion of the cormorant colony on the AMB and elsewhere in the CRE (Lawes et al. 2021b, Evans et al. 2023, *Figure 6*). For example, in 2014, an estimated 333 pairs of double-crested cormorants nested on the AMB, but by 2025 it had increased to 5,310 pairs. The AMB is located approximately 10 Rkm upstream of ESI in the freshwater mixing zone of the CRE—an aquatic environment where fewer marine forage fish and a greater proportion of juvenile salmonids are available relative to the waters surrounding ESI, which is in the marine zone of the CRE (Cramer et al. 2021b, Evans et al. 2022b). Starting in 2021, to estimate predation rates on salmonid smolts by cormorants breeding on the AMB, efforts to recover smolt PIT tags deposited by cormorants nesting on a concrete footing of the bridge (referred to as the North Crib) commenced. The North Crib is an area where PIT tags could be recovered by researchers following the breeding season (see also Evans et al. 2022b). Estimates of per capita (per breeding pair) predation rates derived from a known number of double-crested cormorants nesting on the North Crib (e.g., 673 pairs in 2025) were then extrapolated to account for all double-crested cormorants nesting on the AMB (5,310 pairs in 2025) to generate colony-wide estimates of predation rates (see also Evans et al. 2022b).

Colony-wide predation estimates indicated that cormorants breeding on the AMB in 2025 consumed between 5.5% (0.6–20.0%) of SR sockeye to upwards of 13.9% (5.8–28.7%) of SR subyearling Chinook (*Table 6*). Estimates of predation were similar amongst and between salmon and steelhead, with no statistically significant difference detected in relative comparisons of predation in 2025 (*Table 6*). Results suggest cormorants were consuming smolts in proportion to their relative availability in the estuary, with all salmonid species and age-classes equally susceptible to cormorant predation. It should be noted, however, that estimates of predation were often imprecise due to the process of extrapolating estimates from a subsample of breeding pairs on the North Crib to all breeding pairs on the AMB (see also Methods & Analysis section). Estimates of predation rates on salmonid smolts by cormorants nesting on the AMB in 2025 were similar to those in 2024 and amongst the highest observed since estimates were first generated in 2021 (*Table 6* and *Appendix A*). Increases were presumably associated, in part, with the larger size of the cormorant colony on the AMB during 2023–2025 relative to 2021–2022 (F*igure 6*).

Estimates of per capita (per breeding pair) predation by cormorants on the AMB in 2025 and years past were approximately 2–5 times greater (depending on salmonid population) than average annual per capita predation rates by cormorants that nested on ESI prior to implementation of management actions on ESI (2003–2014; *Appendix A*). For instance, per capita predation rates on SR steelhead by cormorants on ESI were, on average, 0.0006% (0.0005–0.0007%; Lawes et al. 2021b). This was significantly lower than those of cormorants breeding on the AMB in 2025 at 0.0019% (0.0016 –0.0027%). As such, despite the smaller size of the double-crested cormorant colony on the AMB (4,803 pairs on average during 2021–2025) compared with the cormorant colony on ESI (12,787 pairs on average during 2003–2014), colony-wide predation rates by cormorants on the AMB were greater, on average, than those of cormorants that formerly nested on ESI. For instance, average annual colony-wide predation rates on SR yearling Chinook and SR steelhead by cormorants breeding on ESI during 2003–2014 (prior to management on ESI) were 4.6% (4.1–5.3%) and 7.2% (6.3–8.5%), respectively (Lawes et al. 2021b), compared to 6.7% (5.6–9.4%) and 9.5% (6.7–12.6%), respectively, by cormorants on the AMB during 2022–2025 (Evans et al. 2025 and *Appendix A*). Collectively, results indicate that colony location is a key factor associated with cormorant predation on salmonid smolts, and that currently, predation rates by cormorants nesting on the AMB and other sites in the upper CRE and LCR (see *below*) are, on average, higher than the rates observed prior to management when the vast majority of cormorants nested on ESI (see also Cramer et al. 2021b and Evans et al. 2022a for a more detailed description of cormorant predation effects by colony location).

Due to small sample sizes of PIT-tagged smolts tagged and released from Lower Columbia River (LCR) salmonid populations in recent years (since 2020), estimates of predation effects by cormorants breeding on the AMB and other colonies in the CRE are lacking (Evans et al. 2023). Impacts, however, are likely substantial given predation effects on LCR salmonids by cormorants that formerly nested on ESI. For instance, predation effects by cormorants nesting on ESI were, on average, 15.0% (12.2–18.2%) and 27.5% (24.3–30.7%) on ESA-listed LCR coho and Chinook, respectively (Lawes et al. 2021b). Efforts to PIT tag larger numbers of LCR smolts

would be necessary to evaluate predation effects on LCR salmonid populations by cormorants breeding on the AMB and at other locations in the CRE in the future (see also Evans et al. 2023).

For the fourth consecutive year, smolt PIT tags were recovered from a cormorant colony located on transmission towers (TRT) near the town of Troutdale, OR, approximately 45 Rkm downstream of Bonneville Dam. Smolt PIT tags were recovered in the area underneath five transmission towers with breeding cormorants. In 2025, predation estimates ranged from 0.8% (0.2–2.3%) for UCR yearling Chinook to 4.6% (1.5–13.3%) for SR sockeye (*Table 6*). Analogous to predation by cormorants nesting on the AMB colony site, predation rates by cormorants breeding at the TRT were similar across steelhead and salmon species and age-classes (*Table 6*), indicating similar levels of relative susceptibility. Estimates of predation by TRT cormorants in 2025 were similar to those during 2022–2024 and were commensurate with the estimated colony size of 411 pairs in 2025 (*Table 2*). The TRT cormorant nesting site is in the freshwater zone of the CRE and results demonstrate that per capita predation rates were the highest of three cormorant colonies evaluated downstream of Bonneville Dam (ESI in the marine-zone, AMB in the mixing-zone, and TRT in the freshwater zone). For instance, estimates of average annual per capita (per breeding pair) predation rates on SR steelhead by cormorants on ESI (during 2003–2014), AMB (during 2022–2025), and TRT (during 2022–2025) were 0.0006% (0.0005–0.0007%), 0.0018% (0.0009–0.0032%), and 0.0082% (0.0065–0.0160%), respectively. Collectively, these results provide strong evidence that predation effects were significantly higher on a per capita basis the further upstream cormorants nest in the Columbia River. Achieving the goals of the *Estuary Cormorant Management Plan* will require the reduction of cormorant nesting sites in the mixing and freshwater zones of the CRE and LCR and potentially relocating these colonies to less impactful locations, such as ESI, which can sustain up to 5,380 to 5,939 pairs (USACE 2015).

Table 6. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by double-crested cormorants (DCCO) nesting on the Astoria-Megler Bridge (AMB) and the Troutdale Towers (TRT) during 2025. See Appendix A for estimates of predation by these colonies in previous years.

Salmonid	AMB DCCO [1]	TRT DCCO
SR Sockeye	5.5% (0.6–20.7)	4.6% (1.5–13.3)
SR Yearling Chin	9.9% (6.6–15.1)	1.2% (0.6–2.2)
UCR Yearling Chin	10.7% (4.3–20.0)	0.8% (0.2–2.3)
SR Subyearling Chin	13.9% (5.8–28.7)	1.0% (0.2–2.7)
SR Steelhead	9.8% (7.0–14.2)	3.3% (2.1–5.9)
UCR Steelhead	12.7% (3.3–27.2)	3.8% (1.4–9.9)
MCR Steelhead	6.4% (2.2–13.1)	1.0% (0.4–2.4)

[1] *Based on a subsample of all available nests (see Methods & Analysis section).*

Cormorants nesting on select channel markers in the CRE and LRC (Rkm 0–218) were also scanned for smolt PIT tags in 2025. The primary purpose of scanning these sites was to increase sample sizes of tagged fish known to have survived passage to Bonneville Dam for use in mark-recapture-recovery models by this and other BPA funded projects (see also Smolt Survival to Bonneville Dam section). In total, 67 smolt tags were recovered from three channel markers in 2025 (*Table 3*). In 2023 and 2024, 452 and 167 smolt PIT tags, respectively, were recovered from cormorant nests on up to eight channel markers. Due to a lack of information on PIT-tag detection probabilities on channel markers, estimates of predation rates could not be generated. Research to quantify detection probabilities of PIT tags deposited by cormorants on channel markers will be necessary to generate estimates of predation rates in the future (see also Evans et al. 2025).

Finally, cormorants also nested on the Longview Bridge in 2025 (408 pairs; *Table 2*) and a portion of the bridge was scanned for smolt tags. In total, 182 smolt PIT tags were recovered (*Table 3*). Similar to cormorants nesting on channel markers, however, there was no information available on PIT tag detection probabilities at the Longview Bridge. Furthermore, the area scanned for tags on the Longview Bridge was between 20–40 meters below cormorant nests (which were located on bridge support beams). As such, an unknown (but likely small) proportion of tags consumed by cormorants were subsequently deposited on the area that was scanned by researchers for smolt tags. This situation differs from the AMB cormorant colony, where researchers were able to scan a nesting site with a known number of nests (a flat, concrete platform; see *above* for details).

Predation Rates by Other Piscivorous Colonial Waterbirds

Predation/consumption rates of ESA-listed salmonids by gulls and cormorants nesting at colonies in the CPR were often greater than those of terns. Estimates were particularly high at gull colonies, with estimates of consumption > 10.0% on some salmonid populations in 2025, amongst the highest gull consumption estimates documented since system-wide studies of avian predation commenced. It should be noted, however, that gulls are known to consume dead or moribund fish and to kleptoparasite fish from other piscivorous waterbirds, such as terns. Consequently, smolt PIT tag recoveries on gull colonies may be more indicative of consumption rates, rather than predation rates (Cramer et al. 2021a; Evans et al. 2022a, 2022b). Terns, cormorants, and pelicans are strictly piscivorous and are believed to rarely consume dead fish in the wild (Evans et al. 2022a). As such, we refer to gull estimates of fish loss as “consumption rates” and losses to terns, cormorants, and pelicans as “predation rates” (see also Evans et al. 2022a). ESA-listed salmonid predation/consumption rates by gulls, cormorants, pelicans, and herons nesting at select colonies – those scanned for smolt PIT tags (*Table 5*) – in 2025 are provided below by predator species and colony.

California and Ring-billed Gulls - Island 20: Consumption rates of smolts by gulls nesting on Island 20 ranged from 0.2% (0.1–0.4%) for SR subyearling Chinook to 9.4% (6.8–13.8%) for UCR steelhead (*Table 7*). Similar to trends in years past, estimates of consumption were significantly higher for steelhead smolts compared with salmon smolts. Previous studies suggest that higher

gull consumption rates of steelhead smolts compared with salmon smolts are associated with the larger average size and the surface orientation of steelhead smolts relative to salmon, factors that increase the susceptibility of steelhead to plunge-diving (surface oriented) predators like gulls and terns (Evans et al. 2016, Evans et al. 2019, Cramer et al. 2021a, Hostetter et al. 2023). To date, the years with the highest estimates of consumption of steelhead by Island 20 gulls were those of 2015, 2021, and 2025, with estimates of 7.9%, 6.1%, and 9.4%, respectively, on UCR steelhead (*Table 7* and *Appendix A*). These increases may be associated with increases in colony size and/or environmental factors that increase smolt exposure to gull consumption during outmigration (Hostetter et al. 2012, Roby et al. 2016, Payton et al. 2016, Hostetter et al. 2023). For example, the years with the highest estimates of predation/consumption by Island 20 gulls on UCR steelhead coincide with years of below average flows and/or larger colony sizes (Hostetter et al. 2023, Evans et al. 2024a), like those observed in 2015, 2021, and 2025.

California and Ring-billed Gulls - Crescent Island: Consumption rates on salmonid smolts for gulls nesting on Crescent Island were amongst the highest recorded since the gull colony reformed in 2021, ranging from 0.7% (0.1–1.7%) for SR subyearling Chinook to 11.0% (7.4–18.4%) for SR sockeye in 2025 (*Table 7*). Consumption estimates were also substantial on steelhead, with estimates of 8.1% (2.9–11.3%) and 8.4% (5.8–12.1%) on SR and UCR steelhead, respectively (*Table 7*), again demonstrating the greater susceptibility of steelhead to gull consumption relative to most salmon species. Consumption estimates by gulls on Crescent Island in 2025 were higher than those observed during 2021–2024 but similar to those observed during 2008–2020 (*Appendix A*). With the exception of 2025, estimates in recent years (2021–2024), were lower following colony reformation on Crescent Island in 2020 (*Appendix A*). This is likely related, in part, to the smaller size of the gull colony on Crescent Island (average of 3,558 individuals counted during 2021–2024, compared with 7,379 individuals during 2007–2015) and/or the smaller size of the tern colony on Crescent Island, whereby kleptoparasitism rates were also potentially lower during 2021–2024 due to the smaller size of the tern colony in these years (*Table 1*). For instance, if a larger tern colony coincides with higher rates of gull kleptoparasitism, then the number of tags deposited by gulls would also presumably be higher. In 2025, 3,245 gulls (individuals) were counted on Crescent Island and the tern colony was relatively small (147 pairs) compared with the tern colony that existed prior to management (average of 403 pairs), so colony size alone does not explain the high levels of gull consumption observed in 2025 compared with 2021–2024. Factors associated with the high levels of consumption rates by Crescent Island gulls in 2025 are unknown, but may be associated with environmental factors (e.g., flows, turbidity), a lack of alternative food sources, and/or gulls disproportionately foraging near dams (e.g., McNary and Ice Harbor dams), areas where smolts may be more susceptible to gulls (Evans et al. 2016, Hostetter et al. 2023, see also *Miller Rocks* gulls *below*).

California and Ring-billed Gulls - Miller Rocks: Consumption rates on salmonid smolts for gulls nesting on Miller Rocks Island in The Dalles Reservoir, a colony where management actions (active hazing) occurred from late-March to early-May, ranged from 1.4% (0.9–2.3%) for SR yearling Chinook to 11.3% (6.2–19.9%) for UCR steelhead (*Table 7*). Consumption estimates

were also elevated on SR sockeye and steelhead, with an estimated 6.0% (2.8–12.0%) and 9.8% (6.8–14.2%) of available smolts consumed, respectively. Results from previous years with adequate sample sizes indicate that, among available salmon populations, SR sockeye smolts were particularly susceptible to consumption by gulls nesting at Miller Rocks, with estimates often greater than 5% and upwards of 9.1% (5.3–15.1%) in some years past (*Appendix A*). Like results from the Island 20 and Crescent Island gull colonies, however, steelhead smolts were consistently the most susceptible to consumption by Miller Rocks gulls in 2025 and years past (*Appendix A*).

Estimates of smolt consumption rates by gulls nesting at Miller Rocks in 2025 were similar to or higher than those in years past, depending on the salmonid ESU/DPS (*Appendix A*). Starting in 2022, the Confederated Tribes and Bands of the Yakama Nation used active hazing techniques to dissuade gulls from breeding on Miller Rocks during the early portion of the breeding season (T. De Boer, Yakama Nation, pers. comm, see also *Nest Dissuasion Activity* section). Although efforts were successful at preventing colony formation while dissuasion occurred, a colony eventually formed in all three years (2022–2025). In 2025, the colony was slightly larger (4,915 individuals; *Table 2*) than that observed in years past (annual average of 4,527 individuals during 2008–2024; Cramer et al. 2021a, Evans et al. 2022b, Evans et al. 2025). The reason for the slight increase in the size of the Miller Rocks gull colony in 2025 is unknown but maybe related to the abandonment of the gull colony on Anvil Island (part of the Blalocks islands) in John Day Reservoir, located approximately 100 Rkm upstream of Miller Rocks.

Miller Rocks is located in The Dalles Reservoir, 23 Rkm upstream from The Dalles Dam and 18 Rkm downstream from John Day Dam. Evans et al. (2016) observed that gulls disproportionately consumed smolts near dams and hypothesized that smolts may be more vulnerable near dams as a result of (1) increased smolt travel times or delayed migration in the forebay of dams, (2) smolt morbidity or mortality associated with dam passage, or (3) smolts being temporarily stunned or disoriented by hydraulic conditions in the tailrace of dams. Gull consumption of smolts, however, is not limited to foraging near dams, with gulls consuming substantial numbers of smolts in apparent good-condition in open reservoirs and free-flowing sections of the river as well (see Evans et al. 2016 for a detailed discussion). There were also no terns nesting on Miller Rocks, so salmonid smolts were presumably captured and consumed by gulls and not kleptoparasited from terns.

Table 7. Estimated consumption rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by California and ring-billed gulls nesting on Island 20 and Crescent Island in McNary Reservoir and Miller Rocks in The Dalles Reservoir during 2025. NA indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates (see Methods). See Appendix A for estimates of consumption by these colonies in previous years.

ESU/DPS	Island 20	Crescent Is.	Miller Rocks Is.
SR Sockeye	0.9% (0.5–1.9)	11.0% (7.4–18.4)	6.0% (2.8–12.0)
SR Yearling Chinook	0.4% (0.2–0.6)	0.7% (0.1–1.7)	1.4% (0.9–2.3)
UCR Yearling Chinook	NA	NA	2.2% (0.5–6.3)
SR Subyearling Chinook	0.2% (0.1–0.4)	2.5% (0.1–5.8)	2.3% (1.2–7.1)
SR Steelhead	2.8% (2.0–4.1)	8.1% (2.9–11.3)	9.8% (6.8–14.2)
UCR Steelhead	9.4% (6.8–13.8)	8.4% (5.8–12.1)	11.3% (6.2–19.9)

Double-crested Cormorants - Okanogan River Mouth: There was no evidence that cormorants nesting at an arboreal colony site at the mouth of the Okanogan River, located 129 Rkm upstream of RIS on the middle Columbia River, consumed PIT-tagged smolts within the study area (i.e., Rock Island Dam to the Pacific Ocean), with no (zero) tags from smolts last detected passing RIS recovered on the cormorant colony in 2025. As such, predation estimates were presumably to be at or near zero for fish that survived passage to RIS. Smolt PIT tags (n=396), however, were recovered on the colony following the breeding season (*Table 3*), indicating cormorants were consuming juvenile salmonids upstream of RIS. Most tags were from steelhead, coho, and Chinook smolts that were released into tributaries of the middle Columbia River (e.g., Okanogan and Methow rivers). Additional analyses—analyses that were outside the scope of this study—may be warranted to estimate stock-specific predation impacts by cormorants nesting at the mouth of the Okanogan River, a site that has been active since at least 2004 (Cramer et al. 2021a) with PIT tag recovery and detection efficiency data available in recent years (2023–2025).

Double-crested Cormorants - Hanford Island: The cormorant colony on Hanford Island in the Hanford Reach of the middle Columbia River has periodically been scanned for smolt PIT tags since nesting cormorants were first documented at the site in 2007 (Cramer et al. 2021a). Estimates of predation rates on the ESA-listed salmonids smolts have been low, with less than 0.3% of available fish consumed per ESA-listed salmonid population in 2025 (*Table 8*) and less than 0.4% consumed during 2022–2024 (*Appendix A*). In 2025, tags were recovered from UCR yearling Chinook, but sample sizes of available smolts at RIS were too small (< 500) to generate reliable estimates of predation. The smaller size of this cormorant colony (87 pairs; *Table 2*) is presumably responsible, in part, for the lower predation rates on ESA-listed smolts compared with other nearby cormorant colonies like Crescent Island (see *below*). It is worth noting, however, that most of the smolt PIT tags recovered from the Hanford cormorant colony were from non-listed subyearling Chinook from the Upriver Bright population, which accounted for more than 50% of all recovered tags in each year dating back to 2022 (Evans et al. 2025). Recently published research indicated that Upriver Bright Fall Chinook were especially

susceptible to predation by cormorants breeding on Foundation Island and American white pelicans breeding on Badger Island in McNary Reservoir due to the close proximity of these colonies to areas where subyearling Chinook are released from hatcheries (e.g., Priest Rapids Hatchery) and/or where wild fish rear and congregate in the Hanford Reach (Payton et al. 2023). The effects of Hanford Island cormorants on non-listed subyearling Chinook, however, are currently unknown but may be appreciable given the relatively large number of tags recovered on this colony in recent years.

Double-crested Cormorants - Island 20: In 2024, a newly discovered cormorant colony was detected on Island 20 and scanned for smolt PIT tags following the breeding season. The colony was active again in 2025 and was again scanned for PIT tags following the breeding season (*Table 3*). Estimates of predation on ESA-listed salmonids were < 0.1% for all species and age-classes evaluated in 2025 (*Table 8*). Interestingly, and analogous to cormorants nesting on Hanford Island, the majority of smolt tags recovered were subyearling Chinook from the non-listed Hanford Bright Chinook population and smolts originating from the Yakima River. Like data from the cormorant colony at the mouth of the Okanogan River and Hanford Island, additional analyses may be warranted to estimate stock-specific predation using availability data from tributary releases—analyses that were outside the scope of this study.

Double-crested Cormorants - Crescent Island: A large colony (534 breeding pairs) of cormorants was on Crescent Island in 2025 (*Table 2*). The colony rapidly expanded from just 20 pairs in 2022, presumably due to the failure of the Foundation Island cormorant colony located just 9 Rkm upstream. Cormorants had periodically been documented nesting on Crescent Island in years past, but less than 50 pairs were observed in all years prior to 2023 (Cramer et al. 2021a, Evans et al. 2023). In 2025, predation rates were substantial for some ESA-listed populations, with estimates ranging from 1.5% (0.9–2.6%) on SR subyearling Chinook to 5.7% (3.5–9.6%) on SR steelhead. Estimates of predation on SR Sockeye were also high at 4.9% (3.3–8.1%; *Table 8*). At its current size, results suggested that the Crescent Island cormorant colony now poses a greater risk to smolt survival compared with the Foundation Island cormorant colony in years past (*Appendix A*). Similar to 2023 and 2024, cormorants primarily nested in trees on the water's edge. However, since 2024, birds have expanded to interior nesting areas on Crescent Island. Cormorants nesting on Crescent Island in 2025 occupied approximately 20% of the available tree-nesting habitat, suggesting further colony growth is possible in the future.

Double-crested Cormorants - Murdock Towers: In 2024, a newly discovered cormorant colony was detected on transmission towers on Murdock Island in Bonneville Reservoir, and the colony was scanned for smolt PIT tags following the breeding season. In 2025, the colony had 66 breeding pairs that nested across three different towers. Estimates of predation by Tower cormorants in 2025 ranged from 0.3% (0.2–0.5%) on SR yearling Chinook to 1.7% (0.9–3.2%) on SR subyearling Chinook (*Table 8*). Estimates were similar across salmonid species and age-classes, suggesting smolts were equally susceptible to cormorant predation. Estimates in 2025 were similar to, but slightly higher than, those observed in 2024 (*Appendix A*). However, if non-breeding cormorants or other avian predators (e.g., gulls) also deposited tags on the ground underneath the towers (birds that were not documented during aerial surveys), estimates of

predation could be biased high to an unknown degree. As such, results should be interpreted cautiously.

Table 8. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by double-crested cormorants nesting at Hanford Island and Island 20 in the middle Columbia River, Crescent Island in McNary Reservoir, and Murdock Towers in Bonneville Reservoir during 2025. NA indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates (see Methods & Analysis section).

Salmonid	Hanford Island	Island 20	Crescent Island	Murdock Towers
SR Sockeye	< 0.1%	< 0.1%	4.9% (3.3–8.1%)	1.9% (0.8–4.5%)
SR Yearling Chin	< 0.1%	< 0.1%	2.6% (1.7–4.1%)	0.3% (0.2–0.5%)
UCR Yearling Chin	NA	NA	NA	0.8% (0.2–2.2%)
SR Subyearling Chin	< 0.1%	< 0.1%	1.5% (0.9–2.6%)	1.7% (0.9–3.2%)
SR Steelhead	< 0.1%	< 0.1%	5.7% (3.5–9.6%)	0.4% (0.3–0.6%)
UCR Steelhead	0.2% (0.1–0.6%)	< 0.1%	1.5% (0.8–2.6%)	0.6% (0.3–1.5%)

Brandt's Cormorants - Astoria-Megler Bridge: The same methods used to estimate predation by double-crested cormorants on the AMB were used for Brandt's cormorants (BRAC) nesting on the AMB, whereby smolt PIT tags were recovered from concrete footings where a known number of BRAC nested (142 breeding pairs in 2025). Per capita predation rates were then used to generate colony-wide predation rates based on the total number of BRAC nests on the AMB (1,334 pairs in 2025). Although PIT-tag recovery efforts were limited to footings where only BRAC nested, a small number of recovered tags may have been deposited by double-crested cormorants that were nesting on beams above the footings. These tags could bias BRAC predation estimates high to an unknown degree. It should also be noted that only small numbers of smolt PIT tags were recovered from BRAC nesting areas in 2025 (n=124; *Table 3*) which resulted in imprecise estimates of per capita predation rates, especially in cases where sample sizes of available tagged fish passing BON or estimated to survive to BON were also small, like for SR sockeye, SR subyearling Chinook, and UCR spring Chinook. With these caveats in mind, results indicated that even with some double-crested cormorants potentially contributing tags to the scanned areas on the BRAC colony, predation estimates by BRAC on smolts were low for most ESA-listed salmonid species and age-classes evaluated (*Table 9*). Colony-wide predation rates ranged from 0.5% (0.2–1.0%) in SR steelhead to 3.7% (0.7–20.0%) in SR sockeye (*Table 9*). Again, however, results should be interpreted cautiously, particularly in SR sockeye and SR subyearling Chinook, due to small sample sizes and the potential that estimates included some predation by double-crested cormorants. With this caveat in mind, estimates of predation by BRAC in 2025 were similar to, but slightly higher than, those observed during 2022–2024, other years when BRAC nesting areas were scanned for smolt PIT tags (*Appendix A*).

Results suggest that despite a relatively large colony of BRAC on AMB during 2023, 2024, and 2025 (1,224 pairs, 1,201 pairs, and 1,334 pairs, respectively), they posed little threat to smolt

survival in the CRE, especially relative to double-crested cormorants. These results were consistent with those of Cramer et al. (2021b), which observed that predation rates on salmonid smolts by BRAC that formerly nested on ESI were significantly lower than those of double-crested cormorants that nested on ESI. Cramer et al. (2021b) hypothesized that BRAC mainly forage in marine waters that are closer to the Columbia River mouth where non-salmonid prey types (e.g., anchovy, herring, smelt) are more common (Couch and Lance 2004, Peck-Richardson 2017), that the BRAC peak nesting period is typically in June after the peak smolt outmigration period, and that BRAC are slightly smaller (by body mass) and have lower daily food requirements compared with double-crested cormorants (Couch and Lance 2004).

Table 9. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by Brandt's cormorants nesting on the Astoria-Megler Bridge in the Columbia River Estuary during 2025.

Salmonid	Predation Rate [1]
SR Sockeye	3.7% (0.7–20.0)
SR Yearling Chin	0.9% (0.4–1.6)
UCR Yearling Chin	1.0% (0.1–3.6) [2]
SR Subyearling Chin	2.3% (0.6–7.3)
SR Steelhead	0.5% (0.2–1.0)
UCR Steelhead	0.1% (<0.1–2.8) [2]
MCR Steelhead	1.6% (0.1–4.1) [2]

[1] *Based on subsample of all available nests.*

American White Pelicans - Badger Island: In 2025, no (zero) pelicans were observed breeding on Badger Island in McNary Reservoir, the largest historical breeding site for the species in WA (Cramer et al. 2021). A large colony of pelicans, however, was active on Crescent Island, located just 3 Rkm downstream of Badger Island (see *below*). Trail cameras installed on Badger Island by the Yakama Nation observed coyotes (*Canis latrans*) on several different occasions during the breeding season, which was the presumed cause for colony abandonment in 2025 (T. De Boer, Yakama Nation, pers. comm.). Estimates of predation rates on UCR and SR smolts by American white pelicans breeding on Badger Island in years past were low, with estimates ranging from 0.1% (<0.1–0.3%) for SR yearling Chinook to 1.0% (0.2–3.1%) for SR steelhead, depending on the year (*Appendix A*). Results suggested that pelicans breeding on Badger Island posed little threat to the actively migrating ESA-listed UCR and SR smolts.

Unlike terns, gulls, and cormorants, pelicans forage in shallow water less than three meters deep by dipping their bills into the water and scooping prey items (Knopf and Evans 2004). Pelicans have been observed congregating and foraging near diversion structures such as Horn Rapids Dam on the Yakima River, particularly during June when large numbers of subyearling Chinook are migrating (Stinson 2016; Payton et al. 2023). Payton et al. (2023) estimated that

predation rates on non-listed subyearling Chinook from the Upriver Bright population by pelicans nesting on Badger Island were substantial, with upwards of 25% of wild smolts consumed by Badger Island pelicans in some years. Wild Handford Bright subyearling Chinook rear and reside in the middle Columbia River prior to outmigration—behavior that is believed to increase their susceptibility to pelicans breeding on nearby Badger Island (Payton et al. 2023). Recent completed research in the lower Yakima River, a shallow water system that enters the middle Columbia River 28 Rkm upstream of Badger Island, also indicated that pelicans posed a threat to smolt survival in some years, with predation rates on Yakima River origin steelhead, coho, and subyearling Chinook more than 8% of available fish in some years (Evans et al. 2024b). As such, predation rates on smolts by Badger Island pelicans were highly variable depending on the salmonid stock and year, with impacts highest on smolts in tributary systems of the middle Columbia River located near Badger Island (see also Payton et al. 2023 and Evans et al. 2024b).

Unlike gulls and terns, pelicans are also capable of consuming adult-sized salmonids. PIT-tagged adult salmonids ranging in size from 325 mm fork-length (jack Sockeye Salmon) to 770 mm fork-length (adult steelhead) have been consumed by Badger Island pelicans (Roby et al. 2017, Payton et al. 2025). In an investigation of the magnitude of pelican predation on adult sockeye during upstream migration from Bonneville Dam to Priest Rapids (on the middle Columbia River) and Ice Harbor Dam (on the Snake River), Payton et al. (2025) estimated that pelicans breeding on Badger Island consumed between 1,328 and 47,264 adult sockeye annually during 2014–2023. Predation rates on adult sockeye ranged from 1.5% to 8.4% of available sockeye each year, with predation impacts the greatest in years with the largest number of returning sockeye to Bonneville Dam. In addition to adult sockeye, tags from other larger-sized fish species have also been recovered on Badger Island, including white sturgeon, bull trout, and even adult shortnose suckers (*Chasmistes brevirostris*) from the Klamath Basin (U.S. Geological Survey Western Fisheries Research Center, unpublished data). The diverse diet and presence of adult-sized fishes highlight the differences in diet composition between pelicans and other piscivorous colonial waterbirds nesting in the CRB.

American White Pelicans - Crescent Island: An incipient colony of pelicans formed on Crescent Island in 2024, with a peak count of 51 individuals (Evans et al. 2025). The colony rapidly expanded to 2,815 individuals in 2025 (*Table 2*). One likely reason for the rapid growth of the pelican colony on Crescent Island in 2025 was the failure of the pelican colony on nearby Badger Island (see *above*). Similar to predation estimates by pelicans on Badger Island in years past, predation restimates by Crescent Island pelicans on the ESA-listed salmonid populations evaluated were low, ranging from 0.2% (<0.1–0.6%) on SR yearling Chinook to 0.4% (0.1–1.2%) on SR subyearling Chinook (*Table 10*). As noted above for Badger Island pelicans, however, predation effects on other salmonid stocks (e.g., Yakima and Umatilla River salmonids, Upriver Bright Chinook) and age-classes, including adult sockeye, may be substantially higher than those of actively migrating UCR and SR smolts. As such, additional research to quantify Crescent Island pelican predation effects on these and other groups of fish may be warranted.

American White Pelicans - Miller Sands Spit: Estimates of predation rates by pelicans breeding on Miller Sands Spit in the CRE were amongst the lowest of all colonies evaluated in the CRB in 2025 (*Table 10*) and in years past (*Appendix A*). Despite an estimated colony size of 1,168 individuals in 2025 (amongst the largest number counted since the colony formed in 2010; Roby et al. 2021a), only 82 tags from juvenile salmonids were recovered on the colony following the breeding season (*Table 3*), resulting in estimates of predation rates of < 0.1% per salmonid population (*Table 10*). Estimates of predation rates by pelicans on Miller Sands Spit in 2025 were very similar to those in years past, with predation rates of < 0.1% per salmonid ESU/DPS (Evans et al. 2023). Similar to the pelicans breeding on Crescent Island and Badger Island in years past, results suggest that pelicans breeding on Miller Sands Spit pose little threat to the survival of ESA-listed juvenile salmonids from UCR and SR ESUs/DPSs, particularly predation on actively migrating, yearling smolts. It should be noted, however, that of the 82 smolt PIT tags recovered on the Miller Sands Spit pelican colony, 43 (or 52%) were from subyearling Chinook, fish that were potentially still rearing and/or that congregated in shallow water habitats in the CRE (Sebring et al. 2013). Sample sizes of PIT-tagged subyearling Chinook from LCR ESUs/DPSs, were too small to generate reliable estimates of pelican predation. As such, future research may be warranted to determine to what extent pelicans breeding at colonies in the CRE limit the survival of subyearling Chinook or other smolts that originate from LCR tributaries.

Table 10. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by American White Pelicans breeding on Badger Island and Crescent Island during 2025. NA indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates (see Methods & Analysis section).

Salmonid	Crescent Island	Miller Sands
SR Sockeye	0.3% (0.1–1.2)	< 0.1%
SR Yearling Chin	0.2% (<0.1–0.6)	< 0.1%
UCR Yearling Chin	NA	< 0.1%
SR Subyearling Chin	0.4% (0.1–1.2)	< 0.1%
SR Steelhead	0.2% (0.1–1.5)	< 0.1%
UCR Steelhead	0.2% (0.1–1.8)	< 0.1%

Great Blue Herons and Great Egrets - Goose Island, Potholes Reservoir: In 2025, a small colony of herons nested on trees on Goose Island in Potholes Reservoir, where 54 breeding pairs were counted during the breeding season (*Table 3*). In total, only 24 smolt PIT tags were recovered after the breeding season and predation rates on UCR and SR salmonids were estimated to be low to non-existent at < 0.1% per salmonid population (*Table 11*). Unlike terns, which readily commute between Goose Island and the middle Columbia River to forage, herons may not

undertake this journey because the distance (> 30 kilometers) could be prohibitively far, or prey may be sufficiently available locally.

Great Blue Herons and Great Egrets - Hanford Island: In 2025, a small colony of herons also nested on trees on Hanford Island A, where 21 pairs were counted during the breeding season (*Table 3*). A total of 86 tagged fish were recovered following the breeding season, with predation rates less than 0.3% of all salmonid populations evaluated, with the highest estimate observed on UCR steelhead at 0.2% (0.1 –0.5%; *Table 11*). Of the 86 tags recovered, the majority (70 tags or 81%) were UCR Chinook smolts, a species where sample sizes of available smolts were insufficient for predation rate analysis. The overall small number of PIT tags recovered, however, suggests that predation rates on smolts by herons nesting on Hanford Island were low in 2025.

Great Blue Herons and Great Egrets - Island 20: In 2025, PIT tags were also recovered on a recently established heron colony on Island 20, with a total of 832 tags recovered following the breeding season (*Table 3*). This result was surprising given the relatively small size of the heron colony on Island 20 (114 pairs; *Table 3*) and the paucity of tags recovered on other nearby heron colonies, like those on Goose and Hanford islands (see *above*). The majority of tags were not from SR and UCR origin smolts, but rather from fish originating from the Yakima River (320 tags or 38%) or the middle Columbia River downstream of Priest Rapids Dam (409 tags or 49%), fish that were predominately Chinook (639 tags or 77%) or coho (163 tags or 20%) smolts. Results suggest that herons were disproportionately commuting to the lower Yakima River or to areas downstream of RIS to forage on salmon smolts. Research to better understand predation by herons at this and other colonies on Yakima River salmonids or from non-ESA-listed salmonids from the middle Columbia River may be warranted. There was little evidence, however, that herons posed a substantial risk to ESA-listed UCR and SR salmonids in 2025, with all estimates less than 0.1% of available fish (*Table 11*).

Table 11. Estimated predation rates (95% credible interval) on Snake River (SR) and Upper Columbia River (UCR) salmonids (i.e., sockeye, yearling Chinook, subyearling Chinook, and steelhead) by Great Blue Herons and Great Egrets breeding on Goose Island in Pothole Reservoir, Hanford Island and Island 20 in the middle Columbia River during 2025. NA indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates (see Methods & Analysis section).

Salmonid	Goose Island	Hanford Island	Island 20
SR Sockeye	< 0.1%	< 0.1%	< 0.1%
SR Yearling Chin	< 0.1%	< 0.1%	< 0.1%
UCR Yearling Chin	NA	NA	NA
SR Subyearling Chin	< 0.1%	< 0.1%	< 0.1%
SR Steelhead	< 0.1%	< 0.1%	< 0.1%
UCR Steelhead	< 0.1%	0.2% (0.1 –0.5)	< 0.1%

Cumulative Predation & Survival

Based on releases/detections of live PIT-tagged smolts at Lower Granite Dam (LGR) and RIS, avian predation effects were first observed in the river reach located downstream of Lower Monumental Dam (LMN; for SR smolts) and downstream of RIS (for UCR smolts). As such, for SR smolts, results indicated there was no measurable consumption of tagged fish between LGR and LMN associated with the bird colonies included in this study. For UCR smolts, however, predation occurred immediately following release/detection at RIS. Estimates of avian predation/consumption and total mortality (1 - survival) of smolts downstream of RIS and LMN varied considerably by bird species, colony location, and river reach, and are provided *below* by salmonid species and age-class (yearling, subyearling; Chinook only). Results from 2025 were then compared to results in years past to identify trends in predation and survival across river reaches and years dating back to 2008.

Upper Columbia River Steelhead: The cumulative effects of avian predation/consumption (predation by all avian predator species and colonies combined) on UCR steelhead smolts during passage from RIS to the Pacific Ocean were substantial in 2025, with an estimated 34.2% (28.6–40.9%) of smolts consumed (*Figure 8*). Of the avian predator species evaluated, estimated consumption by gull colonies was the greatest at 23.7% (18.9–29.6%), followed by double-crested cormorant predation at 5.3% (3.4–8.6%), tern predation at 4.2% (3.2–5.7%), Brandt's cormorant predation at 0.4% (<0.1–1.4%), heron predation at 0.1% (<0.1–0.5%), and pelican predation at <0.1% (<0.1–0.5%; *Figure 8*). It is important to note that cumulative estimates are based on smolt availability at RIS (or LMN for SR migrants) as opposed to the proportion that survive outmigration to within the foraging range of each downstream bird colony (see *Reach-specific Predation below* for estimates that account for survival during outmigration to each river reach). The cumulative effects of avian predation/consumption on UCR steelhead smolts in 2025 were slightly higher than those during 2022–2024, but lower than those observed in several, but not all, years dating back to 2008 (*Figure 8*). It should also be noted that estimates of cumulative predation/consumption by gull colonies on Island 20 and the Blalock Islands (Anvil and Straight Six) during 2008–2012 and by cormorants on Foundation Island during 2013, 2015–2019, the AMB during 2016–2020, and the TRT during 2012–2021 were not available because these sites were not scanned for smolt PIT tags in these years. As such, cumulative estimates were minimum estimates of predation on UCR and SR ESUs/DPSs in several, but not all, years past (as noted in figure descriptions, e.g., *Figure 8*).

Comparisons of avian predation/consumption to total mortality, which were available during smolt passage from RIS to Bonneville Dam (BON) or LMN to BON (for SR migrants), but not downstream of BON due to a lack of smolt survival estimates in the CRE (see *Methods & Analysis*), indicated that avian predation accounted for 43.4% (33.4–63.0%) of all UCR steelhead mortality sources in 2025. Results suggested that factors other than avian predation were the dominant sources of smolt mortality in 2025. In most years dating back to 2008, however, avian predation/consumption has been the dominant or single greatest source of mortality for UCR steelhead smolts during outmigration to BON (Evans et al. 2016, Payton et al. 2019, Evans et al. 2019, Evans et al. 2022b). For instance, avian predation/consumption has accounted for > 50%

of all UCR steelhead smolt mortality sources during 13 of the last 18 years (*Figure 9*; see also Evans et al. 2022b).

The representative tagging (random and in proportion to the run at-large) of hatchery and wild UCR steelhead (see *PIT-tagging of Upper Columbia River Steelhead* section) at RIS generated a unique dataset to make relative comparisons of predation and survival based on a fish's rear-type (see *Appendix B).*

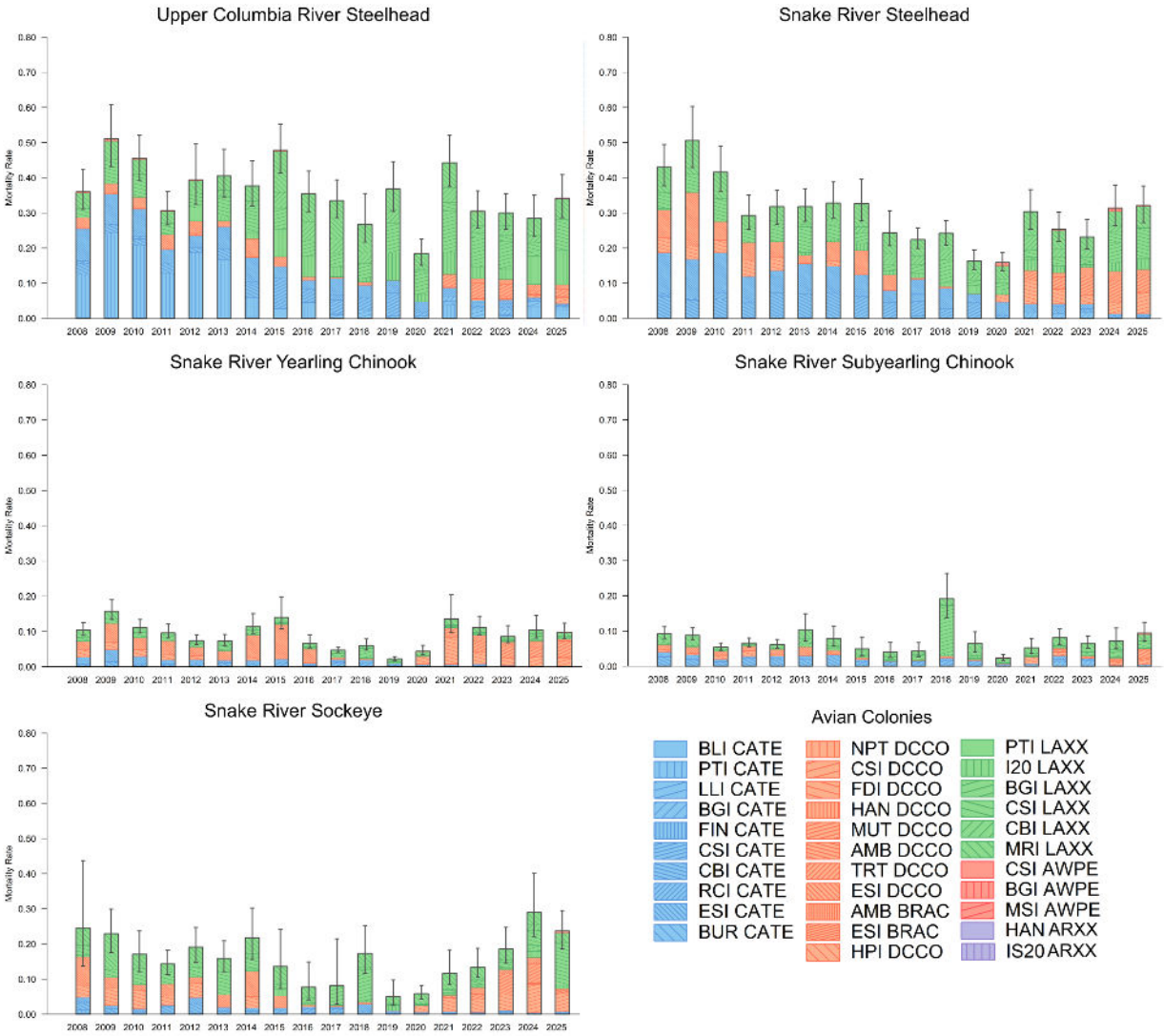


Figure 8. Estimated cumulative predation/consumption for Upper Columbia River steelhead smolts during passage from Rock Island Dam to the Pacific Ocean and for Snake River steelhead, yearling Chinook, subyearling Chinook, and sockeye smolts during passage from Lower Monumental Dam to the Pacific Ocean. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), American white pelicans (AWPE), great blue herons and egrets (ARXX; see Figure 1 for colony locations). Estimates are proportions with error bars representing 95% credible intervals. No estimates of predation were available for IS20 and CBI LAXX during 2008–2012, for FDI DCCO during 2013, 2015–2019, AMB DCCO during 2016–2020, and TRT DCCO during 2012–2021, resulting in minimum estimates of cumulative predation in those years.

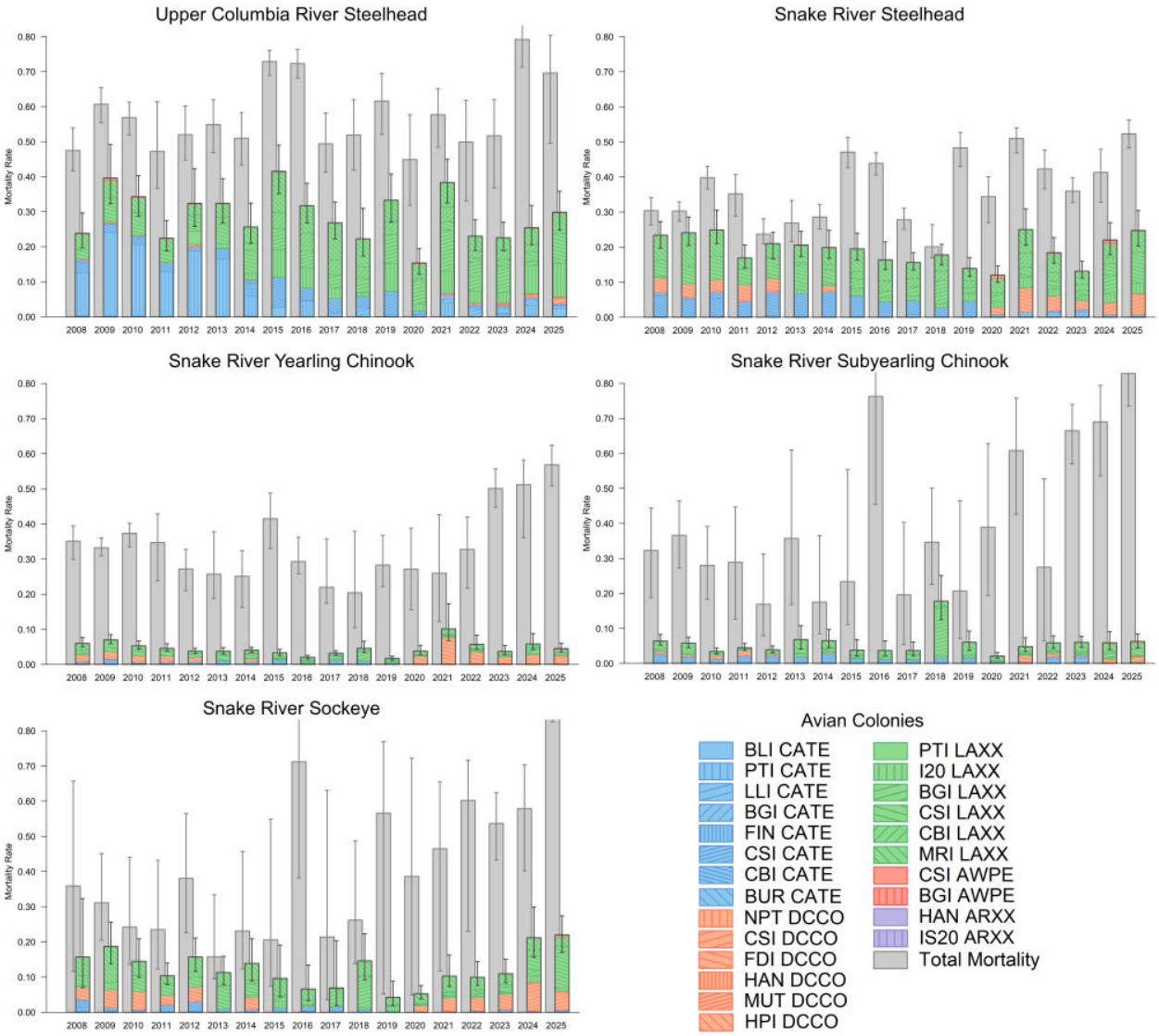


Figure 9. Estimated total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Upper Columbia River steelhead during passage from Rock Island Dam to Bonneville Dam or Snake River steelhead, yearling Chinook, subyearling Chinook, and sockeye during passage from Lower Monumental Dam to Bonneville Dam. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), American white pelicans (AWPE), great blue herons and egrets (ARXX; see Figure 1 for colony locations). Estimates are proportions with error bars representing 95% credible intervals. No estimates of predation were available for IS20 and CBI LAXX during 2008–2012 and FDI DCCO in 2013, 2015-2019, resulting in minimum estimates of predation in those years.

Snake River Steelhead: The cumulative effects of avian predation/consumption on SR steelhead smolts during passage from LMN to the Pacific Ocean in 2025 were similar to those of UCR steelhead, with an estimated 32.1% (27.2–37.7%) of SR steelhead smolts consumed (*Figure 8*). Of the avian predator species evaluated, estimated consumption by gull colonies was the greatest at 17.6% (13.3–21.7%), followed by double-crested cormorants at 12.5% (9.4–16.7%), terns at 1.3% (1.0–1.8%), Brandt's cormorants at 0.3% (0.2–0.5%), pelicans at 0.2% (<0.1–1.5%), and herons at < 0.1% (*Figure 8*). Predation by cormorants on SR steelhead were higher than that of UCR steelhead, especially predation by cormorants that formerly nested on Foundation Island (prior to 2024) and on Crescent Island in 2024 and 2025 (*Figure 8*). The cumulative effects of avian predation/consumptions on SR steelhead smolts in 2025 was slightly higher than that observed during 2021–2024, but consistent with several other years dating back to 2008 (*Figure 8*).

Estimated total mortality of SR steelhead during passage from LMN to BON was 52.3% (48.3–56.3%) in 2025. Estimates of total mortality in 2025 were slightly higher than those in 2022–2024 and amongst the highest observed to-date (*Figure 9*). Comparisons of avian predation/consumption to total mortality indicated that avian predation accounted for 47.5% (38.3–59.6%) of all SR smolt mortality sources during passage from LMN to BON in 2025 (*Figure 9*). Analogous to results on UCR steelhead, avian predation/consumption accounted for > 50% of all SR steelhead smolt mortality sources in 13 of the last 18 years (*Figure 9*; Evans et al. 2022a). As such, avian predation/consumption was the dominant source of mortality during smolt passage from LMN to BON in most, but not all, years. Estimated total mortality of SR steelhead during passage from LGR to LMN was low at 1.9% (1.9–2.1%) of smolts in 2025, indicating that the vast majority of mortality occurred downstream of LMN once smolts were exposed to predation by colonial waterbirds and other sources of mortality (see also Evans et al. 2022a).

Snake River Yearling Chinook: The cumulative effects of avian predation/consumption on SR yearling Chinook during passage from LMN to the Pacific Ocean in 2025 were significantly lower than those of SR and UCR steelhead, with an estimated 9.9% (8.0–12.4%) of smolts consumed (*Figure 8*). Of the predator species evaluated, estimated predation/consumption were the highest by double-crested cormorant colonies at 7.2% (5.5–9.7%). Significantly lower levels of predation/consumption were observed in gulls at 1.8% (1.1–2.8%), Brandt's cormorants at 0.4% (0.2–0.7%), terns at 0.2% (0.1–0.4%), and pelicans at 0.2% (<0.1–0.5%; *Figure 8*). No (zero) SR yearling Chinook PIT tags were recovered on the heron colonies included in the study. The cumulative effects of avian predation/consumption on SR yearling Chinook smolts in 2025 were similar to estimates in year past (*Figure 8*). As noted for UCR and SR steelhead, however, a lack of predation estimates from cormorants on Foundation Island during 2013, 2015–2019, the AMB during 2016–2020, and the TRT during 2012–2021, resulted in minimum estimates of cumulative predation in those years.

Similar to SR steelhead, total mortality of SR yearling Chinook from LGR to LMN was low (1.4% {1.2–3.3%}) but then increased substantially during passage from LMN to BON in 2025 (56.9% {50.8–62.4%}; *Figure 9*). Unlike SR steelhead, however, comparisons of avian

predation/consumption to total mortality in SR yearling Chinook indicated that avian predation/consumption accounted for only 8.0% (6.2–10.9%) of all smolt mortality sources during passage from LMN to BON in 2025 (*Figure 8*). Results are consistent with those in the past and suggest that upstream of BON, predation/consumption on SR yearling Chinook by colonies included in this study posed a much lower risk to smolt survival than that of steelhead, albeit avian predation was a substantial source of SR yearling Chinook smolt mortality in some river reaches and years, particularly cormorant predation both upstream and downstream of BON (see *Reach-specific Predation and Survival* and Evans et al. 2022b for a more detailed discussion).

Snake River Subyearling Chinook: Similar to results in SR yearling Chinook, an estimated 9.4% (7.1–12.5%) of SR subyearling Chinook smolts were consumed by the avian predator species and colonies included in the study during smolt passage from LMN to the Pacific Ocean in 2025 (*Figure 8*). Of the avian predator species evaluated, predation/consumption estimates on SR subyearling Chinook were the greatest by double-crested cormorant colonies at 4.3% (2.9–6.5%), followed by gull colonies at 3.8% (1.7–6.1%), Brandt's cormorants at 0.4% (0.1–1.2%), terns at 0.4% (0.2–0.8%), and pelicans at 0.4% (<0.1–1.2%) (*Figure 8*). Also similar to SR yearling Chinook, no tags from SR subyearling Chinook were recovered on the heron colonies included in the study. The cumulative effects of avian predation/consumption on SR subyearling Chinook smolts in 2025 were similar to those observed in 2024 and in many, but not all, previous years dating back to 2008 (*Figure 8*; Evans et al. 2022a).

Estimates of total mortality in SR subyearling Chinook during passage from LMN to BON were highly variable depending on the year, ranging from 42.8% to 82.8% during 2008–2025, with 2025 being the highest total mortality estimate of SR subyearling Chinook in the study period (*Figure 9;* see also Evans et al. 2022a). This is notably higher than the previous maximum total mortality estimate of 74.8%. Comparisons of avian predation/consumption to total mortality indicated that avian predation accounted for 7.6% (5.4–10.6%) of all SR subyearling Chinook smolt mortality sources during passage from LMN to BON in 2025 (*Figure 9*). The precision of estimates of predation and survival in SR subyearling Chinook have also been highly variable due to fluctuations in the sample sizes of PIT-tagged smolts, with numbers ranging from 2,546 to 34,742 tagged smolts annually (Evans et al. 2022a). It is also worth noting that unlike SR steelhead, SR yearling Chinook, and SR sockeye, a large proportion of SR subyearling Chinook originate downstream of LGR, and results presented herein apply only to those fish that originated upstream of LGR. Finally, results from 2025 were also a record low estimate of total mortality attributed to avian predation/consumption at 7.6% (5.4–10.6%), with a previous range of 9% to 39% from 2008–2024 (*Figure 9*; Evans et al. 2022a).

Snake River Sockeye: The cumulative effects of avian predation/consumption on SR sockeye during passage from LMN to the Pacific Ocean were significantly higher than those on Chinook but comparable to those of steelhead, with an estimated at 23.7% (18.6–29.5%) of smolts consumed in 2025 (*Figure 8*). Of the avian predator species evaluated, estimated predation/consumption was the greatest by gull colonies at 15.3% (11.4–19.7%), followed by double-crested cormorants at 6.3% (4.5–9.7%), and predation estimates of < 1.0% in all other

predator species. For instance, predation estimates on sockeye were significantly lower by tern, pelican, and Brandt's cormorant colonies at just 0.8% (0.4–1.4%), 0.5% (0.1–1.2%), and 0.4% (<0.1%–1.6%) respectively (*Figure 8*). Finally, heron colonies posed little to no threat to SR sockeye, with predation estimates of <0.1% (<0.1–0.2%). The cumulative effects of avian predation/consumption on SR sockeye in 2025 were slightly lower than those in 2024 (29.0% {21.9–40.1%}), which was the highest observed since system-wide studies of predation commenced in 2008 (*Figure 5*). There is also evidence that predation effects, largely from cormorants, steadily increased from 2019 to 2024 before leveling off in 2025 (*Figure 9*). Even though cumulative estimates were based on availability at LMN, predation was the highest by cormorant colonies in the CRE, over 500 Rkm downstream of LMN (see *Reach-specific Predation and Survival* section *below* for estimates of predation after accounting for survival to each river reach).

Estimates of total mortality in SR sockeye in 2025 were amongst the highest observed to-date at 91.1% (82.6–95.1%) during passage from LMN to BON. Comparisons of avian predation/consumption to total mortality indicated that avian predation accounted for 24.3% (18.6–30.7%) of all SR sockeye smolt mortality sources during passage from LMN to BON in 2025 (*Figure 9*). Results from this study suggest that non-avian sources of mortality on SR sockeye were substantial in 2025 and in many, but not, all years past (*Figure 9*). Similar to SR Chinook, however, the highest reach-specific estimates of predation on SR sockeye were downstream of BON in the CRE, a segment of river where estimates of total smolt mortality were unavailable (see *Methods & Analysis* section). Like results from SR subyearling Chinook, fluctuations in the samples sizes of PIT-tagged SR sockeye resulted in imprecise estimates of predation and, especially, survival in years past (*Figure 9*; see also Evans et al. 2022a for a more detailed discussion).

Collectively, results of this study indicated that the cumulative effects of avian predation/consumption on salmonid smolts and the proportion of all sources of smolt mortality that were associated with predation were highly variable across salmonid species, avian predator species, and years. The approach of this study to investigate avian predation on multiple salmonid species that share a common migration corridor revealed several important findings, including that (1) avian consumption was associated with the majority of mortality for steelhead smolts during outmigration, but a relatively small proportion of total mortality for yearling and subyearling Chinook smolts; and (2) the species and colony location of piscivorous waterbirds nesting in the CRB dramatically influenced the magnitude of consumption, with some colonies posing little threat to smolt survival, while others were associated with mortality of a large proportion of the available fish (see also Evans et al. 2019 and 2022b for a more detailed discussion of the cumulative effects of avian predation).

Results from this and several other published studies (e.g., Evans et al. 2016, Evans et al. 2019, Payton et al. 2019, Payton et al. 2020, Evans et al. 2022b) indicate that mortality of both UCR and SR steelhead smolts associated with predation/consumption by piscivorous colonial waterbirds was greater than that from all other mortality sources combined during outmigration to BON in most years. For instance, steelhead smolt losses associated with

predation/consumption by piscivorous colonial waterbirds upstream of BON were greater than the combined direct losses associated with passage through all upstream hydroelectric dams, predation from piscivorous fish, predation by predator species and colonies that were not included in the study, mortality from disease, and all other mortality factors combined. Results from 2025 indicate the predation/consumption of SR sockeye may have leveled off after increasing over the course of the last 5 years (2020–2024). Increased predation on sockeye corresponded with increases in cormorant predation, particularly in the CRE due to cormorants breeding on the AMB and Troutdale Towers (see also Reach-specific Predation/Consumption *below*). These results over the course of the last 18 years of research provide strong evidence that avian predation/consumption is a factor limiting the survival of some salmonid populations that are listed under the U.S. Endangered Species Act, particularly UCR steelhead, SR steelhead, and SR sockeye (see also Evans et al. 2022a).

Reach-specific Predation/Consumption: Reach-specific predation/consumption effects, those that account for the survival of smolts to each downstream dam with PIT tag detection capabilities, indicated that colonial waterbirds were consuming smolts in all river-reaches downstream of RIS and LMN, but that predation/consumption varied by reach, salmonid species, and bird colony. In 2025, avian predation effects on SR salmonids were the greatest during smolt passage from LMN to MCN and from BON to the Pacific Ocean, the furthest downstream reach evaluated. By salmonid species, predation rates in the CRE were the highest for SR sockeye at 18.0% (6.5–43.0%), followed by SR subyearling Chinook at 17.7% (9.1–33.1%), SR steelhead at 15.0% (11.6–20.0%), and SR yearling Chinook at 12.2% (8.8–17.7%) in 2025 (*Figure 10*). Predation in this reach was also substantial for UCR steelhead at 13.6% (6.2–28.3%). Since smolts that have survived outmigration through the Columbia River Power System to the CRE are, on average, more likely to survive to adulthood than those that are yet to complete outmigration, the benefits of managing avian predators in the CRE may be greater on a per fish basis than managing inland avian predators (Roby et al. 2002, Payton et al. 2020, Evans et al. 2022c).

Predation/consumption effects, particularly for UCR and SR steelhead and SR sockeye, were also substantial (>10%) during smolt passage through the middle Columbia and lower Snake rivers to McNary Dam in 2025 (*Figure 10*). For instance, an estimated 23.7% (19.2–29.5%) of UCR steelhead were consumed during passage from RIS to McNary Dam (MCN) in 2025 (*Figure 10*), amongst the highest reach-specific estimates observed since system-wide studies commenced in 2008. Predation/consumption of SR steelhead and SR sockeye was also substantial in this river reach (LMN to MCN) at 18.2% (14.2–23.2%) and 18.5% (13.7–23.6%), respectively, in 2025 (*Figure 10*). Similar to trends in years past, predation/consumption on smolts during passage from MCN to John Day Dam (JDA) and JDA to BON was almost entirely due to gulls in 2025 (*Figure 10*). Gulls consumed between 1–5% of the SR smolts during passage from MCN to John Day Dam (JDA) and between 3–12% of SR smolts during passage from JDA to BON. Predation by gulls in all river reaches evaluated upstream of BON in 2025 were amongst the highest observed to-date.

It is important to note that reach-specific estimates of predation/consumption may differ from ESU/DPS- and colony-specific estimates of predation/consumption (see *Avian Predation Rates* section) because reach-specific estimates partition predation for those colonies capable of foraging in multiple river reaches, while the colony- and ESU/DPS-specific estimates depict the effects of predation on all smolts available to birds at each colony regardless of the river reach. For example, gulls breeding on Miller Rocks can forage on smolts both upstream and downstream of John Day Dam and reach-specific estimates account for this by partitioning predation accordingly (see also Payton et al. 2019 for additional details).

Finally, estimates of reach-specific predation and survival were also generated on a weekly basis, estimates that were used to investigate the additive effects of predation on smolt survival (see *Additive Effects of Predation*). Weekly estimates of predation and survival for UCR and SR smolts in 2025 are provided in *Appendix D*.

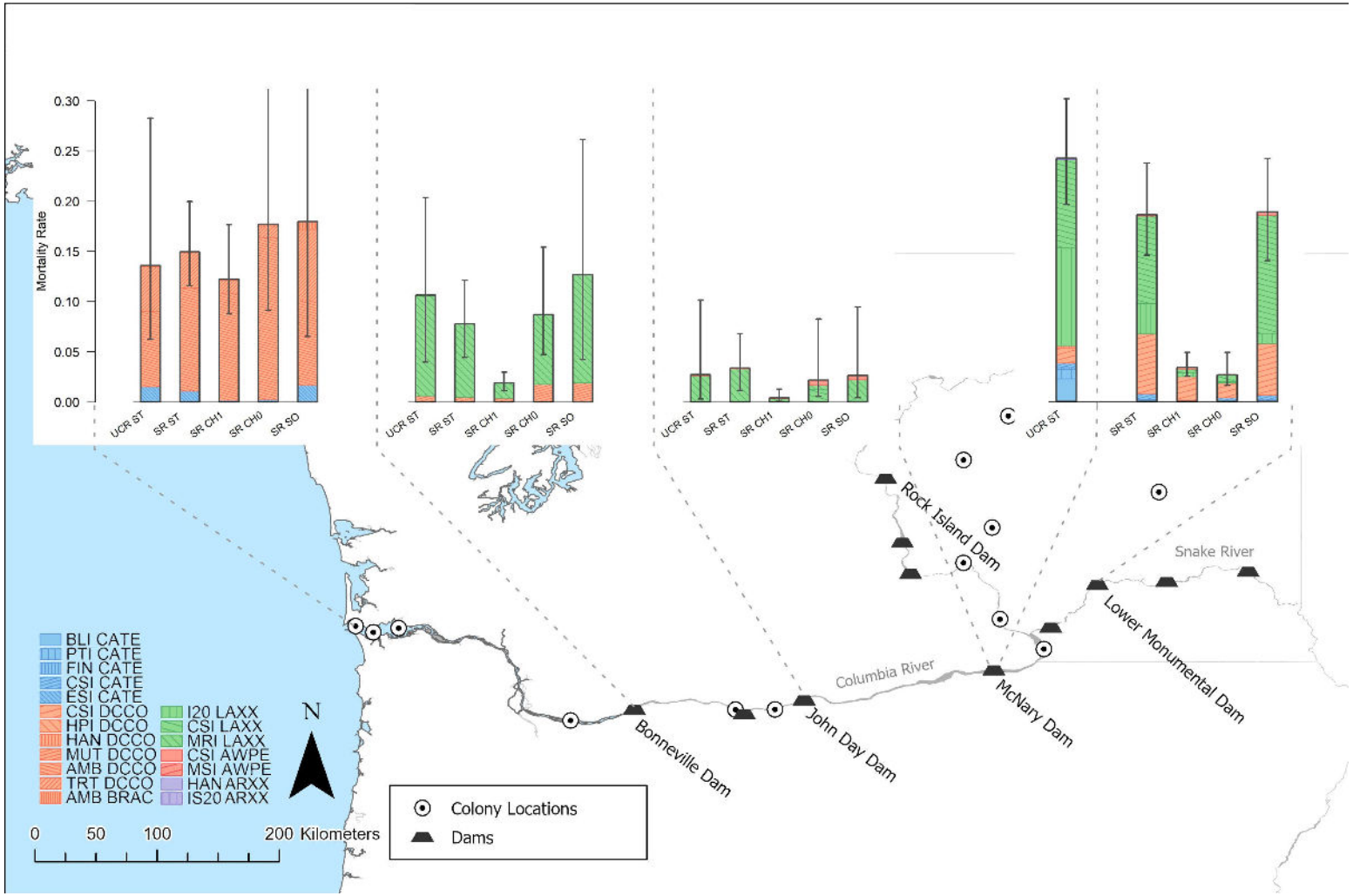


Figure 10. Reach-specific predation/consumption for Upper Columbia River (UCR) steelhead smolts during passage from Rock Island Dam to the Pacific Ocean and for Snake River (SR) steelhead, yearling Chinook (CH1), subyearling Chinook (CH0), and sockeye (SO) during passage from Lower Monumental Dam to the Pacific Ocean in 2024. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), American white pelicans (AWPE), great blue herons and egrets (ARXX; see Figure 1 for colony locations). Estimates are proportions with error bars representing 95% credible intervals.

Additive Effects of Predation

The updated investigation of the additive effects of gull predation/consumption on steelhead smolts indicated that increases in gull predation on smolts between RIS and BON were associated with statistically significant decreases in smolt survival to BON from 2008 to 2025 (*Table 12* and *Figures 11–12*). Results suggest that significantly more steelhead smolts would have survived outmigration to BON in the absence of gull predation upstream of Bonneville Dam. Model comparison via LOO-CV indicated that a full model, including additivity parameters for both gull and tern predation provided the best fit to the data (*Table 12*). By this criterion, the full model performed significantly better than all alternatives. Using the full additivity model, the estimated average difference between observed survival (i.e., survival with gull predation) and baseline survival (i.e., survival in the absence of gull predation) was 0.15 (0.04–0.26) for UCR steelhead after accounting for the predation impacts of terns (*Table 14*). For example, the average annual UCR steelhead survival probability from RIS to BON during 2008–2025 was 0.44 (0.41–0.48) and results of this study suggest that, following the additive relationship measured, estimated average annual survival in the absence of gull predation would have been 0.56 (0.47–0.66).

Table 12. Leave-one-out cross-validation (LOO-CV) comparison of models with different avian predator additivity configurations. Expected log pointwise predictive density (ELPD) values are shown with standard errors (SE). Differences (Diff) are relative to the best model. Models tested included only LAXX additivity, only CATE additivity, LAXX and CATE additivity, and no additivity (null model).

Model	ELPD	ELPD_SE	Diff	SE_Diff	Diff_z
LAXX & CATE	-97463.5	508.2	0.0	0.0	NA
CATE	-97469.5	508.2	6.0	1.8	3.4
LAXX	-97475.9	508.3	12.5	3.5	3.6
null	-97479.6	508.2	16.1	2.3	6.9

Table 13. Average annual predation/consumption probabilities for California and ring-billed gulls (LAXX) and survival probabilities for Upper Columbia River (UCR) steelhead smolts during outmigration from Rock Island Dam to Bonneville Dam during 2008–2025. Also provided are average estimates of the magnitude of the association between predation probabilities and survival probabilities ($\boldsymbol{a}$, additivity) and estimates of the differences ($\boldsymbol{\Phi}^{\Delta}$) in survival probabilities relative to estimated baseline survival probabilities associated solely with gull consumption, associated solely with tern predation, and associated with both predation sources. Values are reported as medians with 95% credible intervals.

Predator	Survival	Predation	a	Φ^{Δ}
LAXX		0.18 (0.17–0.19)	0.88 (0.21–1.57)	0.15 (0.04–0.26)
CATE	0.44 (0.41–0.48)	0.11 (0.10–0.12)	1.51 (0.94–2.10)	0.14 (0.09–0.20)
LAXX & CATE		0.29 (0.27–0.30)	0.99[1] (0.53–1.49)	0.26 (0.14–0.38)

[1] *The a statistic for combined predation was not explicitly computed, but rather imputed from the a statistic for LAXX and CATE independently.*

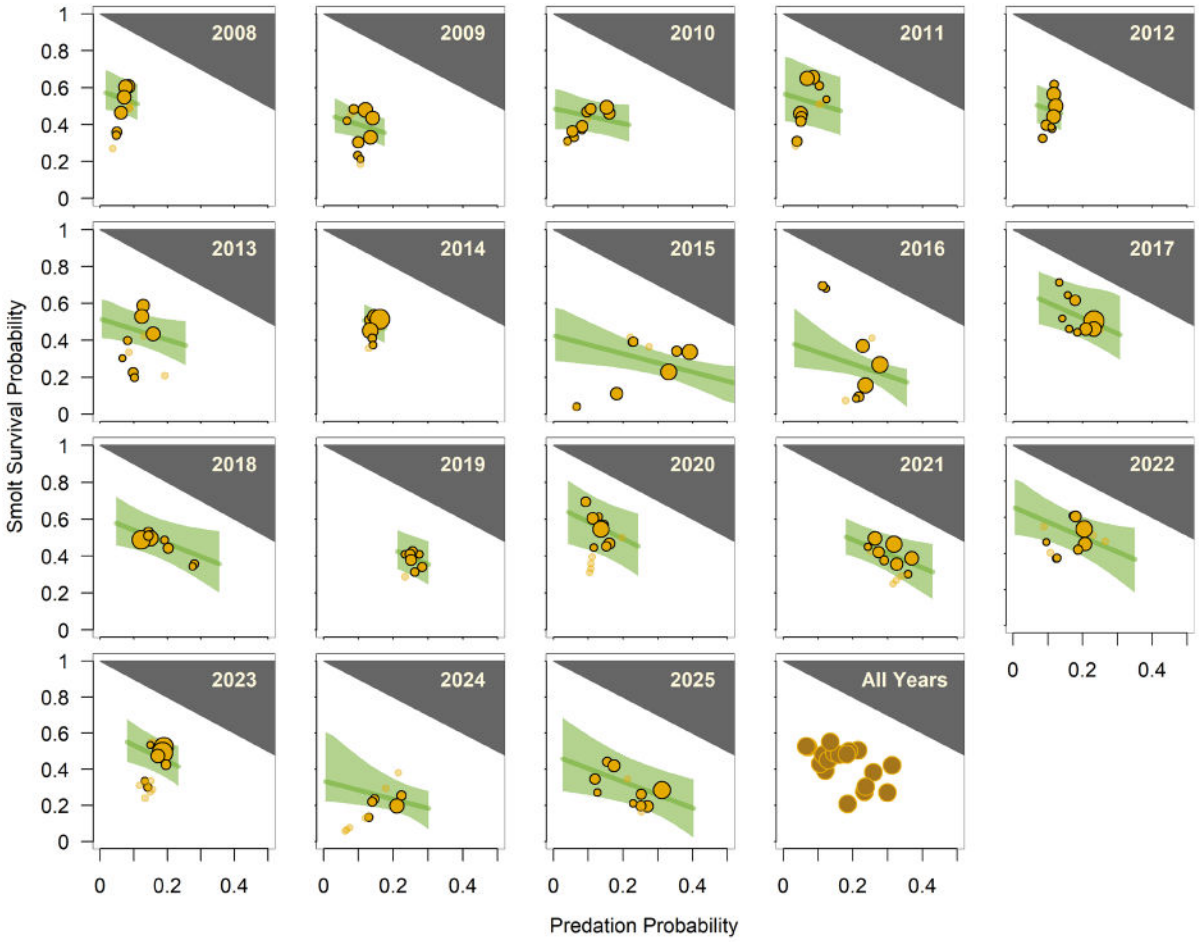


Figure 11. Weekly and annual probability estimates of Upper Columbia River steelhead smolt survival and gull consumption during smolt outmigration from Rock Island Dam to Bonneville Dam during 2008–2025. The size of circles depicts relative numbers of steelhead smolts available each week. Dashed lines represent the best fit estimate of the linear relationship between survival and consumption, and shading denotes 95% credible intervals around the best fit. The grey triangle represents the boundary space where consumption cannot be greater than survival.

Previously published research demonstrated that predation by terns was an additive source of steelhead mortality during smolt outmigration from RIS to BON (Payton et al. 2020). Similar to predation rates by terns, gulls also consumed a substantial proportion of available steelhead smolts in most, but not all, years (*Table 14*). However, despite high annual consumption probabilities by gulls, previous research could not effectively describe a relationship between gull consumption and fish survival. Weekly estimates of gull consumption were often less variable and less precise than estimates of tern predation. Further, the data available for gulls is incomplete with several missing years of recovery data from 2008–2012, years when gull colonies existed, but they were not scanned for fish tags (Evans et al. 2022a). Past analyses of gull consumption have also limited the scale of interest and truncated the dataset to avoid concerns inherent to analyses of larger spatial scales and mortality sources – such as other avian (e.g., tern) and non-avian mortality factors – which may influence and possibly mask the effects of gull consumption on smolt survival. Additionally, these previous analyses attempted

to censor gulls which co-nested with terns (e.g., on Crescent and Goose islands) whereby some proportion of the fish consumed by gulls were presumably captured by terns and stolen by gulls, this consumption was therefore inherently compensatory (Evans et al. 2022a). However, with this updated analytical approach, we have explicitly included tern predation, essentially constructing a new "baseline" survival probability with tern predation included. This allows for a concerted look at the full impacts (to the extent data was available) of gull predation on steelhead smolt survival with tern predation occurring on varying levels.

In the initial investigation of the additive effects of tern predation, the Burnham model was reparametrized, using the statistic, a, to deliberately address the specific question of "what proportion of depredated salmonids would have otherwise survived if not for avian predators". While this approach effectively answered that question of interest, the measure is necessarily tied to the annual rate of baseline survival. Using the original Burnham parameterization – measuring additivity using the statistic, b (where $a = b * S_0$ which implies $b = a/S_0$) – the level of additivity inherent to predation is explicitly independent of variations in probabilities of baseline survival. By assessing the additive effects of gull consumption independent of baseline survival and by calculating additivity measures of tern predation in the same analyses, we can better assess the true "nature" or impact of gull predation/consumption. With this simultaneous estimation approach, we are also able to assess whether there is a relationship between the additivity levels in gulls versus that of terns, which might be expected if kleptoparasitism accounts for a significant number of tags subsequently recovered on gull colonies. In fact, we see little variation in the level of additivity of gull predation/consumption across years in which gulls were nesting among terns and years in which gulls and terns did not nest on any of the same islands (*Figure 13)*. The lack of variation in additivity levels of gulls among years, and a consistent estimate of *b* near 1.0 (i.e., complete additivity) may indicate that gulls are simply taking a random sample of fish from the river. This contrasts greatly to the variation in additivity levels observed in terns, which may be largely explained by the considerable variation in tern colony sizes and locations. Starting with management actions in 2014, terns from Goose Island that foraged in the middle Columbia River moved downstream to the lower Columbia River – closer to the end of the reach under consideration – which increased the level of additivity (i.e., they were eating a much larger proportion of steelhead that would otherwise survive to the end of the reach). This would further help explain the coincidental increase in predation by gulls following the induction of tern management. With the highly super-additive source of steelhead mortality eliminated upstream, a larger proportion of steelhead was then available to gulls downstream and may have consequently resulted in higher predation rates by these other predators. Further model adjustments should be explored to assess whether this relationship between gull predation and survival rates is consistent at different spatial scales and across prey species.

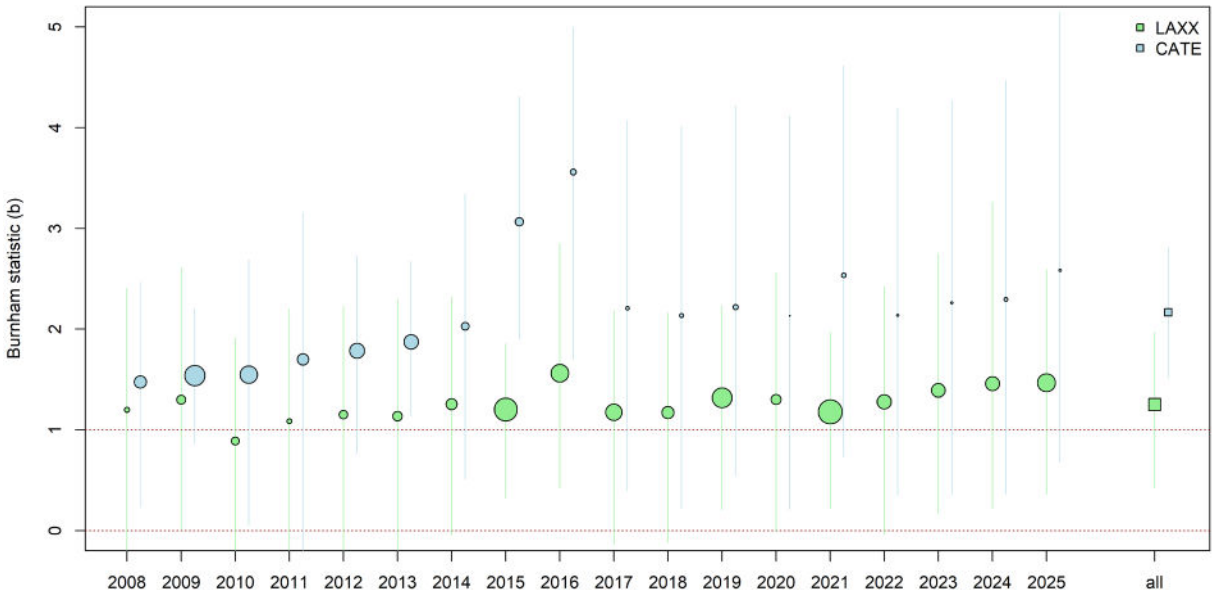


Figure 13. Annual estimates of the Burnham statistics (b) associated with gull predation (green points) and Caspian tern predation (blue points) on Upper Columbia River steelhead smolts during smolt outmigration from Rock Island Dam to Bonneville Dam during 2008–2025. The size of circles depicts relative predation probability in each year. Dashed lines at 0 and 1 represent b-static values associated with completely compensatory and completely additive sources of mortality.

Results suggest that gull consumption was a largely, if not completely, additive source of mortality in UCR steelhead. The average estimate of $\hat{a}$ = 0.88 {0.22–1.57} across all years suggests that most (~88%) smolts potentially would have survived outmigration to BON in the absence of gull predation upstream of BON. Using the b statistic (isolating the additive effects from variations in baseline survival) we consistently observe estimates greater than one, indicating fully additive, if not super-additive, effects of gull predation. Payton et al. (2020) observed a super-additive (a > 1.0) relationship with tern predation on steelhead. Using the b statistic parameterization, we observe an even greater super-additive implication. Super-additivity could be related to the foraging behavior of terns and perhaps gulls, where unsuccessful predation attempts result in some proportion of smolts being injured (lethally- or sub-lethally), losses that are akin to latent mortality or crippling losses in harvest management (Williams et al. 2002, Schaub and Lebreton 2004, Payton et al. 2020). Second, terns and gulls disproportionally consume larger-sized steelhead smolts, smolts that are more likely to survive outmigration (Evans et al. 2014). As such, estimates of avian predation/consumption probabilities reported herein may underestimate the full impact of gull and, especially, tern predation on steelhead smolts during outmigration, and thus produce estimates of super-additivity during the smolt life-stage (Payton et al. 2020).

Previous research indicates that smolts are especially susceptible to gull predation in the tailrace of hydroelectric dams, areas where smolts may be injured or stunned because of dam passage or where turbulent hydraulic conditions may disorient fish, thereby increasing their risk of being depredated by gulls (Ruggerone 1986, Zorich et al. 2011, Evans et al. 2016). The

proportion of smolts consumed by gulls in the current study that were temporarily stunned or moribund in the tailraces of dams were unknown. Gull predation/consumption, however, was not limited to the tailraces of dams, with gulls breeding on Crescent Island and Miller Rocks foraging both upstream and downstream of McNary Dam and John Day Dam, respectively (see *Figure 10*). Evans et al. (2016) documented that the majority (numerically) of the smolts consumed by gulls breeding at Miller Rocks in 2012 and 2014 were consumed in the open reservoirs upstream and downstream of John Day Dam. As such, although gulls breeding at colonies that are in close proximity to dams may disproportionately consume smolts at those dams, they do not exclusively forage at dams. Thus, additional research is warranted to quantify where gull predation/consumption occurs (near dams versus away from dams) and what proportion of smolts were moribund and, ultimately, to what degree gull management at dams is likely to increase smolt survival.

The results of our updated analysis include data on gull predation effects from eight additional years (2008–2012, 2023–2025). Updated results also relied on a reconsideration of some of the underlying assumptions of Payton et al. (2020) to better reflect fundamental properties of the system under consideration, including accounting for the additive effects of tern predation in the same model. The new modelling framework also allows for a more precise understanding of the impacts of each predator species independently. Regardless, difficulties persist in calculating precise estimates of additivity and the ambiguity associated with the interpretation of either the a or b parameterizations are the primary motivation for instead focusing on the differences in survival probabilities with and without predation ($i.e.\ \Phi^{A}$*).* Unlike estimates of additivity, this metric provides management-relevant information with which to quantify the degree to which predation limits fish survival (see also Payton et al. 2020). For instance, the new model results indicate that the average difference in observed and baseline survival across all years attributable solely to gull predation for UCR steelhead was 0.13 (0.03–0.23) while the average difference in observed and baseline survival across all years attributable to gulls and terns combined was 0.26 (0.15–0.37).

The ability to accurately assess the relationship between predation/consumption and survival probabilities depends on several criteria, including the level or magnitude of predation, intra-annual (e.g., weekly) variation in estimates of predation, the salmonid life-stage (smolt or smolt-to-adult) evaluated, and sufficient sample sizes of tagged smolts to generate precise estimates of predation/consumption and survival probabilities (see also Payton et al. 2021). Accounting for annual variation in predation/consumption and survival probabilities and the use of weekly cohorts within years as replicates are also critical components of the modelling framework (Payton et al. 2020, ISAB 2021). In cases where predation/consumption probabilities were low or where baseline mortality was high, larger sample sizes of tagged fish may still be insufficient to identify the relationship between predation and survival probabilities (Payton et al. 2021). However, in these circumstances, where predation probabilities were low, reductions in avian predation presumably would not dramatically increase fish survival. For example, in 2020, when gull consumption probabilities were estimated to be 0.04, reductions in gull consumption would do little to increase smolt and, especially, smolt-to-adult survival.

Results of the additive effects of avian predation/consumption by other avian predator species on other ESA-listed salmonid species (i.e., Chinook and sockeye salmon) and populations are provided in Payton et al. (2021a). It is important to note that the magnitude of avian predation/consumption rates differ dramatically amongst salmonid ESUs/DPSs, as does the potential benefits of managing avian predators to increase salmonid survival in the CRB. For instance, predation effects are generally, but not always, lower on salmon compared with steelhead, so even if the additive effects of predation are high, the potential benefits of managing predators to increase salmon survival maybe substantially less than that of steelhead due to lower levels of predation in salmon. The exception appears to be predation by cormorants on salmon in the CRE, where predation rates are often as great or greater (e.g., for Lower Columbia River ESUs/DPSs; Lawes et al. 2021b) on salmon compared with steelhead. Finally, a more detailed discussion of the modelling approach used in this study is provided in Payton et al. (2020, 2021a) and can also be found in a report by the Independent Scientific Advisory Board (ISAB 2021). The construction and implementation of this multi-predator modelling effort is the first step in a comprehensive look at the additive effects of mortality across the salmonid life cycle. Future analyses will expand upon this investigation looking at other salmonid prey species (e.g., Snake River steelhead, Chinook, and sockeye), additional avian predators (e.g., cormorants, pelicans) and at multiple spatial scales (e.g., partitioning the to BON spatial segment into finer scales and BON to adult return) within the same analyses. The assessment of all predator impacts across multiple spatial scales within a single model could potentially allow managers a much more refined and precise tool with which to address predator management in the future.

MANAGEMENT RECOMMENDATIONS

Based on the results of this study from 2025 and in previous years (Roby et al. 2021a, Evans et al. 2022b, Evans et al. 2023, Collis et al. 2024, Evans et al. 2024a, Evans et al. 2025), the following management recommendations are offered to maximize the benefits to ESA-listed juvenile salmonids associated with management of piscivorous colonial waterbirds in the CRB, while at the same time, minimizing the impacts of management on protected populations of migratory birds. Previous management policy for avian predators in the basin has been shaped by political forces that reflect the diverse values of stakeholders and the differing resource management mandates of federal, state, and tribal agencies. Although we do not expect the scientific results described in this report to change the values and political views of stakeholders, we believe that the best policies are built on a solid foundation of science that informs management. Towards this end, we hope that this report and the recommendations below will make a substantive contribution to that foundation (see also Roby et al. 2021a).

CASPIAN TERNS

The plan to reduce the size (number of breeding pairs) of the tern colony on ESI to reduce predation rates on juvenile salmonids continues to be successful, with predation on SR and UCR

steelhead reduced by upwards of 80% in recent years. The number of terns nesting on ESI, however, has declined dramatically over the course of the last eight years and is now well below the target colony size identified in the management plan (3,125–4,375 pairs). The latest census of the Pacific Flyway breeding population of terns occurred in 2024 and indicated that the tern population has declined by more than 70% since the population peak in 2009, one year after management began in 2008 (McGuire et al. 2025). A decline in the Pacific Flyway population of 50% was identified in the *Caspian Tern Management Plan for the Columbia River Estuary* as a trigger that would prompt adaptive management to reverse the decline and preclude putting the tern population at risk (USFWS 2005, Section 4.2.1.3). Unfortunately, this trigger point was reached following the Pacific Flyway census in 2021 (Lawes et al. 2022), indicating the need for adaptive management to halt the decline. In keeping with the goals and objectives of the *Estuary Tern Management Plan*, adaptive management to restore the tern colony at ESI to the size range identified in the plan should be considered. The colony on ESI has not consistently fledged chicks since 2019, which is likely contributing to the overall population decline. In addition to and/or in lieu of an ESI tern colony of 3,125-4,375 pairs in the future, we recommend alternative nesting sites for terns in Grays Harbor, the Strait of Juan de Fuca, and/or Puget Sound. These alternative sites could complement previously created sites (e.g., in San Franisco Bay and lakes and reservoirs in interior Oregon and California), sites that to-date have not fully absorbed terns displaced from ESI. These coastal regions have a history of supporting large tern colonies, have high connectivity with the tern colonies in the Columbia River basin, and pose a reduced risk to salmonid stocks of conservation concern (Roby et al. 2021b; Collis et al. 2024). Please see Collis et al. (2024) for a more detailed review of the history and efficacy of the *Estuary Tern Management Plan*, including adaptive management recommendations.

There is evidence that tern predation rates on UCR steelhead and, to a lesser degree, SR steelhead, have been reduced and survival has increased as a result of the *Inland Avian Predation Management Plan*. There continues, however, to be high fidelity of terns to managed and unmanaged sites in the CPR. This is evident by the re-establishment of a tern colony on Crescent Island, after management actions that had prevented tern nesting on the site since 2015 were largely discontinued in 2021. The tern colony on Crescent Island has now exceeded the target colony size of 40 pairs for four consecutive years, with an average of 143 pairs during 2022–2025. Recent and dramatic increases in the amount of suitable tern nesting habitat on Crescent Island further increase the likelihood of continue tern nesting, perhaps to levels as high or higher than those observed prior to implementation of management actions on Crescent Island in 2015. Small-scale management efforts, like the placement of woody debris, however, are unlikely to be successful on Crescent Island given the amount of suitable habitat now available to terns. As such, large-scale adaptive management techniques (e.g., the replanting of native vegetation and the use of passive and active dissuasion) to eliminate tern nesting habitat from Crescent Island should be implemented in 2026 and beyond. Terns have also demonstrated high fidelity to Goose Island and surrounding islands in Potholes Reservoir, but adaptive management efforts have been largely successful in preventing or quickly dissuading new or incipient colonies. Without continued adaptive management in Potholes Reservoir, however, large and sustained tern colony(s) are likely to be re-established in the

coming years. Finally, the shift in the distribution of terns from managed colonies to unmanaged colony sites in the CPR (i.e., Badger Island, Shoal Island in Lenore Lake, Goose Island in Banks Lake, and others) supports the hypothesis of strong regional fidelity by nesting terns. Both waterbird ecologists and salmon managers were aware that breaking the attachment that nesting terns have to the region would be challenging, requiring a commitment to perennial adaptive management to assure lasting success, as suggested by the adaptive management provisions included in the IAPMP (USACE 2014, Section 3).

DOUBLE-CRESTED CORMORANTS

Cormorants have largely abandoned ESI as a nesting site and thousands of birds are now nesting further upriver on the Astoria-Megler Bridge, an unintended consequence of the *Double-crested Cormorant Management Plan for the Columbia River Estuary*. The cormorant colony on the Astoria-Megler Bridge is currently the largest breeding colony anywhere in the Pacific Flyway of North America. For the fourth consecutive year, predation rates on ESA-listed juvenile salmonids by cormorants nesting on the Astoria-Megler Bridge are now, on average, higher than those of cormorants that nested on ESI prior to implementation of the Plan. The number of cormorants nesting at other sites in the upper CRE and the Lower Columbia River (LCR) has also increased in recent years, with substantial predation effects by cormorants nesting at the Troutdale Towers in the LCR. Given these findings, management of cormorants in the CRE has not improved smolt survival, as intended, and adaptive management to dissuade cormorants from nesting on the Astoria-Megler Bridge and other sites in upper CRE and LCR are needed to meet the management objectives of the plan. Adaptive management aimed at restoring the cormorant colony on ESI to the target colony size specified in the plan (5,400–5,900 pairs), while greatly reducing or eliminating cormorant colonies upstream of ESI, is recommended to achieve the original goals of the management plan. If achieved, predation rates on steelhead and salmon in the LCR and CRE could be reduced by upwards of 70%, resulting in millions of additional smolts surviving outmigration to the Pacific Ocean each year.

In addition to increasing numbers of breeding cormorants on the LCR downstream of Bonneville Dam, there is now some evidence that cormorants breeding on select colonies in the CPR are also increasing in size. Most notably is the colony on Crescent Island in McNary Reservoir, which has increased from less than 50 breeding pairs during 2022–2023 to over 500 pairs in 2024–2025. Although much of this increase is presumably associated with the failure of a nearby cormorant colony on Foundation Island in McNary Reservoir, predation rates on ESA-listed smolts by Crescent Island cormorants are now greater than those observed when cormorants nested on Foundation Island. Unlike Foundation Island, where habitat is limited, Crescent Island can support an even larger cormorant colony in the future. There are currently no management plans for inland cormorant colonies but continued monitoring of the Crescent Island cormorant colony and other colonies on the Columbia River (e.g., Hanford Island and Murdock Island towers) may be necessary to understand to what degree inland cormorant colonies are limiting smolt survival upstream of Bonneville Dam.

OTHER PISCIVOROUS COLONIAL WATERBIRDS

Predation/consumption on juvenile salmonids by California gulls, ring-billed gulls, Brandt's cormorants, and American white pelicans is highly variable depending on the predator species, colony, salmonid species, and year. Several of the colonies investigated as part of this study posed little threat to smolt survival, while predation/consumption by other colonies, particularly larger-sized gull colonies located on islands in the middle and lower Columbia River, consumed an appreciable proportion of available smolts. Based on the impacts of gulls nesting on Miller Rocks in The Dalles Reservoir on smolt survival, nest dissuasion activities were implemented at that colony and there was some evidence that dissuasion activities had initially reduced the size of the colony and predation/consumption rates. In 2025, however, the gull colony increased in size and predation rates were similar to or higher than those observed prior to the implementation of management actions in 2022. Results of this and other studies indicated that consumption of smolts by gulls from several colonies in the region, particularly those on islands at considerable distances from the Columbia River (e.g., Goose Island in Potholes Reservoir and islands in Lenore Lake), however, pose little threat to smolt survival and may be sites where gulls from managed colonies can relocate. Results from this and other studies also indicate that predation by Brandt's cormorants nesting in the CRE, ring-billed gulls nesting in the CRE, and double-crested cormorants nesting on islands in Potholes Reservoir, Shoal Island in Lenore Lake, and Harper Island in Sprague Lake (all "off-river" cormorant colony locations) pose little threat to the smolt survival; these colonies should not be of concern to fisheries managers and may also be locations where birds from managed colonies in the CRB can relocate.

To-date, there have been no management plans aimed at reducing the size of pelican colonies in the Columbia River basin (CRB). The pelican colony on Miller Sands Spit in the CRE, however, has periodically been passively dissuaded by the disposal of dredge materials on nesting habitat prior to the breeding season, actions that may continue in 2026 and beyond (K. Tidwell, USACE, pers. com.). Predation rates on SR, UCR, and MCR smolts by American white pelicans on Miller Rocks have been low to non-existent (< 1.0% per ESA-listed salmonid ESU/DPS), suggesting pelicans pose little threat to smolt survival in the CRE. Predation by pelicans on Crescent Island (birds that formerly nested on nearby Badger Island), the largest pelican colony in the CRB, were also relatively low on SR and UCR smolts (< 1.0% per ESU/DPS). Research from other studies, however, indicates pelican predation on unlisted salmonid stocks from the middle Columbia River were substantial (> 20.0%) in some years (Payton et al. 2023). Pelicans are also capable of consuming adult-sized fishes, including adult sockeye salmon, and predation effects can be substantial in some years, with upwards of 8.4% (equivalent to more than 47,000 fish) of the adult sockeye run passing Bonneville Dam consumed in some years (Payton et al. 2025). Management aimed at reducing pelican predation at specific hot spots (e.g., fishways, weirs, shallow water habitat, and other pinch points were smolts are more susceptible to predation) could be effective. A better understanding of how biotic (e.g., prey density, fish travel times, and pelican foraging behavior) and abiotic (temperature, flow) factors influence adult sockeye susceptibility to pelican predation is also warranted and could provide important insight into developing effective management actions to reduce pelican predation in the CPR.

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APPENDIX A: PREDATION RATE SUMMARY TABLES

Table A1. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by Caspian terns nesting on Crescent Island and Badger Island in McNary Reservoir, Goose Island in Potholes Reservoir, an unnamed island in northern Potholes Reservoir, islands in Lenore Lake, and Twinning Island in Banks Lake on ESA-listed salmonid populations originating from the Snake River (SR; based on detections at Lower Monumental Dam) and Upper Columbia River (UCR; based on detections at Rock Island Dam) during 2007–2025 for spring (Sp), summer (Su), and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (<500) to generate reliable predation rates. Estimates from 2007–2024 are those previously reported by Evans et al. (2025).

	Crescent Island, McNary Reservoir					
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	0.4% (0.3-0.6)	0.9% (0.4-1.7)	-	-	3.9% (3.1-5.6)	2.5% (1.7-3.8)
N	22,730	2,147		254	17,122	3,782
2008	0.9% (0.7-1.3)	1.6% (1.2-2.3)	-	1.7% (0.6-3.7)	5.9% (4.7-8.5)	2.9% (2.1-4.3)
N	30,142	22,968		767	28,653	8,403
2009	1.5% (1.1-2.2)	1.1% (0.8-1.6)	0.2% (<0.1-1.2)	1.0% (0.5-1.7)	4.6% (3.7-6.6)	2.3% (1.7-3.5)
N	20,679	26,567	738	2,651	52,102	8,025
2010	0.6% (0.4-1.1)	1.3% (1.0-1.9)	0.9% (0.3-2.3)	1.5% (0.5-3.4)	5.5% (4.2-7.9)	1.8% (1.3-2.7)
N	5,790	28,067	929	566	7,913	8,382
2011	0.8% (0.6-1.2)	0.6% (0.5-0.9)	0.5% (0.1-1.2)	0.9% (0.7-1.4)	3.0% (2.3-4.3)	2.4% (1.8-3.6)
N	54,944	46,593	1,567	12,445	53,565	8,002
2012	0.7% (0.5-1.0)	0.6% (0.5-1.0)	0.2% (0.1-0.8)	2.4% (1.5-3.8)	3.1% (2.3-4.5)	1.2% (0.8-2.0)
N	41,258	24,772	1,812	2,884	25,841	6,845
2013	0.7% (0.5-1.1)	0.9% (0.6-1.5)	0.4% (<0.1-1.2)	1.2% (0.5-2.7)	3.5% (2.7-5.1)	2.9% (2.1-4.3)
N	14,859	4,773	992	848	9,696	6,019
2014	0.8% (0.6-1.1)	0.6% (0.4-1.0)	0.7% (0.2-2.1)	1.5% (0.8-2.8)	6.1% (4.8-8.9)	3.4% (2.5-4.8)
N	22,195	6,043	641	1,414	16,599	7,757
2021	<0.1%	<0.1%	<0.1%	-	<0.1%	<0.1%
N	562	1,574	1,546		1,289	8,090
2022	0.1% (<0.1-0.2)	0.2% (0.1-0.4)	-	-	1.3% (0.9-2.1)	0.6% (0.2–1.0)
N	6,492	4,643			4,626	6,214

2023	0.1% (<0.1–0.2)	0.5% (0.2–1.3)	-	-	0.7% (0.4–1.4)	0.8% (0.4–1.6)
N	12,498	1,558			8,809	8,377
2024	<0.1%	0.2% (0.1–0.7)	-	-	0.6% (0.2–1.3)	0.9% (0.4–1.7)
N	1,231	2,198			2,611	3,800
2025	0.3% (0.1–1.5)	1.1% (0.2–4.0)	-	-	0.8% (0.2–2.3)	0.4% (0.2–1.1)
N	2,157	842			3,853	7,671

Badger Island, McNary Reservoir

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2017	<0.1%	<0.1%	<0.1%	-	0.4% (0.2-0.6)	0.5% (0.3-0.8)
N	27,977	9,769	2,681		24,247	7,644
2021	0.8% (0.2-2.4)	0.3% (0.1-0.8)	<0.1%	-	1.5% (0.8-2.9)	1.4% (1.0-2.2)
N	562	1,574	1,546		1,289	8,090
2022	0.2% (0.1–0.4)	0.6% (0.3–1.0)	-	-	2.8% (2.0–4.4)	0.7% (0.4–1.2)
N	6,492	4,643			4,626	6,214
2023	0.2% (0.1–0.4)	1.0% (0.4–2.1)	-	-	1.9% (1.2–3.2)	1.4% (0.9–2.5)
N	12,498	1,558			8,809	8,377

Goose Island, Potholes Reservoir

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	<0.1%	0.3% (<0.1-1.1)	-	-	0.1% (<0.1-0.2)	15.3% (9.8-27.7)
N	22,730	2,147			17,122	3,782
2008	<0.1%	<0.1%	-	0.4% (<0.1-1.6)	<0.1%	11.1% (8.6-16.4)
N	30,142	22,968		767	28,653	8,403
2009	<0.1%	<0.1%	5.5% (2.7-10.7)	0.1% (<0.1-0.4)	0.1% (<0.1-0.1)	22.6% (17.2-33.7)
N	20,679	26,567	738	2,651	52,102	8,025
2010	<0.1%	<0.1%	2.0% (0.7-4.4)	0.3% (<0.1-1.9)	<0.1%	14.6% (11.0-21.8)
N	5,790	28,067	929	566	7,913	8,382
2011	<0.1%	<0.1%	0.6% (0.1-1.9)	<0.1%	<0.1%	12.9% (9.6-19.6)
N	54,944	46,593	1,567	12,445	53,565	8,002
2012	<0.1%	<0.1%	2.6% (1.2-5.4)	0.2% (<0.1-0.6)	0.2% (0.1-0.4)	18.4% (13.5-28.5)
N	41,258	24,772	1,812	2,884	25,841	6,845

2013	<0.1%	0.1% (<0.1-0.4)	2.5% (1.1-5.2)	0.1% (<0.1-1.1)	0.1% (0.1-0.4)	14.8% (11.4-21.6)
N	14,859	4,773	992	848	9,696	6,019
2014	<0.1%	0.1% (<0.1-0.8)	0.6% (0.1-2.2)	0.2% (<0.1-1.1)	<0.1%	2.9% (1.9-5.1)
N	22,195	6,043	641	1,414	16,599	7,757
2021	<0.1%	<0.1%	0.3% (0.1-1.5)	-	<0.1%	3.9% (2.4-6.3)
N	562	1,574	1,546		1,289	8,090
2022	<0.1%	<0.1%	-	-	<0.1%	0.1% (<0.1-0.2)
N	6,492	4,643			4,626	6,214
2023	<0.1%	<0.1%	-	-	<0.1%	0.6% (0.3–1.2)
N	12,498	1,558			8,809	8,377
2024	<0.1%	<0.1%	-	-	<0.1%	1.3% (0.7–2.6)
N	1,231	2,198			2,611	3,800
2025	< 0.1%	< 0.1%	-	-	<0.1%	0.9% (0.5–1.7)
N	2,157	842	-	-	3,853	7,671

Unnamed Island, Northern Potholes Reservoir

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2016	<0.1%	<0.1%	0.1% (<0.1–0.3)	<0.1%	<0.1%	4.1% (2.9-6.3)
N	38,633	5,461	1,956	522	20,729	7,003
2024	<0.1%	<0.1%	-	-	<0.1%	1.7% (1.0–3.2)
N	1,231	2,198			2,611	3,800
2025	<0.1%	<0.1%	-	-	<0.1%	0.1% (<0.1–0.2)
N	2,157	842			3,853	7,671

Lenore Lake Islands, Lenore Lake

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2015	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%
N	4,471	1,393	766	1,262	2,400	7,222
2016	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%
N	38,633	5,461	1,956	522	20,729	7,003
2017	<0.1%	<0.1%	0.3% (0.1-0.8)	-	<0.1%	1.0% (0.6-2.0)
N	27,977	9,769	2,681		24,247	7,644

2018	<0.1%	<0.1%	0.1% (0.1-0.8)	<0.1%	<0.1%	0.8% (0.4-1.7)
N	19,986	8,753	2,090	1,443	19,632	7,511
2019	<0.1%	<0.1%	<0.1%	0.1% (<0.1-0.3)	<0.1%	1.0% (0.6-1.7)
N	18,757	7,501	1,885	1,675	28,813	4,401
2020	<0.1%	<0.1%	0.3% (0.1-0.8)	-	<0.1%	1.0% (0.6-1.5)
N	2,931	1,607	947		1,130	6,843
2021	<0.1%	<0.1%	0.1% (0.1-0.8)	-	<0.1%	0.9% (0.6-1.5)
N	562	1,574	1,546		1,289	8,090
2022	<0.1%	<0.1%	-	-	<0.1%	2.1% (1.4–3.4)
N	6,492	4,643			4,626	6,214
2023	<0.1%	<0.1%	-	-	<0.1%	0.4% (0.2–0.8)
N	12,498	1,558			8,809	8,377
2024	<0.1%	<0.1%	-	-	<0.1%	1.5% (0.9–2.6)
N	1,231	2,198			2,611	3,800
2025	<0.1%	<0.1%	-	-	<0.1%	0.1% (<0.1–0.3)
N	2,157	842			3,853	7,671

Twinning Island or Goose Island, Banks Lake

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2008	<0.1%	<0.1%	-	<0.1%	<0.1%	<0.1%
N	30,142	22,968		767	28,653	8,403
2009	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	0.1% (<0.1-0.3)
N	20,679	26,567	738	2,651	52,102	8,025
2010	<0.1%	<0.1%	<0.1%	0.2% (<0.1-1.2)	<0.1%	0.1% (<0.1-0.3)
N	5,790	28,067	929	566	7,913	8,382
2012	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	0.1% (<0.1-0.3)
N	41,258	24,772	1,812	2,884	25,841	6,845
2014	<0.1%	<0.1%	0.5% (<0.1-7.9)	0.1% (<0.1-0.6)	<0.1%	1.2% (0.3-6.4)
N	22,195	6,043	641	1,414	16,599	7,757
2015	<0.1%	<0.1%	0.2% (<0.1-0.9)	0.1% (0-0.5)	<0.1%	2.6% (1.8-3.9)
N	7,706	3,449	766	1,262	3,601	7,222
2016	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	0.1% (<0.1-0.2)

N	38,633	5,461	1,956	522	20,729	7,003
2024	<0.1%	<0.1%	-	-	<0.1%	0.2% (0.1–0.5)
N	1,231	2,198			2,611	3,800
2025	<0.1%	<0.1%			<0.1%	2.2% (1.5–3.3)
N	2,157	842	-	-	3,853	7,671

Table A2. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by Caspian terns nesting on the Blalock Islands on ESA-listed salmonid populations originating from the Snake River (SR) and Upper Columbia River (UCR) based on detections at McNary Dam during 2007–2020 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable predation rates. Estimates from 2007–2021 are those previously reported by Evans et al. (2022a).

	Blalock Islands, John Day Reservoir					
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	<0.1%	0.1% (<0.1-0.2)	<0.1%	-	0.9% (0.6-1.4)	1.0% (0.6-1.7)
N	74,905	7,374	6,764		7,683	3,111
2008	0.1% (0.1-0.2)	<0.1%	0.1% (<0.1-0.2)	-	0.8% (0.6-1.2)	0.7% (0.4-1.2)
N	27,288	36,857	4,713		15,449	3,399
2009	0.3% (0.2-0.4)	<0.1%	0.2% (0.1-0.5)	<0.1%	0.6% (0.4-0.9)	0.5% (0.3-1.0)
N	60,155	43,461	3,982	2,088	29,877	3,663
2010	0.1% (<0.1-0.1)	<0.1%	0.1% (<0.1-0.1)	0.2% (<0.1-0.6)	0.9% (0.7-1.4)	0.9% (0.6-1.6)
N	52,129	29,587	10,456	1,327	17,806	4,161
2011	0.1% (<0.1-0.1)	0.1% (0.1-0.2)	<0.1%	0.3% (0.1-0.8)	0.1% (0.1-0.2)	0.1% (<0.1-0.3)
N	38,629	41,007	3,981	2,769	16,759	5,155
2013	<0.1%	0.1% (<0.1-0.1)	<0.1%	<0.1%	0.1% (<0.1-0.2)	0.2% (<0.1-0.5)
N	47,685	14,398	6,778	1,213	9,391	2,621
2014	0.1% (0.1-0.2)	0.3% (0.2-0.5)	0.2% (0.1-0.4)	0.4% (0.1-1.1)	0.4% (0.2-0.7)	0.6% (0.3-1.2)
N	41,109	10,293	4,611	1,922	10,389	2,686
2015	1.4% (1.1-2.2)	0.4% (0.4-0.8)	0.9% (0.5-1.5)	1.3% (0.7-2.5)	8.0% (6.0-11.4)	8.2% (5.9-12.4)
N	31,474	4,390	4,921	1,712	6,824	2,056
2016	0.3% (0.2-0.5)	0.6% (0.4-1.1)	0.2% (0.1-0.4)	2.3% (1.2-4.1)	3.9% (3.9-5.7)	3.1% (2.3-4.6)
N	47,573	6,726	11,320	1,095	14,332	7,414
2017	0.9% (0.6-1.3)	0.6% (0.4-1.1)	1.1% (0.7-1.8)	-	3.4% (2.4-5.1)	4.2% (2.7-6.5)
N	17,215	9,230	6,517		5,795	2,536
2018	0.5% (0.3-0.9)	0.7% (0.4-1.4)	0.3% (0.1-0.8)	2.0% (0.4-6.1)	2.5% (1.4-4.5)	2.9% (1.5-5.2)
N	17,963	8,450	5,228	514	3,585	2,228
2019	0.4% (0.2-0.8)	1.3% (0.6-2.5)	0.9% (0.2-2.1)	1.4% (0.4-3.7)	3.0% (1.9-4.7)	5.9% (3.4-10.0)
N	11,225	3,395	2,838	1,167	5,878	1,671
2020	0.1% (<0.1-0.4)	0.2% (0.1-0.8)	0.1% (<0.1-0.5)	-	2.2% (0.7-5.4)	1.3% (0.3-4.2)
N	9,618	3,522	2,422		1,448	707

Table A3. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by Caspian terns nesting on East Sand Island on ESA-listed salmonid populations originating from the Snake River (SR; based on detection at Bonneville Dam), Upper Columbia River (UCR; based on detections at Bonneville Dam), middle Columbia River (MCR; based on detection at Bonneville Dam), and Upper Willamette River (WR; based on detections at Sullivan Dam) during 1999-2020 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable predation rates. Accurate and comparable estimates of predation were not available (NA) in 2019. No estimates of predation were available 2021–2025, although smolt PIT tags were recovered following the breeding season (Table 3). Estimates are those previously reported by Evans et al. (2022a).

	East Sand Island, Columbia River Estuary							
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	UWR Sp Chinook	SR Sockeye	MCR Steelhead	SR Steelhead	UCR Steelhead
2000	4.6% (3.6-6.6)	3.3% (2.1-5.3)	2.2% (1.2-3.8)	-	-	-	10.5% (8.4-15.0)	16.3% (12.8-22.9)
N	11,810	1,323	1,123				10,356	3,100
2001	14.0% (11.1-20.0)	6.4% (4.2-10.0)	13.2% (9.9-19.5)	-	-	15.0% (11.1-21.9)	33.9% (26.3-49.1)	-
N	8,845	807	1,230			872	774	
2002	2.9% (2.3-4.1)	1.7% (1.2-2.6)	2.5% (1.9-3.5)	-	-	-	21.9% (17.6-31.0)	14.2% (10.1-21.3)
N	30,617	4,899	20,493				7,331	561
2003	4.7% (3.7-6.9)	2.7% (2.0-4.0)	3.7% (2.9-5.3)	-	-	-	26.0% (21.0-36.2)	19.0% (15.4-26.9)
N	28,150	6,234	30,723				8,553	27,918
2004	4.8% (3.6-7.0)	1.3% (0.6-2.6)	3.7% (2.9-5.4)	-	-	-	25.8% (19.7-37.3)	14.1% (11.3-19.8)
N	4,816	929	9,533				803	6,040
2005	3.0% (2.2-4.4)	1.3% (0.6-2.6)	2.4% (1.6-3.8)	-	-	-	28.3% (21.6-40.6)	15.1% (11.9-21.6)
N	5,935	1,121	2,518				753	5,610
2006	3.3% (2.4-5.0)	2.5% (1.7-3.9)	3.6% (1.8-6.6)	-	-	-	27.5% (21.0-39.1)	23.4% (18.1-34.1)
N	5,570	4,057	731				1,100	2,064
2007	3.1% (2.5-4.4)	3.4% (2.3-5.3)	1.9% (1.2-3.2)	1.4% (0.8-2.5)	-	18.7% (14.6-26.8)	22.6% (18.2-32.4)	15.7% (12.4-22.6)
N	23,830	2,005	2,268	1,505		2,234	6,391	3,042
2008	2.5% (1.9-3.6)	1.9% (1.5-2.7)	1.7% (1.0-2.9)	4.4% (3.2-6.7)	-	13.5% (10.6-19.2)	14.2% (11.5-19.9)	16.7% (13.1-24.2)
N	11,425	24,136	1,662	2,509		2,291	19,572	2,513
2009	4.7% (3.7-6.9)	2.0% (1.5-2.9)	3.7% (2.5-5.6)	1.7% (1.2-2.7)	1.3% (0.7-2.2)	14.1% (11.1-20.0)	14.5% (11.9-20.1)	20.0% (15.6-29.3)
N	17,396	16,314	2,064	5,573	1,845	2,700	23,311	2,265
2010	3.4% (2.7-4.8)	0.7% (0.5-1.1)	2.9% (2.2-4.3)	1.8% (0.6-4.4)	1.6% (0.8-2.9)	11.9% (9.4-17.4)	14.3% (11.3-20.4)	13.7% (11.0-19.3)
N	*38,441*	*17,974*	*5,972*	*510*	*1,382*	*8,515*	*40,024*	*12,284*
2011	2.5% (1.8-3.6)	0.7% (0.5-1.1)	2.9% (1.4-5.3)	0.9% (0.3-2.0)	0.4% (0.1-1.3)	9.6% (6.6-14.7)	12.0% (9.4-17.3)	9.1% (6.9-13.4)

N	6,557	12,327	704	1,119	826	865	7,028	2,419
2012	2.2% (1.7-3.3)	0.7% (0.5-1.1)	1.2% (0.7-2.1)	0.7% (0.4-1.3)	2.1% (1.2-3.7)	9.4% (6.5-14.4)	10.2% (7.7-14.9)	7.5% (5.6-11.3)
N	17,929	10,742	3,227	3,731	1,457	1,084	4,768	3,357
2013	1.2% (0.8-1.8)	0.9% (0.5-1.6)	0.7% (0.3-1.4)	1.0% (0.5-1.8)	0.8% (0.3-2.0)	9.9% (7.0-15.3)	12.7% (9.6-18.5)	8.9% (6.6-13.4)
N	16,167	4,465	3,112	2,629	1,454	1,865	8,516	4,473
2014	1.1% (0.8-1.7)	1.0% (0.5-1.9)	1.4% (0.7-2.5)	1.2% (0.5-2.5)	1.6% (0.8-3.0)	9.5% (6.5-14.5)	8.6% (6.7-12.5)	11.4% (8.5-16.8)
N	14,828	2,800	2,297	1,587	1,739	1,119	8,812	3,841
2015	2.0% (1.5-2.9)	0.8% (0.4-1.5)	1.9% (1.3-2.9	0.4% (0.1-1.5)	1.6% (1.0-2.6)	7.8% (5.9-11.4)	10.2% (8.2-14.6)	10.5% (8.2-15.0)
N	20,245	2,629	5,943	768	3,311	3,927	16,451	6,004
2016	0.8% (0.6-1.2)	0.7% (0.3-1.3)	1.4% (0.9-2.1)	1.2% (0.4-3.2)	-	8.8% (6.4-13.0)	6.1% (4.8-8.8)	7.5% (5.8-10.7)
N	21,874	2,887	5,939	604		2,086	14,473	8,123
2017	0.8% (0.5-1.2)	0.2% (0.1-0.5)	1.4% (0.9-2.3)	-	-	8.4% (5.6-13.1)	5.3% (3.9-7.7)	6.5% (4.7-9.6)
N	13,151	4,635	4,622			1,069	6,497	3,275
2018	1.4% (1.0-2.1)	1.3% (0.7-2.1)	1.4% (0.9-2.3)	-	4.2% (2.9-6.4)	5.3% (3.8-8.0)	6.9% (5.3-10.2)	6.5% (4.8-9.7)
N	11,174	5,981	3,370		2,546	3,209	9,572	5,322
2019	NA	NA	NA	NA	NA	NA	NA	NA
N								
2020	0.7% (0.5-1.1)	0.3% (0.1-0.7)	0.4% (0.2-0.9)	-	1.1% (0.6-2.2)	5.4% (3.8-7.9)	5.9% (4.5-8.1)	4.5% (3.3-6.4)
N	20,246	3,389	4,895		2,122	3,157	11,868	5,894
2021	NA	NA	NA	NA	NA	NA	NA	NA
N								
2022	NA	NA	NA	NA	NA	NA	NA	NA
N								
2023	NA	NA	NA	NA	NA	NA	NA	NA
N								
2024	NA	NA	NA	NA	NA	NA	NA	NA
N								
2025	NA	NA	NA	NA	NA	NA	NA	NA
N								

Table A4. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by double-crested cormorants nesting on East Sand Island on ESA-listed salmonid populations originating from the Snake River (SR; based on detection at Bonneville Dam), Upper Columbia River (UCR; based on detections at Bonneville Dam), middle Columbia River (MCR; based on detection at Bonneville Dam), and Upper Willamette River (UWR; based on detections at Sullivan Dam) during 2003–2018 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable estimates. Accurate and comparable estimates of predation were not available (NA) in 2019. During 2020, 2021, 2024, and 2025 few, if any, double-crested cormorants successfully nesting on East Sand Island, denoted as no colony (NC). Estimates from 2003–2018 are those previously reported by Evans et al. (2022a).

	East Sand Island, Columbia River Estuary							
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	UWR Sp Chinook	SR Sockeye	MCR Steelhead	SR Steelhead	UCR Steelhead
2003	1.7% (1.2-2.7)	1.1% (0.7-2.0)	1.4% (0.9-2.1)	-	-	-	1.9% (1.2-3.0)	1.5% (1.0-2.4)
N	28,150	6,234	30,723				8,553	27,918
2004	5.1% (3.3-8.5)	1.9% (0.6-4.7)	4.7% (3.2-7.6)	-	-	-	3.6% (1.4-8.0)	7.4% (5.1-11.8)
N	4,816	929	9,533				803	6,040
2005	4.8% (3.2-7.9)	3.6% (1.8-6.9)	4.5% (2.8-7.8)	-	-	-	4.3% (2.0-8.6)	5.5% (3.7-8.8)
N	5,935	1,121	2,518				753	5,610
2006	5.2% (3.5-8.5)	2.7% (1.6-4.6)	4.7% (2.2-9.5)	-	-	-	13.1% (8.2-22.7)	4.7% (2.8-8.2)
N	5,570	4,057	731				1,100	2,064
2007	1.7% (1.1-2.7)	1.6% (0.7-3.3)	2.7% (1.5-5.1)	1.0% (0.3-2.6)	-	2.8% (1.5-5.2)	3.5% (2.3-5.8)	3.4% (2.1-6.1)
N	23,830	2,005	2,268	1,505		2,234	6,391	3,042
2008	3.5% (2.4-5.5)	2.6% (1.9-4.2)	3.6% (2.0-6.6)	3.3% (1.9-5.8)	-	14.0% (9.5-23.2)	14.7% (10.6-23.2)	6.2% (4.0-10.4)
N	11,425	24,136	1,662	2,509		2,291	19,572	2,513
2009	6.8% (4.9-10.7)	4.5% (3.2-7.1)	2.7% (1.5-4.9)	1.4% (0.8-2.4)	5.7% (3.5-9.8)	14.9% (10.3-23.8)	16.6% (12.0-25.7)	7.2% (4.7-12.0)
N	17,396	16,314	2,064	5,573	1,845	2,700	23,311	2,265
2010	5.3% (3.9-8.4)	3.9% (2.7-6.1)	3.3% (2.3-5.4)	4.2% (1.6-9.2)	2.6% (1.3-4.9)	8.2% (5.8-13.1)	7.5% (5.5-12.0)	6.8% (4.9-10.6)
N	38,441	17,974	5,972	510	1,382	8,515	40,024	12,284
2011	4.3% (2.9-6.9)	1.9% (1.3-3.1)	5.6% (2.9-10.8)	0.4% (0.1-1.5)	4.8% (2.4-9.1)	7.8% (4.6-14.0)	5.3% (3.7-8.5)	11.4% (7.8-18.6)
N	6,557	12,327	704	1,119	826	865	7,028	2,419
2012	3.7% (2.6-6.0)	2.6% (1.8-4.2)	2.1% (1.2-3.7)	0.6% (0.3-1.3)	3.7% (2.0-6.9)	3.3% (1.7-6.4)	4.9% (3.2-8.1)	6.5% (4.3-10.8)
N	17,929	10,742	3,227	3,731	1,457	1,084	4,768	3,357
2013	3.6% (2.5-5.7)	2.2% (1.3-3.7)	3.0% (1.8-5.3)	1.0% (0.4-2.0)	3.3% (1.8-6.2)	2.1% (1.0-4.1)	2.5% (1.7-4.0)	3.4% (2.2-5.7)
N	16,167	4,465	3,112	2,629	1,454	1,865	8,516	4,473

2014	8.5% (6.1-13.2)	2.4% (1.5-4.2)	6.1% (3.9-10.1)	1.8% (0.9-3.6)	4.5% (2.7-7.7)	6.4% (3.7-10.7)	7.8% (5.6-12.0)	10.4% (7.3-16.3)
N	14,828	2,800	2,297	1,587	1,739	1,119	8,812	3,841
2015	14.5% (10.5-22.4)	8.7% (6.0-14.0)	8.3% (5.9-12.9)	2.4% (0.9-5.2)	2.4% (1.5-4.1)	12.4% (8.8-19.2)	12.8% (9.3-19.6)	10.5% (7.6-16.2)
N	20,245	2,629	5,943	768	3,311	3,927	16,451	6,004
2016[a]	5.1% (3.7-8.1)	2.1% (1.1-3.9)	3.5% (2.3-5.7)	0.4% (0-2.1)	-	2.7% (1.4-4.9)	6.8% (4.8-10.7)	5.1% (3.6-8.2)
N	21,874	2,887	5,939	604		2,086	14,473	8,123
2017[a]	0.7% (0.4-1.1)	0.1% (0-0.2)	0.4% (0.1-0.8)	-	-	0.7% (0.1-2.1)	0.4% (0.2-0.8)	1.4% (0.8-2.7)
N	13,151	4,635	4,622			1,069	6,497	3,275
2018[a]	0.5% (0.3-0.8)	0.9% (0.5-1.6)	0.6% (0.3-1.2)	-	0.9% (0.5-1.9)	0.4% (0.1-1.0)	0.5% (0.3-0.9)	0.7% (0.4-1.4)
N	11,174	5,981	3,370		2,546	3,209	9,572	5,322
2019	NA	NA	NA	NA	NA	NA	NA	NA
N								
2020	NC	NC	NC	NC	NC	NC	NC	NC
N								
2021	NC	NC	NC	NC	NC	NC	NC	NC
N								
2022	NA	NA	NA	NA	NA	NA	NA	NA
N								
2023	NA	NA	NA	NA	NA	NA	NA	NA
N								
2024	NC	NC	NC	NC	NC	NC	NC	NC
N								
2025	NA	NA	NA	NA	NA	NA	NA	NA
N								

Table A5. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by California and ring-billed gulls nesting at Crescent and Badger islands in McNary Reservoir, Island 20 in the middle Columbia River, and Goose Island in Potholes Reservoir on ESA-listed salmonid populations originating from the Snake River (SR; based on detections at Lower Monumental Dam) and Upper Columbia River (UCR; based on detections at Rock Island Dam) during 2007–2025 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates. Estimates from 2007–2024 are those previously reported by Evans et al. (2025). Results from Goose Island gulls were available in select years only (2012, 2020, 2022) but gulls nested on Goose Island in all years during 2007–2025. No gulls nested on Badger Island in 2025.

	Crescent Island, McNary Reservoir					
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	0.6% (0.3-1.1)	0.6% (0.1-2.3)	-	-	4.1% (2.7-6.5)	5.9% (3.5-10.1)
N	22,730	2,147			17,122	3,782
2008	0.9% (0.5-1.4)	0.6% (0.3-1.1)	-	1.7% (0.2-6.1)	4.0% (2.8-5.9)	3.0% (1.8-4.8)
N	30,142	22,968		767	28,653	8,403
2009	0.9% (0.5-1.6)	0.7% (0.4-1.1)	0.8% (<0.1-4.7)	2.6% (1.1-5.4)	4.7% (3.4-6.8)	7.5% (5.0-11.4)
N	20,679	26,567	738	2,651	52,102	8,025
2010	1.4% (0.7-2.8)	0.3% (0.2-0.6)	0.5% (<0.1-3.3)	0.9% (<0.1-5.4)	5.1% (3.3-7.9)	7.8% (5.3-11.7)
N	5,790	28,067	929	566	7,913	8,382
2011	1.1% (0.7-1.7)	0.7% (0.4-1.2)	0.4% (<0.1-2.4)	1.6% (0.9-2.8)	3.4% (2.4-5.1)	3.6% (2.2-5.9)
N	54,944	46,593	1,567	12,445	53,565	8,002
2012	1.1% (0.7-1.7)	0.6% (0.3-1.2)	1.3% (0.3-4.0)	1.3% (0.4-3.2)	5.2% (3.5-8.0)	4.7% (2.9-7.8)
N	41,258	24,772	1,812	2,884	25,841	6,845
2013	1.0% (0.5-1.8)	0.8% (0.2-2.0)	0.6% (<0.1-3.6)	2.8% (0.6-8.3)	5.8% (3.8-8.9)	6.1% (3.8-9.7)
N	14,859	4,773	992	848	9,696	6,019
2014	1.1% (0.6-1.8)	0.3% (<0.1-0.9)	2.1% (<0.3-7.5)	3.1% (1.1-7.0)	5.5% (3.8-8.2)	6.8% (4.6-10.4)
N	22,195	6,043	641	1,414	16,599	7,757
2021	1.9% (0.2-8.6)	0.2% (<0.1-2.2)	0.7% (0.1-3.0)	-	2.8% (0.7-8.6)	2.9% (1.7-5.0)
N	562	1,574	1,546		1,289	8,090
2022	0.1% (0.1–0.5)	0.1% (<0.1–0.7)	-	-	2.4% (1.2–4.3)	1.3% (0.6–2.5)
N	6,492	4,643			4,626	6,214
2023	0.2% (0.1–0.6)	1.0% (0.2–3.7)	-	-	1.3% (0.6–2.5)	1.8% (1.0–3.4)
N	12,498	1,558			8,809	8,377

2024	1.3% (0.2–4.4)	0.7% (0.1–2.5)	-	-	2.3% (0.9–5.1)	2.5% (1.2–4.8)
N	1,231	2,198			2,611	3,800
2025	0.3% (0.1–1.6)	0.7% (0.1–5.6)	-	-	8.7% (3.0–15.6)	7.7% (4.6–12.4)
N	2,157	842			3,853	7,671

Badger Island, McNary Reservoir

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2015	0.1% (<0.1-0.5)	0.1% (0-0.9)	0.5% (<0.1-3.3)	1.1% (0.2-4.8)	2.9% (1.3-6.4)	5.2% (3.2-9.2)
N	7,706	3,449	766	1,262	3,601	7,222
2016	0.2% (0.1-0.4)	<0.1%	0.9% (0.1-3.3)	1.2% (0.1-7.7)	1.1% (0.6-1.9)	4.3% (2-13.8)
N	38,633	5,461	1,956	522	20,729	7,003
2017	0.2% (<0.1-0.4)	0.4% (0.1-1.0)	0.6% (0.1-2.1)	-	1.0% (0.6-1.8)	1.3% (0.6-2.6)
N	27,977	9,769	2,681		24,247	7,644
2018	1.0% (0.5-1.8)	1.0% (0.4-2.1)	1.1% (0.2-4.1)	4.0% (1.1-9.5)	4.3% (3-6.7)	4.8% (2.8-8.0)
N	19,986	8,753	2,090	1,443	19,632	7,511
2019	1.2% (0.6-2.1)	1.9% (0.8-8.4)	3.6% (1.1-8.8)	3.1% (0.9-8.3)	5.6% (3.9-8.4)	10.9% (6.7-17.7)
N	18,757	7,501	1,885	1,675	28,813	4,401
2020	1.0% (0.2-3.1)	1.0% (0.3-6.1)	0.6% (0.1-4.0)	-	9.2% (2.0-18.4)	4.9% (1.7-9.3)
N	2,931	1,607	947		1,130	6,843
2021	0.7% (0-3.4)	1.8% (0.1-5.4)	0.5% (0-1.8)	-	9.7% (1.3-20.9)	8.6% (3.6-14.7)
N	562	1,574	1,546		1,289	8,090
2022	1.3% (0.3–3.0)	0.2% (0.1–0.8)	-	-	3.4% (1.6–7.5)	6.6% (3.3–11.4)
N	6,492	4,643			4,626	6,214
2023	0.9% (0.3–1.7)	0.6% (0.1–3.2)	-	-	2.6% (1.0–4.9)	5.8% (2.1–9.9)
N	12,498	1,558			8,809	8,377
2024	0.4% (0.1–1.7)	0.1% (0.1–0.6)	-	-	1.1% (0.4–3.6)	0.7% (0.3–1.3)
N	1,231	2,198			2,611	3,800

Island 20, Middle Columbia River

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2013	0.3% (0.1-0.7)	0.1% (<0.1-0.7)	0.5% (<0.1-3.2)	0.6% (<0.1-3.6)	0.7% (0.3-1.4)	1.4% (0.6-2.7)
N	14,859	4,773	992	848	9,696	6,023

2014	0.2% (0.1-0.5)	0.2% (<0.1-0.9)	0.8% (<0.1-5.1)	0.4% (<0.1-2.3)	0.6% (0.3-1.1)	1.6% (0.8-3.1)
N	22,195	6,043	641	1,414	16,599	7,757
2015	0.3% (0.1-0.8)	0.1% (0-0.8)	0.6% (<0.1-3.7)	NA	2.4% (1.2-4.5)	7.9% (5.3-12.0)
N	7,706	3,449	766	1,262	3,601	7,222
2016	0.2% (0.1-0.4)	<0.1%	0.2% (<0.1-3.8)	0.9% (<0.1-5.8)	1.2% (0.7-2.0)	5.7% (3.7-8.9)
N	38,633	5,461	1,956	522	20,792	7,003
2017	0.2% (<0.1-0.4)	0.2% (<0.1-0.6)	0.1% (<0.1-1.0)	-	1.7% (1.1-2.6)	3.0% (1.8-4.9)
N	27,977	9,769	2,681		24,247	7,644
2018	0.1% (<0.1-0.3)	0.3% (0.1-0.8)	0.2% (<0.1-1.3)	0.8% (0.1-2.8)	1.3% (0.8-2.1)	1.1% (0.5-2.0)
N	19,986	8,793	2,090	1,443	19,632	7,511
2019	0.2% (0.1-0.4)	0.2% (0.1-0.7)	0.2% (0.1-1.5)	0.2% (<0.1-1.5)	2.8% (1.9-4.2)	7.4% (4.8-11.7)
N	18,757	7,501	1,885	1,675	28,813	4,401
2020	0.3% (<0.1-1.3)	0.1% (<0.1-1.4)	0.1% (<0.1-1.8)	-	1.6% (0.3-4.9)	2.1% (1.1-3.6)
N	2,931	1,607	947		1,130	6,843
2021	1.4% (0.2-6.2)	0.1% (<0.1-1.1)	0.5% (0.1-2.1)	-	4.8% (2.1-9.5)	6.1% (4.1-9.3)
N	562	1,574	1,546		1,289	8,090
2022	0.5% (0.2–1.3)	0.1% (0.1–0.6)	-	-	1.0% (0.4–2.1)	3.4% (2.1–5.6)
N	6,492	4,643			4,626	6,214
2023	0.2% (0.1–0.5)	0.7% (0.1–2.7)	-	-	0.7% (0.3–1.3)	2.8% (1.8–4.6)
N	12,498	1,558			8,809	8,377
2024	1.0% (0.2–3.6)	0.6% (0.1–1.9)	-	-	1.0% (0.3–2.8)	8.5% (5.6–13.2)
N	1,231	2,198			2,611	3,800
2025	0.6% (0.1–1.9)	0.7% (0.2–3.3)	-	-	2.4% (1.2, 4.4)	9.4% (6.5–14.1)
N	2,157	842			3,853	7,671
			Goose Island, Potholes Reservoir			
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2012	<0.1%	<0.1%	1.3% (0.3-4.0)	<0.1%	0.1% (<0.1-0.3)	2.8% (1.1-5.6)
N	41,258	24,772	1,812	2,884	25,841	6,845
2020	<0.1%	<0.1%	<0.1%	-	<0.1%	<0.1%
N	2,931	1,607	947		1,130	6,843
2022	<0.1%	<0.1%	-	-	<0.1%	<0.1%

N	6,492	4,643		4,626	6,214

Table A6. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by California and ring-billed gulls nesting Miller Rocks Island in The Dalles Reservoir and the Blalock Islands in the John Day Reservoirs on ESA-listed salmonid populations originating from the Snake River (SR) and Upper Columbia River (UCR) based on detections at McNary Dam during 2007–2025 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable estimates. Estimates from 2007-2024 are those previously reported by Evans et al. (2025).

	Miller Rocks Island, The Dalles Reservoir					
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	1.2% (0.8-1.7)	3.3% (2.1-5.4)	2.2% (1.3-3.8)	-	9.9% (6.9-14.6)	8.7% (5.5-13.9)
N	74,905	7,374	6,764		7,683	3,111
2008	2.3% (1.5-3.4)	2.7% (1.9-4.0)	1.6% (0.8-3.2)	-	9.2% (6.6-13.4)	7.2% (4.5-11.6)
N	27,288	36,857	4,713		15,449	3,399
2009	2.2% (1.5-3.2)	3.7% (2.6-5.4)	2.5% (1.2-4.6)	9.1% (5.3-15.1)	9.8% (7.1-14.3)	7.4% (4.6-11.8)
N	60,155	43,461	3,982	2,088	29,877	3,663
2010	1.8% (1.2-2.7)	1.0% (0.6-1.5)	1.7% (1.0-2.9)	4.1% (1.6-8.8)	9.1% (6.5-13.2)	7.0% (4.4-11.2)
N	52,129	29,587	10,456	1,327	17,806	4,161
2011	0.8% (0.5-1.3)	0.6% (0.4-1.0)	1.0% (0.3-2.3)	2.0% (0.8-4.2)	5.0% (3.5-7.6)	3.5% (2.0-5.9)
N	38,629	41,007	3,981	2,769	16,759	5,155
2012	0.6% (0.4-1.0)	0.7% (0.4-1.3)	1.5% (0.8-2.9)	6.3% (3.6-10.9)	4.6% (3.0-7.2)	7.2% (4.4-11.6)
N	40,168	25,017	6,800	2,492	8,840	3,804
2013	1.1% (0.7-1.7)	2.4% (1.6-3.8)	1.9% (1.0-3.4)	6.4% (3.0-12.5)	6.4% (4.4-9.7)	11.7% (7.4-18.4)
N	47,685	14,398	6,778	1,213	9,391	2,621
2014	1.0% (0.7-1.6)	1.8% (1.0-3.0)	1.3% (0.6-2.7)	4.4% (2.1-8.3)	5.3% (3.6-8.0)	6.1% (3.5-10.4)
N	41,109	10,293	4,611	1,922	10,389	2,686
2015	1.7% (1.1-2.6)	2.6% (1.4-4.6)	3.5% (2.1-6.0)	7.4% (4.1-13.1)	9.7% (6.6-14.6)	13.2% (8.3-21.1)
N	31,474	4,390	4,921	1,712	6,824	2,056
2016	1.2% (0.8-1.9)	1.0% (0.4-2.1)	2.5% (1.6-4.0)	6.4% (2.9-12.8)	6.7% (4.6-9.9)	10.1% (7.0-15.2)
N	47,573	6,726	11,320	1,095	14,332	7,414
2017	0.6% (0.3-1.1)	0.8% (0.4-1.7)	2.2% (1.2-3.9)	-	7.0% (4.3-11.0)	6.9% (3.8-12.0)
N	17,215	9,230	6,517		5,795	2,536
2018	0.4% (0.2-0.9)	2.1% (1.2-3.5)	1.2% (0.5-2.4)	7.0% (2.3-16.4)	3.3% (1.8-6.0)	8.3% (4.8-13.9)

N	17,963	8,450	5,228	514	3,585	2,228
2019	0.9% (0.5-1.7)	3.2% (1.7-5.8)	2.4% (1.1-4.9)	5.9% (2.6-11.8)	5.1% (3.2-8.1)	6.5% (3.4-11.8)
N	11,225	3,395	2,838	1,167	5,878	1,671
2020	1.1% (0.5-2.0)	1.9% (0.8-3.9)	3.1% (1.4-6.2)	-	4.4% (1.8-9.2)	2.5% (0.5-8.0)
N	9,618	3,522	2,422		1,448	707
2021	1.2% (0.7-2.2)	2.8% (1.1-6.3)	0.9% (0.2-2.8)	-	13.9% (9.1-21.8)	12.4% (6.2-23.1)
N	9,816	1,711	1,822		2,739	799
2022	0.8% (0.2–2.3)	1.9% (0.8–4.1)	1.2% (0.1-6.4)	-	5.7% (2.7–11.8)	5.4% (1.9–19.3)
N	3,278	3,645	617		1,634	554
2023	2.0% (1.0–3.5)	4.7% (2.4–8.9)	2.6% (1.1-5.4)	-	6.7% (3.5–12.4)	9.9% (4.9–18.7)
N	5,701	2,182	1,603		1,468	819
2024	0.5% (0.2–1.2)	1.5% (0.3–4.4)	1.4% (0.2–4.7)	-	5.2% (2.2–10.8)	-
N	9,074	1,771	1,246		1,802	
2025	1.6% (0.7–3.1)	5.1% (1.4–13.1)	2.2% (0.5–6.3)	7.6% (2.8–17.2)	9.7% (5.3–17.3)	-
N	6,324	721	1,329	732	1,948	

Blalock Islands, John Day Reservoir

Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2013	0.1% (<0.1-0.2)	0.3% (0.1-0.8)	0.4% (0.1-1.2)	1.9% (0.4-5.6)	1.2% (0.6-2.3)	1.2% (0.3-3.1)
N	47,685	14,398	6,778	1,213	9,391	2,621
2014	0.2% (0.1-0.4)	0.4% (0.1-0.9)	0.4% (0.1-1.1)	1.0% (0.2-2.7)	1.5% (0.9-2.5)	2.5% (1.2-4.9)
N	41,109	10,293	4,611	1,922	10,389	2,686
2015	0.2% (0.1-0.4)	0.7% (0.2-1.7)	0.6% (0.2-1.5)	1.4% (0.4-3.7)	2.6% (1.6-4.2)	6.8% (3.9-11.4)
N	31,474	4,390	4,921	1,712	6,824	2,056
2016	0.1% (0.1-0.3)	0.5% (0.2-1.2)	0.2% (0-0.5)	3.7% (1.4-8.2)	3.5% (2.4-5.2)	6.3% (4.4-9.2)
N	47,573	6,726	11,320	1,095	14,332	7,414
2017	0.2% (0.1-0.5)	0.2% (<0.1-0.5)	0.2% (0-0.7)	-	2.5% (1.5-4.2)	5.2% (3-8.8)
N	17,215	9,230	6,517		5,795	2,536
2018	0.2% (0.1-0.4)	0.3% (0.1-0.8)	0.6% (0.2-1.5)	2.2% (0.3-8)	3.3% (1.8-5.8)	3.1% (1.5-6)
N	17,963	8,450	5,228	514	3,585	2,228
2019	02% (0.1-0.5)	0.3% (0.1-1.1)	0.3% (0.1-1.2)	1.4% (0.3-4.2)	1.6% (0.8-2.8)	1.5% (0.4-3.8)

N	11,225	3,395	2,838	1,167	5,878	1,671
2020	0.1% (<0.1-0.4)	0.1% (<0.1-0.7)	0.1% (<0.1-0.9)	-	2.6% (0.7-6.8)	3.2% (0.7-9.6)
N	9,618	3,522	2,422		1,448	707
2021	0.5% (0.2-1.0)	0.2% (<0.1-0.3)	0.9% (0.2-2.7)	-	5.7% (3.3-9.6)	7.9% (3.6-15.8)
N	9,816	1,711	1,822		2,739	799
2022	0.4% (0.1–1.4)	0.3% (0.1–1.2)	1.0% (0.1–4.9)	-	5.5% (2.8–10.2)	5.2% (1.5–13.0)
N	3,278	3,645	617		1,634	554
2023	0.4% (0.1–1.7)	0.5% (0.1–1.8)	1.0% (0.1–4.9)	-	3.6% (1.6–7.4)	3.2% (1.0–8.2)
N	5,701	2,182	1,603		1,468	819
2024	0.5% (0.2–1.1)	0.7% (0.1–2.7)	1.5% (0.4–4.1)	-	1.7% (0.4–4.7%)	-
N	9,074	1,771	1,246		1,802	

Table A7. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by double-crested cormorants nesting at Crescent Island in McNary Reservoir, Foundation Island in McNary Reservoir, Hanford Island in the middle Columbia River, and islands in Lenore Lake on ESA-listed salmonid populations originating from the Snake River (SR; based on detections at Lower Monumental Dam) and Upper Columbia River (UCR; based on detections at Rock Island Dam) during 2007–2025 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable estimates. The Foundation Island colony was active in 2013 and during 2015–2019 but was not scanned for smolt PIT tags, so estimates in those years were unavailable. The Foundation Island colony failed in 2024 and 2025. The Lenore Lake colony was active in all years but was not scanned for smolt PIT tag from 2007–2018 and 2021 and 2025, so estimate in those years were unavailable. Estimates from 2007–2024 are those previously reported by Evans et al. (2025).

Crescent Island, McNary Reservoir						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2023	0.4% (0.2-0.8)	0.1% (<0.1-0.6)	-	-	0.4% (0.2-0.8)	0.2% (0.1-0.4)
N	12,498	1,558			8,809	8,377
2024	2.5% (1.4-4.8)	0.4% (0.1-1.8)	-	-	2.6% (1.1-6.4)	0.7% (0.2-1.9)
N	1,231	2,198			2,611	3,800
2025	1.8% (0.7-3.8)	0.9% (0.1-3.0)	-	-	5.4% (3.3-9.9)	1.5% (0.9-2.9)
N	2,157	842			3,853	7,671
Foundation Island, McNary Reservoir						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2007	1.5% (1.0-2.4)	1.9% (1.0-3.6)	-	-	4.7% (3.4-7.5)	< 0.1%
N	22,730	2,147			17,122	3,782
2008	2.0% (1.4-3.1)	0.9% (0.6-1.4)	-	2.4% (1.0-5.3)	4.7% (3.4-7.3)	0.3% (0.1-0.6)
N	30,142	22,968		767	28,653	8,403
2009	1.8% (1.3-2.9)	1.1% (0.7-1.7)	0.2% (<0.1-1.4)	4.3% (2.7-7.1)	3.6% (2.6-5.6)	0.2% (0.1-0.5)
N	20,679	26,567	738	2,651	52,102	8,025
2010	2.4% (1.5-3.9)	1.1% (0.7-1.7)	0.2% (<0.1-1.2)	3.8% (1.5-8.3)	3.6% (2.5-5.9)	0.2% (0.1-0.4)
N	5,790	28,067	929	566	7,913	8,382
2011	1.4% (1.0-2.4)	1.2% (0.8-1.9)	0.5% (0.1-1.8)	1.6% (1.0-2.8)	4.3% (2.9-7.0)	0.3% (0.1-0.6)
N	54,944	46,593	1,567	12,445	53,565	8,002
2012	0.9% (0.6-1.5)	0.6% (0.3-1.1)	0.5% (0.1-1.9)	4.1% (2.3-7.6)	2.4% (1.5-3.9)	0.5% (0.2-1.1)

N	41,258	24,772	1,812	2,884	25,841	6,845
2014	1.1% (0.5-2.1)	0.6% (0.2-1.9)	1.0% (<0.1-6.2)	2.8% (0.7-8.0)	1.8% (0.9-3.4)	0.2% (<0.1-0.8)
N	22,195	6,043	641	1,414	16,599	7,757
2020	2.5% (1.0-5.8)	0.8% (0.1-3.6)	0.1% (<0.1-2.4)	-	4.0% (1.2-10.7)	0.1% (<0.1-0.3)
N	2,931	1,607	947		1,130	6,843
2021	4.4% (0.8-15.6)	0.1% (<0.1-0.6)	<0.1%	-	3.0% (0.8-9.3)	<0.1%
N	562	1,574	1,546		1,289	8,090
2022	3.7% (2.0–7.0)	1.1% (0.4–2.7)	-	-	3.5% (1.9–6.5)	0.5% (0.1–1.2)
N	6,492	4,643			4,626	6,214
2023	2.1% (1.1–4.6)	1.3% (0.2–4.9)	-	-	3.1% (1.6–6.4)	0.3% (0.1–0.9)
N	12,498	1,558			8,809	8,377
Hanford Island, Middle Columbia River						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2018	<0.1%	<0.1%	0.2% (0.1-0.7)	<0.1%	<0.1%	0.2% (0.1-0.4)
N	19,986	8,753	2,090	1,443	19,632	7,511
2020	<0.1%	<0.1%	0.1% (<0.1-0.9)	-	<0.1%	<0.1%
N	2,931	1,607	947		1,130	6,843
2021	0.1% (<0.1-1.6)	<0.1%	0.1% (<0-0.5)	-	0.1% (<0.1-0.8)	0.3% (0.1-0.7)
N	562	1,574	1,546		1,289	8,090
2022	<0.1%	<0.1%	-	-	<0.1%	0.2% (<0.1-0.4)
N	6,492	4,643			4,626	6,214
2023	<0.1%	<0.1%	-	-	<0.1%	0.5% (0.1–1.9)
N	12,498	1,558			8,809	8,377
2024	<0.1%	<0.1%	-	-	<0.1%	0.2% (0.1–0.9)
N	1,231	2,198			2,611	3,800
2025	<0.1%	<0.1%	-	-	<0.1%	0.2% (0.1–0.7)
N	2,157	842			3,853	7,671
Lenore Lake Islands, Lenore Lake						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2017	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%

N	27,977	9,769	2,681	304	24,247	7,644
2019	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%
N	18,757	7,501	1,885	1,675	28,813	4,401
2020	<0.1%	<0.1%	<0.1%	-	<0.1%	<0.1%
N	2,931	1,607	947		1,130	6,843
2022	< 0.1%	<0.1%	-	-	<0.1%	0.1% (<0.1-0.2)
N	6,492	4,643			4,626	6,214

Table A8. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by double-crested cormorants nesting at Murdock Towers in Bonneville Reservoir on ESA-listed salmonid populations originating from the Snake River (SR; based on detections at John Day Dam) and Upper Columbia River (UCR; based on detections at John Day Dam) for spring (Sp) summer (Su) and Fall run fish during 2024 – 2025. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable estimates. Estimates from 2024 are those previously reported by Evans et al. (2025).

	Murdock Towers, Bonneville Reservoir					
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2024	0.3% (0.1–1.2%)	0.9% (0.3–2.3%)	0.9% (0.5-5.0)	-	0.7% (0.1–3.3%)	-
N	3,408	2,284	933		1,191	
2025	0.5% (0.2–1.1)	4.6% (2.4–8.8)	0.8% (0.2–2.2)	-	0.9% (0.3–2.4)	-
N	3,453	795	1,177		1,071	

Table A9. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by double-crested cormorants nesting at Astoria-Megler Bridge on ESA-listed salmonid populations originating from the Snake River (SR; based on detection at Bonneville Dam), Upper Columbia River (UCR; based on detections at Bonneville Dam), and middle Columbia River (MCR; based on detection at Bonneville Dam) during 2022–2025 for spring (Sp), summer (Su), and Fall run fish, when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts interrogated passing dams were too small (< 500) to generate reliable estimates. Although the Astoria-Megler Bridge was scanned for smolt PIT tags in 2021, ESU/DPS-specific estimates were not available due to small sample sizes of known ESA-listed smolts within experimental plots on the bridge (see Evans et al. 2022b). Estimates by salmonids species, however, were reported (see Evans et al. 2022b, 2023).

Astoria-Megler Bridge, Columbia River Estuary							
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	MCR Steelhead	SR Steelhead	UCR Steelhead
2022	4.9% (2.6-8.1)	3.1% (2.1-7.9)	5.2% (2.0-10.3)	6.6% (1.7-14.7)	7.4% (2.1-15.5)	7.2% (3.5-12.0)	8.6% (3.2-15.1)
N	14,345	2,069	3,390	1,677	1,791	7,935	3,294
2023	7.1% (4.1–11.3)	3.8% (1.1–7.9)	8.2% (4.2–13.8)	14.3% (6.3–24.4)	6.8% (1.5–14.4)	10.9% (6.2–17.4)	10.4% (5.1–17.7)
N	17,539	3,738	6,238	2,904	2,245	10,670	5,272
2024	4.4% (2.6–10.8)	2.8% (1.8–7.6)	7.1% (2.8–13.3)	9.1% (2.8–18.7)	10.9% (5.0–20.8)	9.9% (4.9–17.3)	6.3% (1.5–13.8)
N	14,171	2,462	4,874	2,117	4,333	7,623	2,427
2025	7.6% (4.0–12.6)	6.3% (0.2–18.5)	10.7% (4.3–20.0)	-	6.4% (2.2–13.1)	10.7% (4.8– 18.8)	12.7% (3.3– 27.2)
N	13,147	923	3,494		4,172	6,954	1,541
Troutdale Towers, Lower Columbia River							
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	MCR Steelhead	SR Steelhead	UCR Steelhead
2022	2.4% (1.3–4.3)	0.7% (0.1–2.0)	1.7% (0.4–3.6)	4.4% (1.2–9.5)	3.2% (0.8–7.0)	2.8% (1.2–5.0)	3.2% (1.2–6.1)
N	14,345	2,069	3,390	1,677	1,791	7,935	3,294
2023	2.9% (1.2–8.4)	0.5% (0.1–1.9)	1.5% (0.6–4.6)	2.1% (0.8–8.1)	3.7% (1.3–12.6)	3.1% (1.3–9.4)	2.5% (1.1–6.7)
N	17,539	3,738	6,283	2,904	2,245	10,670	5,272
2024	3.2% (1.9–6.9)	2.4% (1.0–5.7)	2.7% (1.4–6.4)	3.0% (1.4–6.4)	4.8% (2.6–9.3)	3.8% (2.2–6.9)	5.9% (3.2–12.1)
N	14,171	2,462	4,874	2,117	4,333	7,623	2,427
2025	1.1% (0.4–1.8)	0.5% (0.1–2.6)	0.8% (0.2–2.3)	-	1.0% (0.4–2.4)	2.9% (0.9–4.0)	2.8% (0.6–4.9)
N	13,147	923	3,494		4,172	6,954	1,541

Table A10. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by American White Pelicans on Badger Island and Crescent Island in McNary Reservoir on ESA-listed salmonid populations originating from the Snake River (SR; based on detections at Lower Monumental Dam) and Upper Columbia River (UCR; based on detections at Rock Island Dam) during 2020–2025 for spring (Sp) summer (Su) and Fall run fish when applicable. The Badger Island colony did not form in 2025, while the Crescent Island colony rapidly expanded in size that year. A dashed line indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates. Estimates from 2020–2024 are those previously reported by Evans et al. (2025).

Badger Island, McNary Reservoir						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2020	0.1% (<0.1-0.9)	0.4% (0.1-1.6)	0.2% (<0.1-1.3)	-	0.4% (0.1-4.2)	0.3% (0.1-1.0)
N	2,931	1,607	947		1,130	6,843
2021	0.2% (<0.1–1.4)	0.9% (<0.1–2.9)	< 0.1%	-	1.0% (0.1–7.7)	0.4% (0.1–2.4)
N	562	1,574	1,546		1,289	8,090
2022	0.4% (0.1–1.3)	0.3% (0.1–0.9)	-	-	0.9% (0.1–2.8)	0.7% (0.2–2.6)
N	6,492	4,643			4,626	6,214
2023	0.1% (<0.1–0.3)	0.2% (0.1–1.1)	-	-	0.6% (0.1–1.6)	0.7% (0.2–2.7)
N	12,498	1,558			8,809	8,377
2024	0.1% (<0.1-0.3)	0.2% (0.1-1.1)	-	-	1.0% (0.2-3.2)	0.2% (0.1-0.7)
N	1,231	2,198			2,611	3,800

Cresent Island, McNary Reservoir						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead
2024	< 0.1%	< 0.1%	-	-	< 0.1%	< 0.1%
N	1,231	2,198			2,611	3,800
2025	0.1% (<0.1–0.3)	0.6% (0.1–2.8)	-	-	0.2% (<0.1–2.6)	0.2% (0.1–1.8)
N	2,157	842			3,853	7,671

Table A11. Number of available PIT-tagged smolts (N) and annual predation rates (95% credibility intervals) by American White Pelicans on Miller Sands Spit in the Columbia River Estuary on ESA-listed salmonid populations originating from the Snake River (SR), Upper Columbia River (UCR), and Middle Columbia River (MCR) based on detections at Bonneville Dam during 2021–2025 for spring (Sp) summer (Su) and Fall run fish when applicable. A dashed line indicates that sample sizes of PIT-tagged smolts were too small (< 500) to generate reliable estimates. Estimates from 2020–2023 are those previously reported by Evans et al. (2023). No estimate were available in 2024, but a colony existed that year.

	Miller Sands Spit, Columbia River Estuary						
Year	SR Sp/Su Chinook	SR Fall Chinook	UCR Sp Chinook	SR Sockeye	SR Steelhead	UCR Steelhead	MCR Steelhead
2021	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%
N	20,246	3,389	4,895	2,122	11,868	5,894	3,157
2022	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%
N	14,345	2,069	3,390	1,677	7,935	3,294	1,791
2023	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%
N	17,539	3,738	6,238	2,904	2,245	10,670	5,272
2024 N	NA	NA	NA	NA	NA	NA	NA
2025	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%	< 0.1%
N	13,147	923	3,494		4,172	6,954	1,541

APPENDIX B: PREDATION AND SURVIVAL OF UPPER COLUMBIA RIVER HATCHERY AND WILD STEELHEAD SMOLTS

The analysis presented herein was originally included in the 2022 Avian Predation Annual Report (Evans et al. 2023) and has been updated with results from 2023–2025. References for citations are provided in the Literature Cited section above.

One of the primary purposes of randomly selecting (regardless of size, condition, or rear-type) Upper Columbia River (UCR) steelhead smolts for tagging at Rock Island Dam (RIS) was to ensure that the sample of fish used to estimate predation and survival probabilities was representative of all steelhead smolts in the population at-large (tagged and untagged). Upper Columbia River steelhead smolts were also tagged in proportion to the run passing RIS, with more fish tagged when more fish were available in-river. This approach eliminated the need to weight estimates of predation and survival based on the number and run-timing of steelhead smolts passing RIS each year (see *PIT-tagging of Upper Columbia River Steelhead* section for details). Data regarding the rear-type (hatchery, wild), external condition, and size (fork length) of steelhead smolts – coupled with survival and predation probabilities – provided a unique opportunity to investigate the relationship between individual fish characteristics and susceptibility to bird predation. Presented herein is an analysis of the odds of survival and predation of UCR steelhead smolts based on a fish's rear-type using all available data since tagging commenced at RIS in 2008.

Methods

Steelhead smolts PIT-tagged or recaptured (previously tagged) at RIS were classified as being either hatchery or wild. Hatchery fish were classified by the absence of an adipose fin or by characteristics associated with hatchery-rearing practices, including the erosion of pectoral, pelvic, or dorsal fins or a PIT-tag where the original tagger classified the fish as hatchery or a coded wire tag (2013–2016, 2023–2025) that indicated the fish was of hatchery origin (see *PIT-tagging of Upper Columbia River Steelhead* section for additional details). Wild fish were classified by the presence of fully intact fins, with no tags indicating hatchery origin. It is possible, however, that some hatchery fish also met the criteria of being a wild fish and were thus misclassified as wild. As such, a likely small but unknown number of tagged fish could have been misclassified as wild.

Weekly and annual survival and predation probabilities (proportion of available fish) were generated using the joint mortality and survival (JMS) model of Payton et al. (2019) for the two cohorts (hatchery, presumed wild) of steelhead tagged/recaptured and released at RIS. Full details of the JMS model are provided in Payton et al. (2019; see also *Methods & Analysis* and *Cumulative Predation & Survival* sections). In brief, for use in this analysis, cohort was defined with specific parameters for survival, predation, and recapture, with,

> $\boldsymbol{\Theta}^{c}$**,** defined to be a DxJ matrix where $\theta^{c}_{d,j}$ represented the probability (from release) that a fish from release cohort r succumbed to depredation by colony *d* for $d \in$

$\{1,2,\ldots,D-1\}$ or some other cause of mortality for $d = D$, is the reach of river immediately preceding recapture opportunity j

and

$\boldsymbol{p^c}$, defined as a J-length vector where p_j^c represented the probability that a fish from release cohort r alive at recapture opportunity *j* was successfully recaptured (reseen)

and a common probability of tag recovery among cohorts was assumed with,

$\boldsymbol{\gamma}$, defined as a D-length vector where γ_d represented the probability of recovering a fish from any cohort which died due to depredation by colony *d* for $d \in \{1,2,\ldots,D-1\}$, and γ_{15} = 0 represented the lack of recovery opportunity for fish which died from all other unspecified causes.

Letting *m* represent the final recapture opportunity at which the fish was seen (with $m = 0$ representing a fish never reseen following release) the combined likelihood was then,

$$L = \prod_r \left(\prod_{j \le m} \left({p^c}_j^{y_j} * \left(1 - p_j^c\right)^{(1-y_j)} \right) * \prod_d {\chi^c_{m+1,d}}^{r_d} \right);$$

where,

$$\chi^c_{j,d} = \begin{cases} \theta^c_{j,d} * \gamma_d + \left(1 - p^c_{j+1}\right) * \chi^c_{j+1,d}; & for\ d \in 1,\ldots,D-1 \\ \sum_{i=1}^{D} \theta^c_{j,i} * (1-\gamma_i) + \left(1 - p^c_{j+1}\right) * \chi^c_{j+1,i}\,; & for\ d = D \end{cases}$$

To evaluate if survival and predation probabilities of tagged steelhead differed by rear-type, the odds of predation was compared among the cohorts across weeks and years and noted any differences that were statistically significant, defined as instances in which the log of the odds-ratio between the two cohorts had negligible overlap with zero (i.e., >95% of the posterior distribution of the difference lay above or below zero).

Comparisons of survival and predation probabilities by rear-type were investigated based on the avian predator species (Caspian terns [CATE], California and ring-billed gulls [LAXX], and double-crested cormorants [DCCO], or all birds combined) and the river reach (RIS to McNary Dam [MCN]), MCN to Bonneville Dam [BON]) and BON to the Pacific Ocean) where predation occurred (see also *Methods & Analysis* and *Cumulative Predation and Survival* sections).

Results & Discussion

Most UCR steelhead smolts captured, tagged/recaptured, and released at RIS were hatchery-reared, comprising 71.6–80.8% of all tagged fish sampled during 2008-2025 (*Table B.1*). Ratios of hatchery to wild fish were also relatively consistent throughout the study period (*Table B.1*).

Table B.1. Rear-type (hatchery, wild) of PIT-tagged Upper Columbia River steelhead smolts at Rock Island Dam during 2008–2025.

	Rear-type	
Year	No. (%) Hatchery	No. (%) Wild
2008	5,373 (73.9%)	1,898 (26.1%)
2009	5,150 (72.4%)	1,964 (27.6%)
2010	5,387 (73.1%)	1,978 (26.9%)
2011	5,961 (76.9%)	1,795 (23.1%)
2012	5,107 (76.1%)	1,605 (23.9%)
2013	4,284 (72.7%)	1,609 (27.3%)
2014	5,686 (74.2%)	1,977 (25.8%)
2015	5,105 (72.2%)	1,964 (27.8%)
2016	4,965 (73.4%)	1,799 (26.6%)
2017	5,776 (77.7%)	1,660 (22.3%)
2018	5,261 (72.7%)	1,980 (27.3%)
2019	3,201 (72.8%)	1,196 (27.2%)
2020	4,895 (71.6%)	1,946 (28.4%)
2021	6,366 (78.7%)	1,724 (21.3%)
2022	4,579 (73.7%)	1,638 (26.3%)
2023	6,172 (73.7%)	2,205 (26.3%)
2024	3,038 (79.9%)	762 (20.1%)
2025	6,201 (80.8%)	1,470 (19.2%)

Predation/consumption probabilities by rear-type varied by avian predator species, river reach, and year (2008–2025; *Figure B.1-B.3*). Results indicated that hatchery fish were more likely to be consumed than their wild counterparts by CATE and LAXX breeding at colonies that foraged on smolts between RIS and MCN (*Figure B.1*). Differences were statistically significant when data from all weeks and years were considered. Of the two predator species evaluated, relative differences were often greater or more pronounced in LAXX compared with CATE. Taken together, predation/consumption probabilities by both CATE and LAXX (i.e., all birds combined) on hatchery steelhead averaged 23.0% (21.6–24.5%) compared with 12.6% (11.5–13.9%) on presumed wild steelhead. There was also some evidence that wild fish were more likely, on average, to survive outmigration from RIS to MCN compared to hatchery fish (*Figure B.1*). Differences in survival between hatchery and wild steelhead were commensurate with the relative difference in predation/consumption probabilities on hatchery and wild steelhead in most, but not all, years.

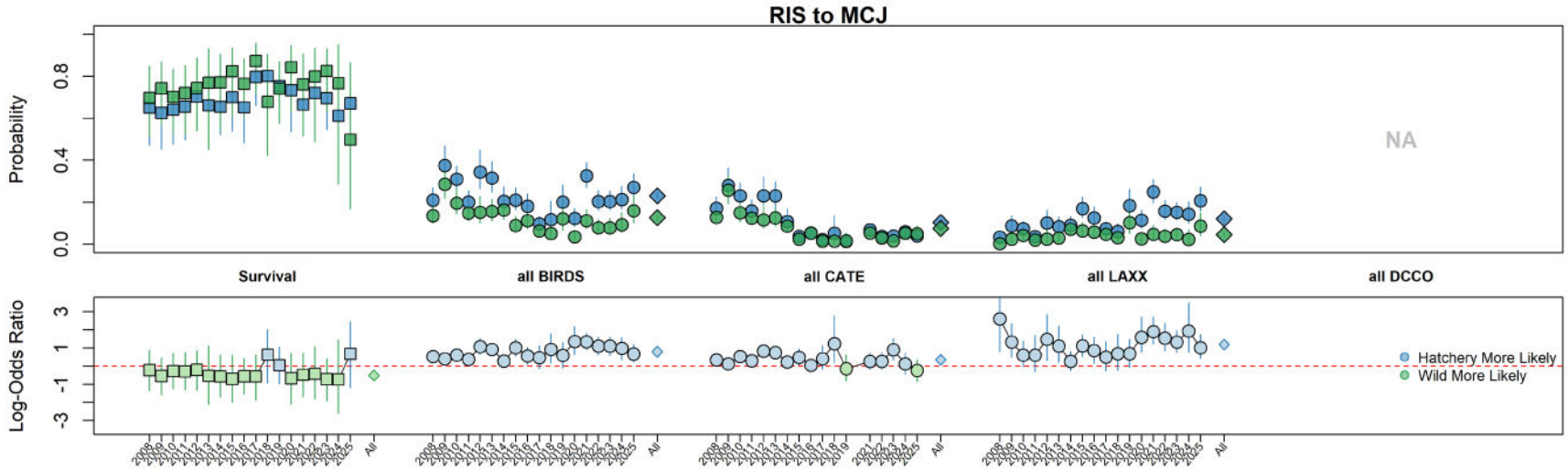


Figure B.1. Relative comparisons of annual survival and predation/consumption probabilities (proportion of available fish) of Upper Columbia River steelhead by rearing-type (hatchery, wild) to predation by colonies of Caspian terns (CATE), California and ring-billed gulls (LAXX), and double-crested cormorants (DCCO) during smolt passing from Rock Island Dam (RIS) to McNary Dam (MCN) during 2008–2025 (upper panel). Statistical comparisons (lower panel) represent the log odds ratio of survival or predation, with values < 0 indicating greater odds for wild fish and values > 0 indicating greater odds for hatchery-reared fish. Error bars represent 95% credible intervals, with uncertainty ranges overlapping 0 associated with relative differences that were not statistically significant. NA denotes that predation/consumption or survival probabilities were not available for relative comparisons.

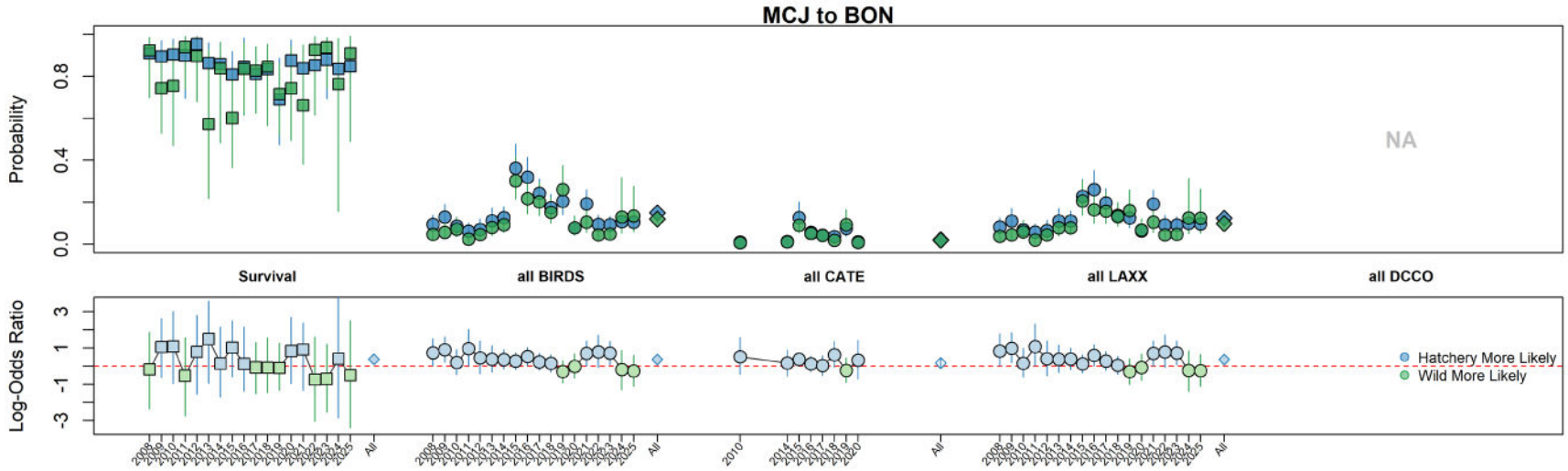


Figure B2. Relative comparisons of annual survival and predation/consumption probabilities (proportion of available fish) of Upper Columbia River steelhead by rearing-type (hatchery, wild) to predation by colonies of Caspian terns (CATE), California and ring-billed gulls (LAXX), and double-crested cormorants (DCCO) during smolt passing from McNary Dam (MCN) to Bonneville Dam (BON) during 2008–2025 (upper panel). Statistical comparisons (lower panel) represent the log odds ratio of survival or predation/consumption, with values < 0 indicating greater odds for wild fish and values > 0 indicating greater odds for hatchery-reared fish. Error bars represent 95% credible intervals, with uncertainty ranges overlapping 0 associated with relative differences that were not statistically significant. NA denotes that predation/consumption or survival probabilities were not available for relative comparisons.

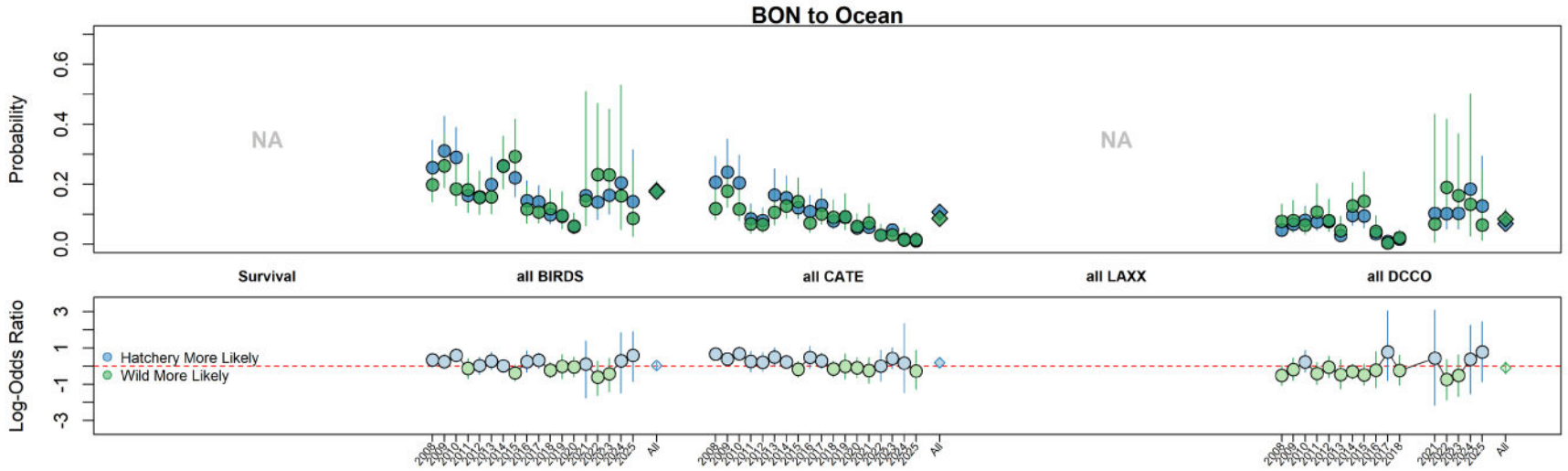


Figure B3. Relative comparisons of annual survival and predation/consumption probabilities (proportion of available fish) of Upper Columbia River steelhead by rearing-type (hatchery, wild) to predation by colonies of Caspian terns (CATE), California and ring-billed gulls (LAXX), and double-crested cormorants (DCCO) during smolt passing from Bonneville Dam (BON) to the Pacific Ocean during 2008–2025 (upper panel). Statistical comparisons (lower panel) represent the log odds ratio of survival or predation/consumption, with values < 0 indicating greater odds for wild fish and values > 0 indicating greater odds for hatchery-reared fish. Error bars represent 95% credible intervals, with uncertainty ranges overlapping 0 associated with relative differences that were not statistically significant. NA denotes that predation/consumption or survival probabilities were not available for relative comparisons.

Although there were active colonies of DCCO and AWPE that foraged on UCR steelhead between RIS and MCN (e.g., Hanford Island, Foundation Island, and Badger Island), predation probabilities were low (< 0.01). Due to these low predation rates, adequate data to investigate relative differences in predation by rear-type due to DCCO and AWPE predation in this river reach were not available. Furthermore, birds from these colonies do not pose a substantial threat to UCR steelhead smolt survival upstream of Bonneville Dam.

There was some evidence that hatchery-reared steelhead smolts were more likely be predated/consumed than wild steelhead during smolt outmigration from MCN to BON (*Figure B2*). Relative differences, however, were less consistent than those observed upstream of MCN, with the occasional finding that wild fish were more likely to be predated/consumed than their hatchery counterparts, depending on the predator species and year. The magnitude of difference in predation/consumption probabilities based on a fish's rear-type were also consistently smaller (less pronounced) than that observed during smolt passage between RIS and MCN, with predation probabilities from all birds during smolt outmigration from MCN to BON averaging 15.0% (13.7–16.5%) and 12.0% (10.5–13.8%) for hatchery and wild smolts, respectively. There was no evidence that the odds of survival from MCN to BON were greater for wild smolts compared with hatchery-reared smolts when data from all weeks and years were considered.

There was no consistent evidence that CATE and DCCO breeding at colonies downstream of BON disproportionately consumed hatchery and wild steelhead smolts, with both rear-types equally susceptible to bird predation in the Columbia River estuary (*Figure B3*). There was limited evidence that DCCO disproportionately consumed wild smolts in some years (e.g., 2018, 2022–2023) but results were not statistically significant when data from all years are considered. Taken together (all birds) the odds of predation by rear-type were nearly indistinguishable between hatchery and wild steelhead downstream of BON. For instance, average annual predation/consumption probabilities by all birds were 17.9% (16.2–20.0%) and 17.6% (15.2–21.0%) for hatchery and wild smolts, respectively. Annual trends in CATE and DCCO predation also followed very similar patterns across the 18-year study period (*Figure B3*). Due to a lack of PIT tag detection sites downstream of East Sand Island in the lower Columbia River estuary, estimates of smolt survival to the Pacific Ocean were not available (see also *Methods & Analysis* and *Cumulative Predation and Survival* sections for details).

These results are consistent with those of several other studies and indicate that steelhead susceptibility to colonial waterbird predation/consumption was associated with a fish's rear-type for some, but not all, predator species and colonies in the Columbia River basin (Hostetter et al. 2012, Payton et al. 2016, Hostetter et al. 2023). When differences were observed, they often indicated higher levels of predation on hatchery-reared steelhead. Hostetter et al. (2012) also observed that hatchery-reared steelhead from the Snake River were more susceptible to CATE predation than wild steelhead, differences that were attributed to the larger average size of hatchery steelhead and to possible behavioral differences. For instance, hatchery-rearing systems may select for individuals that are more surface oriented, less able to endure sustained swimming, and naive to predators relative to their wild counterparts (as reviewed by Hostetter

et al. 2023). In the present study, UCR hatchery-reared steelhead were, on average, larger than wild steelhead (hatchery: 199.8 ± 23.4 mm; wild: 179.6 ± 27.4 mm fork length).

There was some evidence that differences in the relative susceptibility of UCR hatchery and wild steelhead to CATE and LAXX predation/consumption decreased (lessened or diminished) during smolt passage from RIS to the Pacific Ocean. For instance, relative differences in steelhead predation by rear-type were less pronounced and often not statistically different for CATE foraging on UCR steelhead downstream of MCN, even though differences were apparent upstream of MCN (i.e., following tagging and release at RIS). This may be due, in part, to changes in the size distribution of available hatchery and wild smolts to predators downstream, with larger-sized smolts disproportionally removed by plunge-diving predators upstream, functionally changing the length distribution of surviving steelhead to below BON. Additional research is warranted to better understand to what degree the disproportionate predation/consumption of larger-sized steelhead (both hatchery and wild) by some upstream CATE and LAXX colonies influenced the size distribution and subsequent smolt and smolt-to-adult survival of UCR steelhead.

Unlike predation by CATE and LAXX, there was no evidence that DCCO disproportionately consumed UCR hatchery-reared steelhead compared with their wild counterparts. Similarly, Hostetter et al. (2012) found no evidence of a difference in the relative susceptibility of Snake River hatchery and wild steelhead to DCCO predation during smolt passage from Lower Monumental Dam to MCN. Unlike CATE and LAXX, which are surface feeders, DCCO are pursuit-diving predators. Also, unlike CATE and LAXX, there is no evidence that DCCO disproportionately consumed larger size smolts (Hostetter et al. 2012, Roby et al. 2016), with smolts of all lengths equally susceptible to DCCO predation at both estuary and inland colony locations. This result emphasizes that predator-specific interactions can be dynamic and complex, intricacies that should be considered when evaluating the over-all effects of predation on prey populations (see also Hostetter et al. 2023 for a more detailed discussion).

APPENDIX C: PREDATION AND SURVIVAL MODELLING FRAMEWORK FOR UPPER COLUMBIA RIVER STEELHEAD AND YEARLING CHINOOK

Summary

The primary goal of this study was to estimate survival and avian predation on PIT-tagged yearling Chinook and steelhead smolts that were passively recaptured (detected) at the Rocky Reach Dam juvenile bypass facility (RRJ) and the Lower Wenatchee PIT Barge (LWB) in the Middle Columbia River. Predation and survival of smolt cohorts (defined by their species, hydrologic unit [HUC] of origin, and year) were estimated using a Joint Mortality and Survival (JMS) model, with the passive recapture locations of RRJ and LWB considered "virtual-release" locations (i.e., the spatial starting point of the analysis). After initial detection at RRJ/LWB, survival and predation were estimated during smolt outmigration to McNary Dam (MCN), Bonneville Dam (BON), and the Pacific Ocean. Of the candidate models evaluated, the best performing iteration assumed that likelihoods of recapture were equal among all fish within a species, facilitating the development of conclusive estimates of differences among cohorts. This "common resight" model also greatly improved the precision of estimates of survival and predation, enhancing the interpretability of the results.

Predation and survival (proportion of available fish) were estimated for cohorts with > 499 tagged fish available, the minimum number needed to generate reliable estimates of predation. Cohorts of steelhead from up to five HUCs at RRJ from 2021–2025 and one HUC at LWB in 2024 were available for analysis. For yearling Chinook, up to two HUCs at RRJ from 2021–2025 were assessed and one HUC at LWB from 2024–2025. Results indicated that avian predation probabilities on steelhead during passage from virtual-release to MCN were highly variable, with point estimates of predation ranging from 0.064 to 0.352 (equivalent to 6.4% to 35.3%), depending on the cohort and year. Avian predation on yearling Chinook from virtual-release to MCN was consistently low (~0.020) in most years and cohorts evaluated but ranged up to 0.050 for some cohorts and years. Despite low rates of predation, results provided significant differences among HUC-specific predation rates in all years but one. Estimated probabilities of survival from virtual-release to MCN were considerably variable across species and years. Survival differences among HUC-cohorts were more pronounced for steelhead, with moderate to strong differences in most years, than for yearling Chinook, with little to no evidence of differences.

A secondary goal of the study was to investigate how changes in future tagging studies could improve the precision and accuracy of predation and survival estimates, primarily through increased tagging or recapture opportunities. Currently, smolts passively detected at RRJ and LWB are previously tagged fish, a methodology statistically referred to as "convenience-sampling". Consequently, certain demographic groups may be over- or underrepresented relative to the population at large and, unlike with probability-sampling methods (e.g., simple random, cluster, stratified), the probability of including individuals from different demographics in the sample are unknown, making it difficult to make statements about the population at large. Therefore, prior to utilizing data from a convenience sample, re-weighting adjustments should be included in estimate calculations to account for differences in the sample composition. To address this, we conducted a simulation study, using hatchery- and wild-reared

delineations of fish as an illustrative example, to investigate the degree to which different demographic compositions influenced predation and survival estimates and the effectiveness that re-weighting adjustments can have in developing results.

Results of the simulation study indicated that applying a compositional correction (e.g., post hoc re-weighting to reflect the approximate proportion of wild fish in the population at large) was essential to generate unbiased estimates of predation. Additionally, increasing sample sizes without accounting for the compositional correction produced estimates that were centered at the wrong value. When the sample was 97% hatchery fish and no re-weighting was applied (i.e., aligned with the demographics of the currently available data at RRJ/LWB), predation estimates were persistently biased upward toward the hatchery predation rate. Critically, this bias did not diminish meaningfully as sample size increased. For survival estimation, increasing the proportion of wild fish in the tagged sample proved far more effective than increasing total sample size alone. For instance, a sample of 2,000 fish that is 10% wild consistently outperformed a sample of 4,000 fish at 3% wild in both bias and credible interval coverage. While no single sample size guaranteed uniform performance in all conditions, results suggest that an ideal sample includes practical targets of 4,000 to 8,000 fish in a virtual-release sample (i.e., detected at RRJ or LWB) with moderately sufficient representation of the demographics of the full population (e.g., >10% wild fish) combined with consistent application of re-weighting correction. These findings underscore that attempted inference from convenience-sampled data requires deliberate correction for compositional differences in the sample versus the target population, including rear-type but also presumably other factors like tributary of origin, migration timing, abundance, and fish size. If left unaddressed, misrepresentation of these other factors carries the same potential to bias population-level estimates of predation and survival as that demonstrated by differences in rear-type.

Methods

Study Area: Yearling Chinook and steelhead smolts were tagged and released into tributaries of the middle Columbia River with some proportion subsequently detected (passively recaptured) alive at downstream sites with PIT tag integration capabilities or recovered dead on bird colonies (*Figure C1,* see also *Figure 1* in the Methods & Analyses section for bird colony locations). Live recapture sites included the Rocky Reach Dam juvenile bypass facility (RRJ), Lower Wenatchee Barge (LWB), Rock Island Dam juvenile fish trap (RIS), McNary Dam juvenile bypass facility (MCN), John Day Dam juvenile bypass facility (JDA), Bonneville Dam juvenile bypass facility and corner collector (BON), and a pair trawl and/or pile dike antennas in the Columbia River estuary (EST). Tags from dead fish were recovered on up to 24 bird colonies.



Figure C1. Release sites of PIT-tagged yearling Chinook and steelhead subsequently detected alive at Rocky Reach Dam juvenile bypass (RRJ) or the Lower Wenatchee Barge (LWB) during 2021–2025. The size of each circle indicates the average percent contribution each site had to the total number of detections at RRJ or LWB.

Data Compilation: Yearling Chinook and steelhead captured, tagged, and released into tributaries of the middle Columbia River were identified by querying Hydrologic Unit Code (HUC), species, rear-type, and age-class (Chinook only) in the PIT Tag Information System (PTAGIS). Fish released into tributaries between of each year (2021–2025) were included in the analysis, fish that were assumed to be out-migrating in that year (detected at downstream sites between 1 March and September 30). Fish available to be detected at RRJ include fish originating from the Methow (HUC 17020008), Chief Joseph (HUC 17020005), Upper Columbia-Entiat (HUC 17020010), Okanogan (HUC 17020006), and Similkameen (HUC 17020007) basins. Only fish originating from the Wenatchee (HUC 17020011) were available for recapture at LWB. For hatchery Chinook, age classification (yearling, subyearling) was determined using the FPC Hatchery Release Database (https://fishpassagecenter.shinyapps.io/HatcheryQuery/). For wild Chinook, age classification was based on the size of the fish at tagging/release, with yearlings classified as those fish > 80 mm (fork-length) through mid-May, > 90 mm from mid-May to 1 June, and > 140 mm from 1 June to mid-June, and > 150 mm after mid-June (see also Mid-C Smolt Monitoring Protocol 2009). A mark (i.e., release site), recapture (i.e., passive detection at PIT array) and recovery (i.e., found dead on a bird colony) history was then created for each individual fish.

Survival & Predation Modelling: Rocky Reach Dam or Lower Wenatchee Barge were used as starting-points or virtual-release sites for estimates of predation and survival, thus only individuals detected alive at RRJ/LWB were included in the analysis. Weekly and annual survival and predation probabilities (proportion or percentage of available fish) were generated using the joint mortality and survival (JMS) model of Payton et al. (2019) for each species (steelhead, yearling Chinook) and HUC, collectively referred to as “cohorts”. The JMS model jointly or simultaneously estimated survival and mortality due to avian predation on each cohort across multiple spatial and temporal scales using a state-space, hierarchical Bayesian modelling framework. As a comparison, all fish of a species passing RRJ

irrespective HUC was additionally derived, referred to as "single-cohort" estimates. Full details of the JMS model, including model formulations and assumptions, are provided in Payton et al. (2019; see also Methods & Analysis, Cumulative Predation & Survival section).

To construct the most precise and informative estimates of predation and survival specific to each cohort, the inclusion of a stronger assumption with respect to resighting individuals downstream was investigated. To this end, two implementations of the JMS model were compared: (a) the fully "independent" model, where the fate-probabilities (those of predation, survival, and other [unmeasured] mortality) and the resight-probabilities (those of recapture and recovery) were assumed to be mutually independent among all cohorts and (b) the "common resight" model where resight-probabilities of all fish of the same species were equal. By assuming equal likelihood of downstream recapture, the precision of the fate-probabilities (namely survival and predation probabilities in downstream reaches) is increased considerably, especially for cohorts with limited sample sizes. To determine which modelling approach (independent-resight or common-resight) best explained the observed data, leave-one-out cross-validation (LOO-CV; Vehtari et al. 2017) was used to rank the two candidate models. LOO-CV provides a measure of out-of-sample predictive performance while accounting for model complexity, with greater expected log pointwise predictive density (ELPD) values indicating better model fit. The difference in ELPD (ΔELPD) and associated standard errors were calculated to assess the relative support for each modelling approach. When expected log pointwise predictive density (ELPD) differences are "well-behaved" they follow a normal distribution, and the standardized differences (z-score) can therefore be interpreted approximately as normal distribution z-scores for statistical comparison. Probability differences among HUC-cohorts were calculated directly with the simulated draws from posterior distributions. Bayesian p-values for these differences were defined as the minimum proportion of the number of simulated draws from the posterior distribution that were greater than or less than zero.

Simulation Study: Release-recapture-recovery data was simulated from samples across a factorial design of sample sizes, which included 500, 1,000, 2,000, 4,000, 8,000, 16,000 and 32,000 tagged fish, with samples composed of 3%, 10%, 20%, and 30% wild fish. Simulations were parameterized using the probabilities derived from the five years (2021–2025) of empirical data described above using rear-type as the delineating cohort variable rather than HUC, capturing natural year-to-year variation in both the overall rates and the magnitude of rear-type differences. Again, LOO-CV was used to determine whether to use the common-resight model in which probabilities of survival, predation, and "other" (non-avian) mortality were assumed to vary by rear-type, while probabilities of recapture and the recovery of consumed tags were assumed equal across rear-types. Each parameter/year combination was replicated ~100 times. For each simulated dataset, probabilities of survival and predation were then estimated with the JMS model using two methods: (1) assuming equal probabilities across all individuals in the population ("single-cohort"); and (2) assuming survival and predation probabilities differ among cohorts of wild-reared and hatchery-reared fish. For method (2) aggregate population level estimates were derived by taking a weighted average of the cohort specific estimates assuming a population of which 20% were wild-reared, used as an approximation of the true population composition for the purposes of

relative comparisons. Performance was assessed using mean bias (distance of the point estimate from the true value), precision (width of the 95% highest density credible interval [HDI]), and coverage probability (proportion of simulations where the 95% HDI contained the true parameter value).

Results

Sample Sizes: The number of yearling Chinook passively detected at RRJ during 2021–2025 ranged from 6,214 in 2025 to 13,357 in 2021. For steelhead, numbers ranged from 17,539 in 2025 to 25,020 in 2021 (*Table C1*). At LWB, full and consistent detections were available in 2024 and 2025 only, with samples of 1,825 and 4,146 yearling Chinook and 323 and 561 steelhead, respectively. Most detections for both yearling Chinook and steelhead at RRJ originated from the Methow basin (*Table C2* and *Figure C2*). Further, a large portion of detections at RRJ come from just a few individual release sites, with the vast majority being of hatchery origin (see also Supplementary Materials).

Table C1. Sample sizes of PIT-tagged yearling Chinook and steelhead smolts initially detected at the Rocky Reach juvenile bypass facility (RRJ) and Lower Wenatchee Barge (LWB) that were subsequently detected alive at PIT tag arrays located at Rock Island Dam (RIS), McNary Dam (MCN), John Day Dam (JDA), Bonneville Dam (BON), the Columbia River estuary (EST) or recovered dead on avian colonies (Avian).

Year	Species	From RRJ	From LWB	RIS	MCN	JDA	BON	EST	Avian
2021	CH1	13,357	--	34	251	131	1,191	25	138
2022	CH1	11,401	--	30	61	113	581	119	101
2023	CH1	7,539	--	48	93	208	660	166	77
2024	CH1	9,571	--	22	111	70	550	191	109
2025	CH1	6,214	--	18	83	44	333	86	55
2021	ST	25,020	--	507	261	174	3,008	157	1,857
2022	ST	24,202	--	325	180	291	1,495	295	1,312
2023	ST	21,153	--	686	233	911	1,738	575	1,205
2024	ST	19,348	--	155	132	188	862	368	1,067
2025	ST	17,539	--	347	132	138	611	237	804
2024	CH1	--	4,146	10	43	43	197	55	40
2025	CH1	--	1,825	5	21	25	67	15	22
2024	ST	--	561	5	6	3	24	12	22
2025	ST	--	323	4	2	0	2	0	17

Table C2. Average percent of yearling Chinook (CH1) and steelhead (ST) detections at Rocky Reach juvenile bypass by release HUC from 2021–2025.

Species	Release HUC	Watershed	Average % of Species Detections
CH1	17020008	Methow	87.1
CH1	17020005	Chief Joseph	10.8
CH1	17020010	Upper Columbia-Entiat	2.1
ST	17020008	Methow	58.2
ST	17020006	Okanogan	24.5
ST	17020005	Chief Joseph	11.3
ST	17020007	Similkameen	3.7
ST	17020010	Upper Columbia-Entiat	2.3

Model Comparison: In comparing the relative fitness between the two approaches (i.e., the fully "independent" model and the "common-resight" model), significant statistical evidence was found that a model with common recapture and recovery probabilities across HUCs explained variations in the observed data more effectively for both smolt species (ELPD-difference-z-scores [roughly normally distributed] of 3.8 and 3.5 for steelhead and yearling Chinook, respectively (*Table C3*).

The two modeling approaches produced similar trends in survival and predation across years and populations, but differed notably in precision and, in some cases, point estimates. Estimates of survival to McNary Dam were significantly impacted by low probabilities of recapture in the juvenile bypass at McNary Dam in recent years and, as such, it is these estimates that are most significantly impacted by the common-resight model's added assumption of equal recapture probabilities for all fish in the same week regardless of HUC. Using the common-resight model reduced average cohort-specific credible interval width across all years up to 26.5% for steelhead and 54.8% for yearling Chinook released from RRJ. Differences in point estimates were most pronounced in reach-specific survival estimates and varied by species and HUC. Point estimates for cohorts with exceedingly limited sample sizes were found to differ substantially between models in some years, highlighting the sensitivity of results to modeling assumptions. The variation in point estimates reinforces the importance of considering the implications of all values in each reported credible interval when assessing statistical results. All estimates reported herein are from RRJ/LWB to MCN and are results from the common-resight model unless otherwise noted. All other reach specific estimates (e.g., MCN to BON) can be found in the Supplementary Materials (*Table S1*).

Table C3. ELPD comparison of the relative fitness between the independent modelling approach and "common-resight" approach for all comparable groups of steelhead and yearling Chinook passively detected at RRJ.

Species	Model	ELPD	SE ELPD	ΔELPD	SE ΔELPD	z-score
Steelhead	Common resight	-97000.6	553.1	--	--	--
	Independent	-97034.6	553.1	34.0	9.0	**3.8**
Yearling Chinook	Common resight	-26651.0	291.3	--	--	--
	Independent	-26664.1	291.3	13.0	3.8	**3.5**

Steelhead

For the years assessed, recapture samples of steelhead at RRJ and LWB that were of sufficient size to produce reliable estimates of predation (n > 499) originated from the Methow (17020008), Okanogan (17020006), and Chief Joseph (17020005) in all years (2021 through 2025); from the Similkameen (17020007) in 2021, 2022, and 2023; from Upper Columbia-Entiat (17020010) in 2024; and from the Wenatchee (17020011) in 2024.

Predation: The aggregate likelihood of predation by all avian predators was fairly constant over the years evaluated, with the notable outlier of 2021 where estimates were highest for all cohorts (*Figure C2*). Results suggest strong statistical evidence ($p < 0.001$) of significant differences in levels of predation among HUC-cohorts for every year evaluated (*Table S2*). Methow steelhead were generally the most susceptible to predation, with estimates of predation from all avian predators between RRJ and MCN consistently the highest or second highest among all HUC-cohorts, ranging from 0.200 (0.165–0.242) to 0.339 (0.278–0.404) (or predation of 20.0% to 33.9% of available fish). While reliable estimates of survival and predation for Upper Columbia-Entiat steelhead were available only in 2024 (due to small sample sizes in all other years {n < 499}), these steelhead consistently had the lowest estimated predation of any HUC-cohort (0.049 [0.019–0.112] to 0.163 [0.057–0.371] prior to MCN); nearly half that of the predation impacts seen on the highest predated cohort in each year. A sufficient sample size to estimate predation for steelhead detected alive at LWB was only achieved in 2024 and was comparable to the average of all the RRJ-cohorts. Estimates of annual predation for the entire release as a single-cohort largely reflected the predation estimates of the Methow, which represented 52% to 63% of all previously tagged steelhead seen at RRJ from 2021 to 2025. These fish were also 85% and 99% hatchery-reared and therefore, as previous research indicates, more susceptible to some avian predators (terns, gulls) in the CRB.

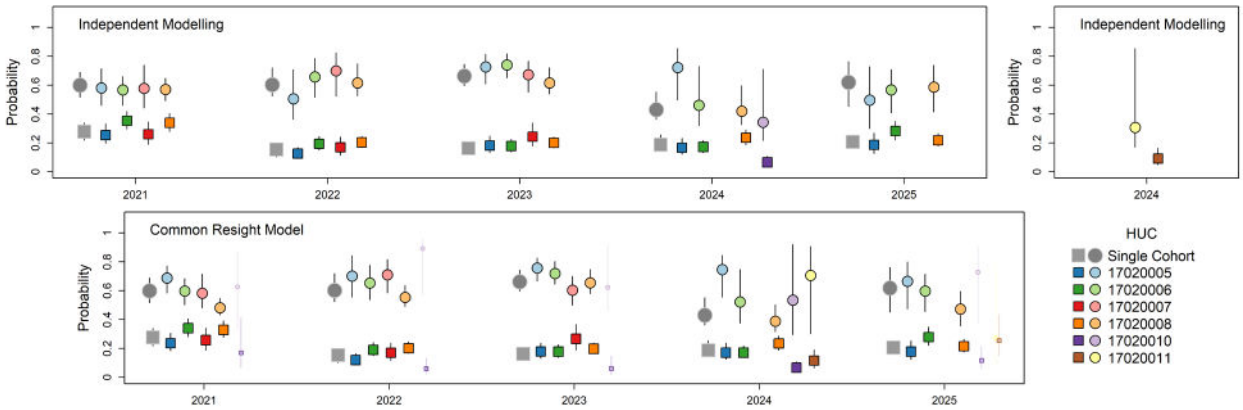


Figure C2. Predation and survival estimates for steelhead passively detected at the Rocky Reach juvenile bypass for fish originating from the Methow (17020008), Okanogan (17020006), Chief Joseph (17020005), Similkameen (17020007), Upper Columbia-Entiat (17020010) HUCs or at the Lower Wenatchee Barge for steelhead originating from the Wenatchee (17020011; data available in 2024 only). The "Single Cohort" estimates represent probabilities of all HUCs combined. Circles represent estimates of survival, squares represent estimates of predation, and line segments represent 95% credible intervals. Smaller, transparent points indicate years with sample sizes of less than 500, which were considered unreliable estimates, but were included here due to statistical leverage provided by the common-resight model.

Survival: Estimates of the likelihood of survival to MCN were similar among cohorts within a year and even among years (around 0.600 or 60%), with the notable exception of 2024. Statistical evaluation of differences in survival likelihoods among cohorts within each year only found strong evidence ($p < 0.001$) of differences between two or more cohorts in two of the five years evaluated, although there was moderate evidence of a few other differences (*Table S2*). The most notable year of inter-cohort variation in survival probabilities was 2024, where survival among cohorts varied considerably, ranging from 0.393 (0.326–0.511) for Methow-origin fish to 0.746 (0.549–0.841) for those originating from Chief Joseph. While no obvious patterns emerged among years and cohorts, steelhead originating from Methow tended to have survival probabilities slightly below their counterparts from other HUCs, which could largely be associated with the heightened levels of predation experienced by hatchery fish. The probabilities are particularly affected by the choice in model assumptions (independent v. common-resight), which is most readily displayed with the disparity in survival estimates associated with Entiat origin fish, whereby the independent model estimated survival of 0.343 (0.216–0.709) compared to the estimate of 0.542 (0.296–0.925) provided by the common-resight model. The credible intervals associated with both of these estimates, however, were relatively uninformative due to the width of credible intervals in both estimates. The Entiat origin Chinook detected at RRJ were exclusively wild-reared fish, as yearling Chinook reared and released from the Entiat National Hatchery are summer run only. There is speculation as to whether wild-reared fish have a different likelihood of entrainment (recapture) in juvenile bypasses than their hatchery counterparts. If this was indeed the case, the assumption of equal recapture would be invalid and consequently estimates of survival to MCN would be biased (note: this would not affect estimates of predation from RRJ). This is also discussed further *below*.

In the assessment of the entire release as a single-cohort, the estimates of survival probability again largely reflected the estimates of Methow origin fish, the dominant demographic group in the sample. As survival probabilities were less variable among cohorts this was less impactful than what was seen with predation estimates, again with the notable exception of the anomalously low estimated survival of Methow steelhead in 2024.

Yearling Chinook

For the years assessed, recapture samples of yearling spring Chinook at RRJ and LWB that were of sufficient size to produce reliable estimates of predation and survival rates (n < 499) originated from the Methow (17020008) in all years (2021 through 2025); from the Chief Joseph (17020005) in 2021 through 2024; and from the Wenatchee (17020011) in 2024 and 2025.

Predation: Avian predation on yearling Chinook from virtual release at RRJ or LWB to MCN was consistently low (~ 0.020 or 2%) in most years (*Figure C3*). Despite these comparatively lower probabilities compared to steelhead, estimates in yearling Chinook provided one of the clearest HUC-specific cohort signals in the dataset, with Chief Joseph origin fish predation probabilities 2–5 times higher than Methow origin fish (e.g., Chief Joseph 0.050 vs Methow 0.009 to MCN in 2023). Methow Chinook experienced low and relatively stable predation (0.010–0.030 to MCN), though with a slight increase in 2024 and 2025. Differences in annual estimates of predation were significantly different between these two cohorts for three of the four comparable years (2022–2024; p<0.001). Predation estimates for yearling spring Chinook detected alive at LWB were also relatively low, with estimates of 0.018 (0.009–0.032) in 2024 and 0.021 (0.010–0.044) in 2025.

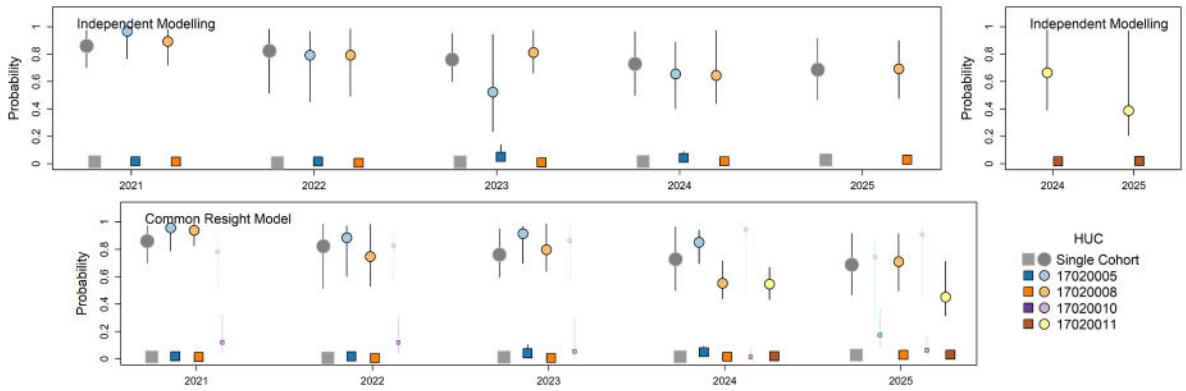


Figure C3. Predation and survival estimates for yearling Chinook detected alive at Rocky Reach juvenile bypass for groups of yearling Chinook originating from the Methow (17020008), Chief Joseph (17020005), and Upper Columbia-Entiat (17020010) rivers or at the Lower Wenatchee Barge for yearling Chinook originating from the Wenatchee (17020011; data available in 2024 only). The "Single Cohort" estimates represent probabilities of all Rocky Reach cohorts combined. Circles represent estimates of survival, squares represent estimates of predation, and line segments represent 95% credible intervals. Smaller, transparent points indicate years with sample sizes of less than 500, which were considered unreliable estimates, but were included here due to statistical leverage provided by the common-resight model.

Survival: Estimates of the likelihood of survival from RRJ to MCN generally decreased over the five years of the study. Probabilities were relatively similar in the first three years of the study, but there was strong evidence of a difference in survival in 2024, with 0.837 (0.653–0.943) of Chief Joseph origin Chinook estimated to survive to MCN compared to just 0.517 (0.414–0.668) of those from the Methow. The survival to MCN probabilities for Methow origin Chinook rebounded some to 0.705 (0.471–0.916) in 2025, but this was still less than all survival estimates from 2021 to 2023. While the focus of this analysis has been constrained to the estimates associated with the river reach from virtual-release (RRJ/LWB) to MCN, it is worth noting the abnormally low estimates of survival to-BON from Chief Joseph origin Chinook of 0.255 (0.170–0.370) and 0.216 (0.147–0.315) in 2023 and 2024, respectively (see Supplemental Material). For comparison, estimates for Methow origin fish were 0.476 (0.377–0.576) and 0.285 (0.211–0.403). This may imply that the fates of fish from these different origins were more similar than what is implied by the to-MCN estimates alone, with the disparities in to-MCN survival probabilities driven in some part by low recapture probabilities at MCN and/or some failure of the related assumptions of the model (e.g., similar arrival time at MCN for smolts leaving RRJ in the same week).

Survival estimates for yearling Chinook detected at LWB were extremely imprecise (and uninformative) with estimates from LWB to MCN of 0.664 (0.392–0.975) in 2024 and 0.388 (0.206–0.969) in 2025. The precision to be gained from common-resight model is considerable, somewhat improving the interpretability of these results, with respective survival estimates of 0.546 (0.433–0.670) and 0.451 (0.315–0.713). However, the assumption needed to include this cohort in the common-resight dataset is much more tenuous: that all fish passing LWB in a given week have a similar downstream recapture probability as their counterparts passing RRJ in the same week. The tenuous nature of this assumption, combined with the suboptimal credible interval widths, indicates the need to bolster sample sizes of available fish at LWB in the future, especially sample size of wild fish (see Simulation results below).

Simulation Study

Fundamental to the simulation was the construction of sample datasets that reflect the passive sampling method at RRJ or LWB as a starting point for mark–recapture–recovery analysis of fish migrating through downstream river reaches. In an archetypical sampling study, every individual in a population has an equal chance of being included, optimally accomplished through “simple random sampling”. However, achieving a truly random sample through field studies can be logistically challenging and costly. Consequently, the detection capabilities of RRJ and LWB are appealing options as “virtual release” sites at which samples of fish can be collected without active handling or tagging. However, fish detected at RRJ and LWB are from previously tagged individuals, making them what is known statistically as a convenience sample. Convenience sampling may reduce labor costs and simplify the logistics of field efforts by creating a sample from individuals that are most readily available to be included in a dataset (e.g., previously tagged fish). However, this inherently introduces bias, whereby estimates ostensibly associated with the whole population are generally dominated by information from the more disproportionately tagged cohorts. These concerns are made evident in the above comparisons of “single-cohort” predation and survival estimates with HUC-specific cohort estimates whereby, in many years, single-cohort estimates largely reflected those of the Methow-specific cohort, which marginalizes

or masks variation from other HUCs to varying degrees. More critically, some cohorts may have little to no opportunity to be included in the convivence sample (i.e., they have no "representation" in the sample despite being part of the population at large). To discuss predation impacts on and survival outcomes of a whole population of fish, it is essential to have a sample that represents the whole population of fish in question. To better understand these concerns, one pertinent covariate, rear-type, was explored to evaluate the implications of such possible sample imbalances.

Inputs: Simulated release groups were constructed to reflect the empirical steelhead release data assessed in the first part of this study. The vast majority of smolts last seen alive at RRJ or LWB were of hatchery origin (*Table C4, Table C5*). Hatchery fish represented an average of 94.8% of all yearling Chinook last seen alive at RRJ from 2021–2025. For steelhead, hatchery fish contributed 95.8% of the RRJ sample. Similarly, at the LWB, hatchery origin fish contributed on average 89.9% of yearling Chinook detections and 97.2% of steelhead detections from 2024–2025. The differences between hatchery and wild fish parameters from which populations were simulated were substantial and consistent across years. Results indicated that wild fish survive to MCN at significantly higher rates than hatchery fish in four of five years, with the wild–hatchery gap in survival ranging from about 2 percentage points (2024, where rates were nearly identical) to almost 24 percentage points (2021). The predation difference was even more striking, with wild fish experience predation rates 3–8 times lower than hatchery fish across all years. In most years, hatchery predation rates were 0.150 to 0.280 (or 15–28%), while wild predation rates range from only 0.020 to 0.110 (or 2–11%) during passage to MCN.

Table C4. Percentages of wild and hatchery reared yearling Chinook (CH1) and steelhead (ST) passively detected at the Rocky Reach juvenile bypass (RRJ) from 2021–2025 and Lower Wenatchee Barge (LWB) from 2024–2025.

Year	Species	% Wild RRJ	% Hatchery RRJ	% Wild LWB	% Hatchery LWB
2021	CH1	4.4	95.6	--	--
2022	CH1	3.1	96.9	--	--
2023	CH1	8.6	91.4	--	--
2024	CH1	7.4	92.6	13.8	86.2
2025	CH1	2.7	97.3	6.5	93.5
2021	ST	3.2	96.8	--	--
2022	ST	3.3	96.7	--	--
2023	ST	3.3	96.7	--	--
2024	ST	8.6	91.4	3.7	96.3
2025	ST	2.7	97.3	1.9	98.1

Table C5. Simulated steelhead probabilities of survival and predation from RRJ to MCN by rear type and overall (based on aggregate empirical data).

Year	Annual Survival Probability			Annual Predation Probability		
	Overall	Hatchery	Wild	Overall	Hatchery	Wild
2021	0.602	0.555	0.791	0.245	0.280	0.107
2022	0.617	0.602	0.676	0.137	0.164	0.028
2023	0.690	0.677	0.743	0.145	0.175	0.022
2024	0.437	0.442	0.420	0.165	0.195	0.046
2025	0.601	0.587	0.656	0.179	0.209	0.060

Model Selection: Simulations applied the "common resight" parameterization analogous to the method described above, here assuming equal recapture probabilities across rearing types. This assumption is grounded in both empirical model comparison and practical necessity. Using LOO-CV on the empirical detection data at RRJ, the common resight model was compared against an independent model that allows recapture probabilities to differ between hatchery and wild fish (*Table C6*). For steelhead, the independent model was favored by an ELPD difference of 10.6 (SE = 7.8, z = 1.4), and for yearling Chinook by 14.1 (SE = 5.9, z = 2.4). While these differences nominally favor the independent model, the common resight model is more parsimonious and neither calculated difference exceeds conventional thresholds for decisive model preference, with the steelhead comparison well within the range attributable to sampling “noise”. Furthermore, under the current sample composition (~ 3% wild), the rear-type-specific recapture rate estimates for wild fish are derived from very few detections and are imprecise and using them as the basis for generating simulated data would propagate that imprecision into the simulation framework itself.

Table C6. ELPD comparison of the relative fitness between the independent modelling approach and “common resight” approach for all comparable groups of steelhead and all comparable groups of yearling Chinook detected passing RRJ.

Species	Model	ELPD	SE ELPD	ΔELPD	SE ΔELPD	z-score
Steelhead	Independent	-98556.4	558.4	--	--	--
	Common resight	-98567.0	558.6	10.6	7.8	1.4
Yearling Chinook	Independent	-20725.7	253.7	--	--	--
	Common resight	-20739.9	253.8	14.1	5.9	2.4

A qualitative summary of the ratios of hatchery to wild fish suggested that the ratios are not consistent from release to initial detection at RRJ and LWB (*Table C7)*, indicating that there may be sampling biases at RRJ and LWB. Specifically, hatchery fish consistently comprise a larger proportion of detections at RRJ than their proportion at release into upstream HUC cohorts, with increases ranging from 0.6 to 4.8 percentage points, depending on species and year. For yearling Chinook at RRJ, hatchery representation increased by an average of 2.6% from release, while steelhead showed an average increase of 3.1%. At LWB, increases were more modest (averaging 0.8% and 1.3% for Chinook and steelhead, respectively), though Wenatchee steelhead showed the highest overall hatchery representation at detection at LWB (>

96% in both years). To investigate a mechanism to explain this change in sample composition, the differences in recapture rates at RRJ and/or LWB from fish detected at BON were tested using the Cochran-Mantel-Haenszel (CMH) Test (*Table C8*). For both species, results indicated that hatchery fish have significantly higher detection rates at RRJ, with a stronger effect for steelhead than yearling Chinook (steelhead: CMH $\chi^2 = 472.11$, $p < 0.001$; Chinook: CMH $\chi^2 = 28.02$, $p < 0.001$). At LWB, detection rates did not differ significantly between hatchery and wild Chinook (CMH $\chi^2 = 2.06$, $p = 0.15$), though statistical power was limited with only 18 wild fish detected.

These differences in recapture rates may account for some of the change in the hatchery to wild ratios from release to detection at RRJ, though several other factors (e.g., survival, run timing, facility operations, etc.) may also be influential. One potential explanation related to run-timing is that Rocky Reach detection operations generally begin on April 1 of each year, after some wild fish have already begun out-migrating. In 2024, Rocky Reach operations began in March and subsequently detected 20 out-migrating steelhead, out of 1,658 total wild steelhead detected at RRJ that year. This further raises concerns about the representation of the Rocky Reach sample, as these early migrants have a probability of being included in the sample of zero. Additionally, hatchery fish may have an advantage over wild fish during downstream migration due to their larger average size (Feldhaus et al. 2016, Beckman et al. 1998). However, other studies suggest that wild fish tend to have greater survival than hatchery fish (Irvine et al. 2013; and results of this study). Behavioral differences and differences in detection timing relative to seasonal river conditions could also affect recapture rates (Gale et al. 2004, Dial et al. 2024, Chasco et al. 2021). Regardless of the mechanism, if detection rates differ between hatchery and wild fish, survival and predation estimates grouped at the HUC level are confounded.

Table C7. Sample size (N) and percent hatchery origin fish at release and detection at Rocky Reach Juvenile Bypass (RRJ) and the Lower Wenatchee Barge (LWB). All hatchery fish that were released during the calendar year of out-migration were included, while wild fish released from January 1 to September 30 were included.

Year	Species	N Release	H% at Release	N RRJ	H% at RRJ	N Release	H% at Release	N LWB	H% at LWB
2021	CH1	31,221	93.6	13,357	95.6	--	--	--	--
2022	CH1	31,604	94.6	11,401	97.0	--	--	--	--
2023	CH1	33,059	90.1	7,539	91.4	--	--	--	--
2024	CH1	31,853	87.9	9,571	92.6	74,573	88.1	4092	87.4
2025	CH1	30,685	94.9	6,213	97.4	71,675	91.2	1825	93.5
2021	ST	70,851	92.0	25,020	96.8	--	--	--	--
2022	ST	75,553	93.5	24,202	96.7	--	--	--	--
2023	ST	71,639	91.9	21,153	96.7	--	--	--	--
2024	ST	68,460	90.8	19,348	91.4	36,774	96.1	560	96.3
2025	ST	68,015	95.1	17,539	97.3	36,875	95.8	323	98.1

Table C8. Results of the Cochran-Mantel-Haenszel (CMH) Test comparing the recapture rates yearling Chinook (CH1) and steelhead (ST) that were detected at BON and RRJ and/or LWB by rear type (i.e., Hatchery [H] and Wild [W]). No (zero) steelhead initially detected at the LWB were subsequently detected at BON.

Species	Release	χ^2	DF	p-value	95% CI	Odds Ratio	Interpretation
CH1	RRJ	28.02	1	<0.001	(0.37–0.63)	0.48	~2.1x odds of detecting hatchery versus wild at RRJ
CH1	LWB	2.06	1	0.151	(0.91–2.47)	1.50	No significant difference
ST	RRJ	472.11	1	<0.001	(0.18–0.24)	0.21	~4.8x odds of detecting hatchery versus wild at RRJ

Given these findings, the assumption of equal recapture probabilities among rearing types downstream warrants further scrutiny. While the LOO-CV comparisons yield z-scores fall short of convincing evidence thresholds, failing to reject a null hypothesis is never a sufficient conclusion, especially given the small number of wild fish in the empirical yearling Chinook dataset. If wild fish are in fact recaptured at different rates, then the cohort-specific survival and predation estimates themselves would be affected, not just the re-weighting step. In the worst case, a systematic difference in recapture probability that is absorbed into the survival or predation estimates could introduce a bias that compositional re-weighting alone cannot correct. The direction and magnitude of such bias would depend on whether wild fish are more or less detectable than hatchery fish and on the specific model structure used for estimation.

Predation Estimate Performance: Results indicated that the most consequential analytical decision was whether to apply compositional correction (i.e., re-weighting) to account for the difference between the rear-type distribution in the tagged sample and the rear-type distribution in the population (tagged and untagged). When all tagged fish were analyzed as a single undifferentiated group (the single-cohort approach), predation estimates were persistently biased upward across every sample size, every sample composition, and every simulated year due to estimates essentially reflecting the hatchery predation rate, which was consistently higher than the predation rate for the population as a whole. Critically, this bias did not diminish meaningfully as sample sizes increased. For instance, at a sample of 500 fish with 3% wild composition, the mean absolute bias in predation to MCN was 0.047 and precision (defined as the width of the 95% credible interval) was 0.209 (*Figure C4*). At 32,000 fish, mean absolute bias decreased only to 0.020, an apparent asymptotic floor, and precision narrowed to 0.068. The credible intervals thus became more precise but centered around a systematically wrong value. Coverage probability (the proportion of simulations in which the 95% credible interval contained the true population value) never exceeded 83% under the single-cohort approach, even at the largest sample sizes, well below the nominal 95% target (*Figure C5*).

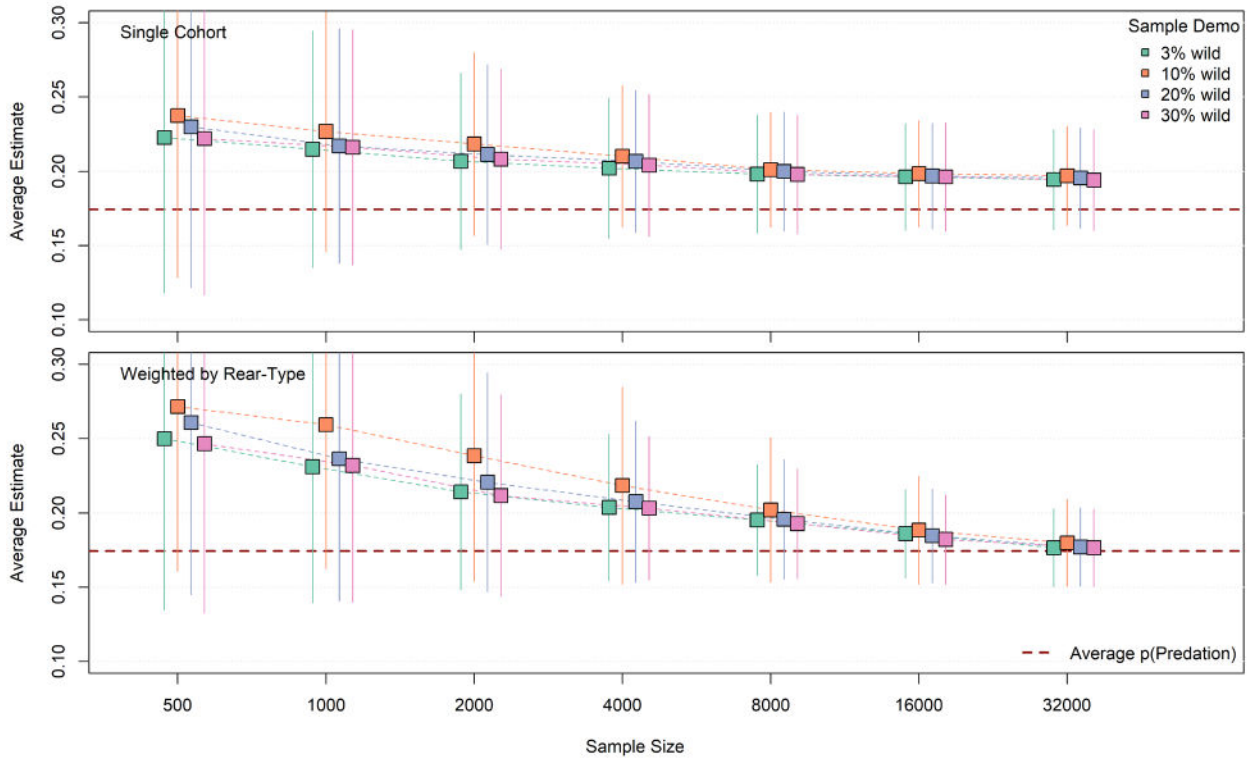


Figure C4. "Average" estimate and credible interval width for estimates of predation from release to McNary Dam plotted by simulated sample sizes and delineated by sample demographics (i.e., the percentage of wild fish in each simulated sample). Circles represent average estimates (across all years and simulations) and line segments represent the average width of 95% credible intervals centered around the average estimate value.

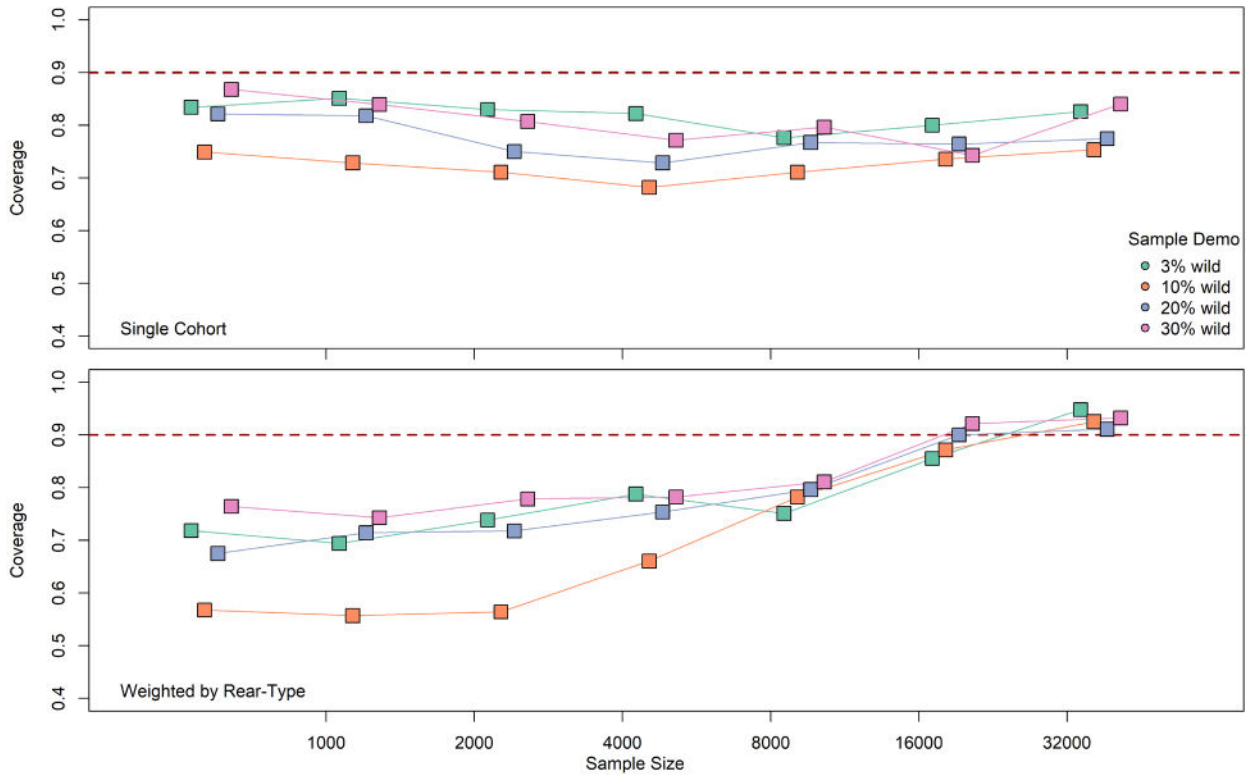


Figure C5. Simulated coverage probability of 95% credible intervals for population-level predation estimates (to McNary Dam) by sample size, delineated by sample composition, under two estimation approaches: single-cohort approach (top) and rear-type-weighted approach (bottom; reweighted for 0.20 true wild proportion). The dashed red line indicates a nominal 90% coverage target.

When rear-type-specific estimates were developed separately and then re-weighted to a presumed 20% wild population proportion, the apparent bias floor was eliminated. Under the corrected approach with 3% wild composition, bias decreased steadily from 0.074 at n = 500 to 0.007 at n = 32,000, and coverage probability rose from 74% to 95%. However, the correction introduced additional estimation variance at small-to-moderate sample sizes because the wild-cohort estimate (based on very few fish) was amplified by the re-weighting factor. At a 3% wild sample, each wild-reared individual effectively balances ~ 6.7 hatchery-reared fish in the correction, substantially inflating the variance contributed by the wild-cohort estimate. Consequently, the corrected approach did not outperform the single-cohort approach in terms of mean absolute bias until sample sizes reached approximately 8,000 fish (with 3% wild composition). Below that threshold, the noise introduced by amplifying a sparse wild-cohort estimate exceeded the compositional bias it was correcting. The sample size at which correction begins to outperform the single-cohort approach was sensitive to the proportion of wild fish in the sample. With 30% of the sample composed of wild fish, the correction outperformed the single-cohort approach at n = 4,000 rather than n = 8,000. This occurred because a higher wild proportion yields a more precise wild-cohort estimate and requires a smaller re-weighting factor, which together reduce the variance penalty associated with

correction. That is to say, the benefit of correction is realized with smaller total sample sizes when wild fish constitute a larger share of the tagged sample.

Survival Estimate Performance: In contrast to predation, survival estimates were considerably less sensitive to analytical approach and the wild proportion in the sample. Across the five simulated years, the absolute difference between hatchery and wild survival probabilities ranged from 0.022 to 0.237 (compared to 0.137 to 0.173 for predation) and the direction of the survival difference was inconsistent. Such inconsistency meant that compositional bias in survival estimates was smaller in magnitude and did not accumulate in a single direction. As a result, the single-cohort approach for survival did not exhibit the same pronounced bias "floor" observed with the predation estimation. At n = 32,000, the single-cohort survival estimate had a mean absolute bias of 0.028 and coverage of 91%, compared to 0.038 bias and 87% coverage for the corrected estimates. For survival, the correction actually increased bias relative to the single-cohort approach at most sample sizes due to the "noise" from the smaller wild cohort being amplified by the correction without eliminating a substantial compositional bias (as the bias for survival was small to begin with). Both approaches yielded credible interval widths in the range of 0.21 to 0.47, reflecting the inherent difficulty of estimating survival to MCJ with high precision irrespective of sample composition. These results suggest that for survival estimation through a single reach, the single-cohort approach may be adequate, provided that the survival difference among cohorts remains modest. However, in years resembling 2021 (where the difference in survival was 0.237), the correction would become more important for survival as well.

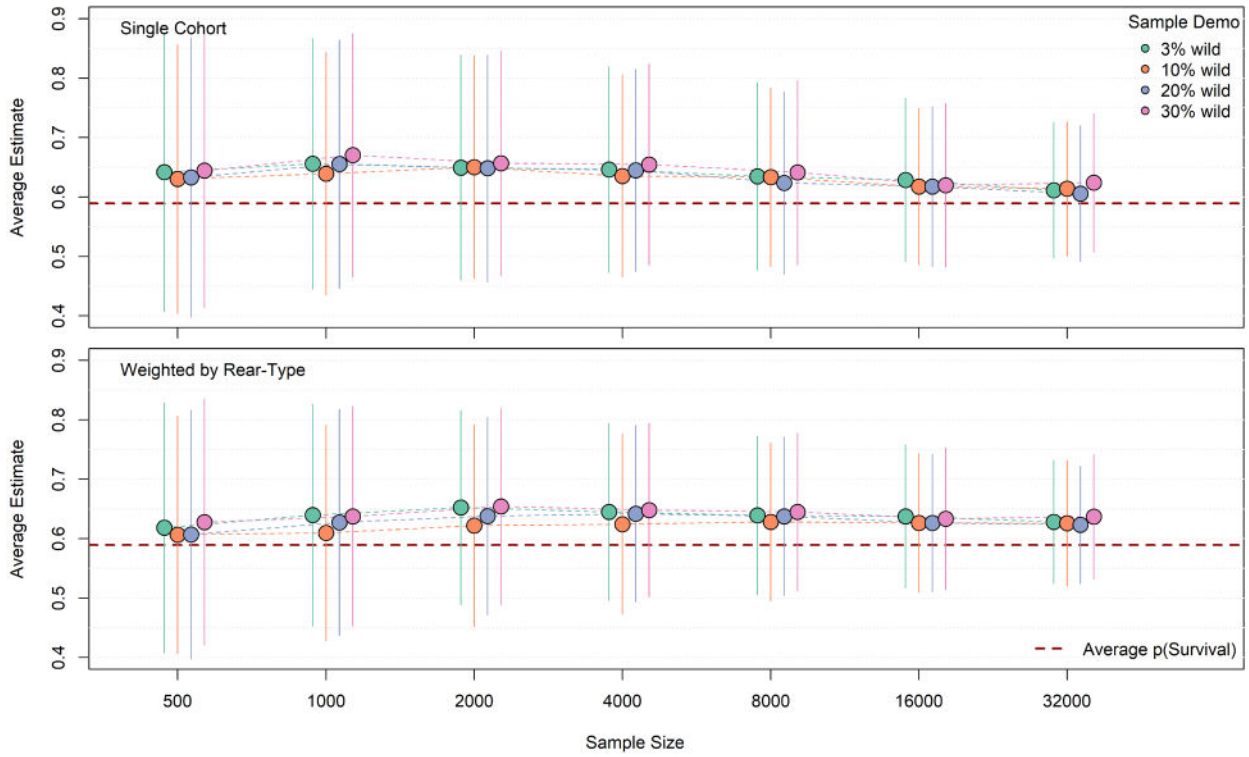


Figure C6. "Average" estimate and credible interval width for estimates of survival from release to McNary Dam plotted by simulated sample sizes and delineated by sample demographics (i.e. the percentage of wild fish in each simulated sample). Circles represent average estimates (across all years and simulations) and line segments represent the average width of 95% credible intervals centered around the average estimate value.

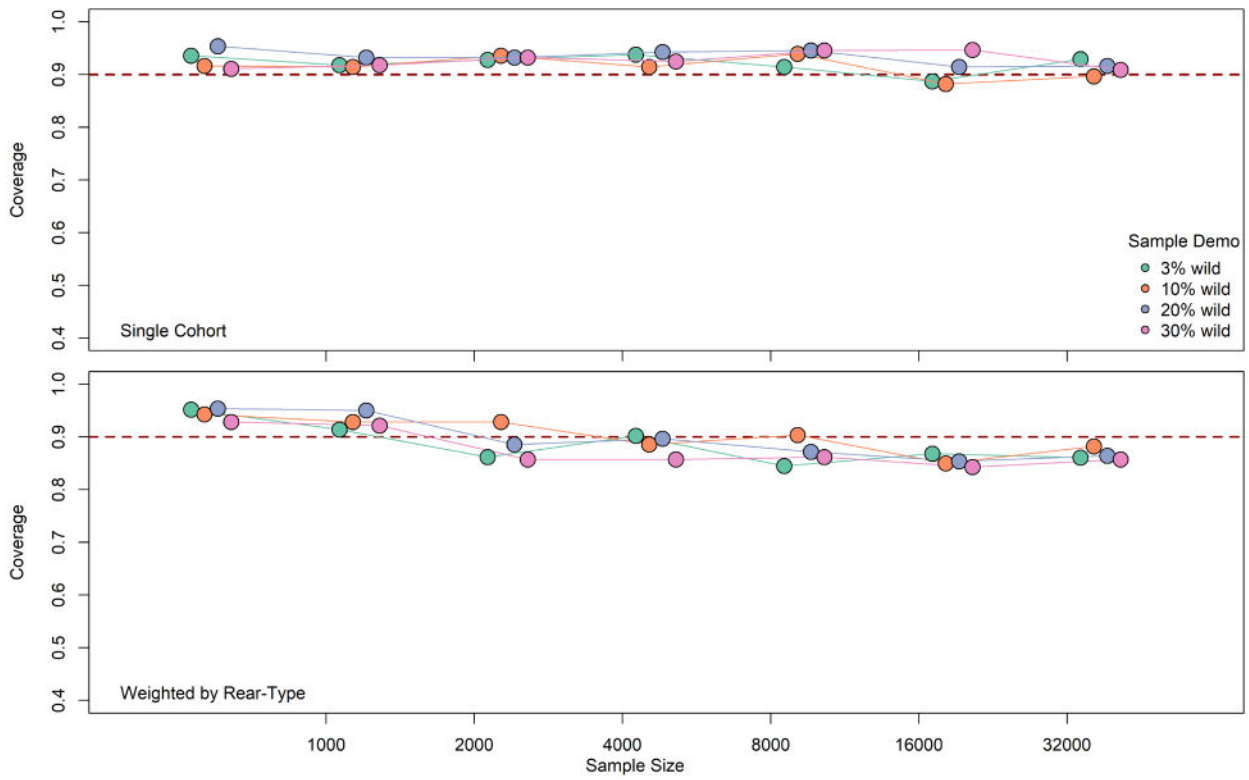


Figure C7. Simulated coverage probability of 95% credible intervals for population-level survival estimates (to MCJ) by sample size, delineated by sample composition, under two estimation approaches: single-cohort approach (top) and rear-type-weighted approach (bottom; reweighted for 20% true wild proportion). The dashed red line indicates a nominal 90% coverage target.

Generalizing Results: It is crucial to note that rear-type is but one probable covariate among numerous others (e.g., tributary of origin, hatchery of origin, migration timing, body size, etc.) that may be associated with meaningful variation in rates of predation and survival and that are inherently unevenly represented under a convenience sampling approach, the degree to which is unknown. Passive PIT-tag detections at the release sites constitute a convenience sample, and a foundational principle of survey sampling is that convenience samples do not afford design-based inference about a target population without additional assumptions or adjustments (Cochran 1977, Lohr 2010, Thompson 2012). Furthermore, the dynamics observed herein (persistent compositional bias in the single-cohort estimates, the variance penalty of correction at small cohort sizes) are not unique to the hatchery-wild distinction. They are properties of any correction applied to any covariate for which the tagged sample does not reflect the population at large (tagged and untagged fish). For instance, when fish from a particular tributary experience systematically lower predation than the population average and that tributary is underrepresented in the tagged sample, the same upward bias observed here for rear-type would arise, and the same re-weighting framework would be needed to address it. Moreover, this study assumed the correction factor (the true population proportion of wild fish) was known and correctly specified at 20%. In practice, however, this proportion would be selected based on empirical data (if available), and the

choice of correction factor represents a source of uncertainty not captured in the simulation. Reliable estimates with respect to proportions of fish of different rear-types, HUC origins, RRJ/LWB passage times, and other factors may require additional monitoring programs, programs that carry their own sampling error and likely vary across years. The sensitivity of estimates to the assumed correction value, and the propagation of uncertainty in that value through to the final population-level estimates, warrants separate investigation. The present results should in-fact be understood as a best-case scenario in which the correction is applied, with perfect knowledge of the population composition. An active probability-based sampling design at or near the virtual release site could be considered as a method of assessing what a representative sample of the population looks like in order to construct more precise corrections. Nonetheless, results herein demonstrate that ignoring known compositional imbalances produces estimates with irreducible bias, bias that does not diminish with additional sampling. The framework demonstrated here for rear-type provides a template for potentially addressing these and other imbalances, especially if additional data and ancillary information to support those corrections become available in the future.

Key Findings & Recommendations

Based on the empirical results and simulation study findings, we offer the following recommendations for improving future estimates of predation and survival using passive detected fish at sites in the Middle Columbia River.

1. Covariates were associated with variation in predation and survival. The initial analysis of this study demonstrated significant differences in rates of survival and predation in fish of different species, HUCs, and years. For predation probabilities, there was also evident of differences by rear-type (hatchery, wild). Other factors or covariates (run-timing, abundance, fish size) are also presumably associated with variation in predation and survival but were not explicitly considered in this analysis.

2. Consider integrating active sampling data collection. Active sampling at or near the virtual release site should be considered as a method to complement the passive tagging infrastructure. Capturing a simple random sample of out-migrating smolts at the release/recapture site would directly characterize the demographic composition of the migrating population including rear-type, tributary of origin, migration timing, and whatever other covariates are deemed critical; thereby providing empirical grounding for the re-weighting corrections applied to the tagged sample.

3. Balance practical sample sizes with representation. Based on this and other studies, reliable estimates of predation and survival often require a minimum of 500 and 5,000 PIT-tagged smolts, respectively, but ultimately depend on recovery and recapture probabilities, the spatial- and temporal-scale under consideration, model assumptions (e.g., independent versus common-resight probabilities), and other factors. While results from this study suggest that estimates of predation from sample sizes of > 499 individuals can be reliable with a post-hoc compositional

correction (i.e., re-weighting) to account for biases introduced by convenience sampling, ideal sample size should be a minimum 4,000 to 8,000 fish in a virtual-release sample (like RRJ or LWB) in tandem with moderate representation of the full population (e.g., >10% wild-reared fish) during field sampling.

4. Prioritize increased representation of the tagged sample. For predation, improving sample composition (e.g., increasing wild fish in the sample) is more cost-effective than simply tagging more fish at current compositions. Results of this study demonstrated that, at practical sample sizes, efforts to tag and release more wild fish would yield greater improvements in bias reduction, per fish tagged, than maintaining current proportional sampling.

5. Consider more advanced modelling techniques: Virtual-release samples can be substantially enlarged without additional tagging or field effort. This can be accomplished by modeling survival and travel time from upstream tagging points to the RRJ/LWB detection sites. An example of this type of model was provided in Appendix C of the 2023 Final Avian Predation Annual Report comparing RRJ detected fish to those tagged and released from Rock Island Dam (RIS), in which travel time and survival were explicitly modelled from RRJ to RIS. The suggestion here is to instead now explicitly model travel time and survival from point of tagging/release upstream of RRJ/LWB to their passive detection at these recapture points.

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APPENDIX D: REACH-SPECIFIC WEEKLY PREDATION AND SURVIVAL OF UPPER COLUMBIA RIVER AND SNAKE RIVER SMOLTS

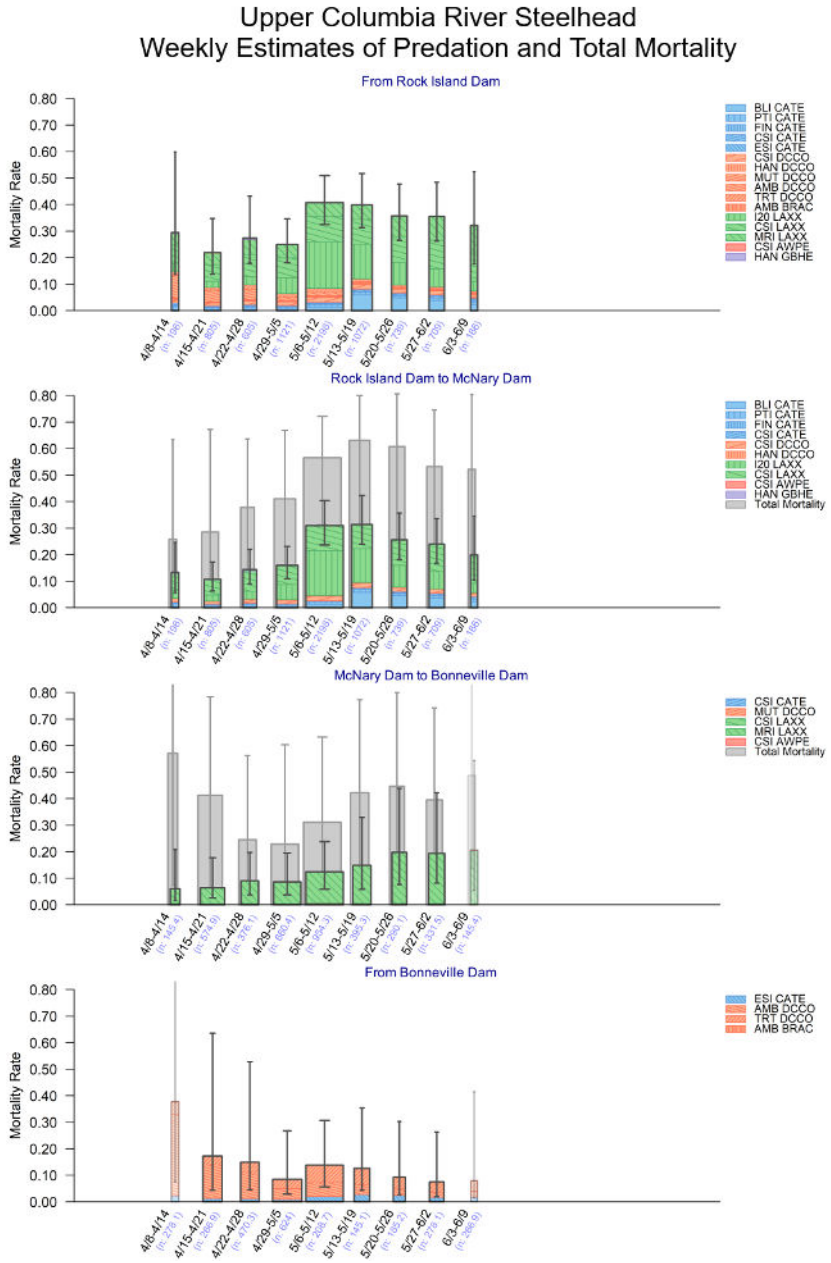


Figure D1. Estimated weekly total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Upper Columbia River steelhead during smolt passage from Rock Island Dam to the Pacific Ocean (top), Rock Island Dam to McNary Dam, McNary Dam to Bonneville Dam, and following passage at Bonneville Dam. Bar widths are proportional to sample abundance at Rock Island Dam and only weeks with > 99 PIT-tagged smolts were included. Bars are transparent for reach/weeks combinations for which the JMS model estimated less than < 100 surviving smolts entered the reach downstream of Rock Island Dam. Estimates are proportions with error bars representing 95% credible intervals. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), and American white pelicans (AWPE; see Map 1 for colony locations).

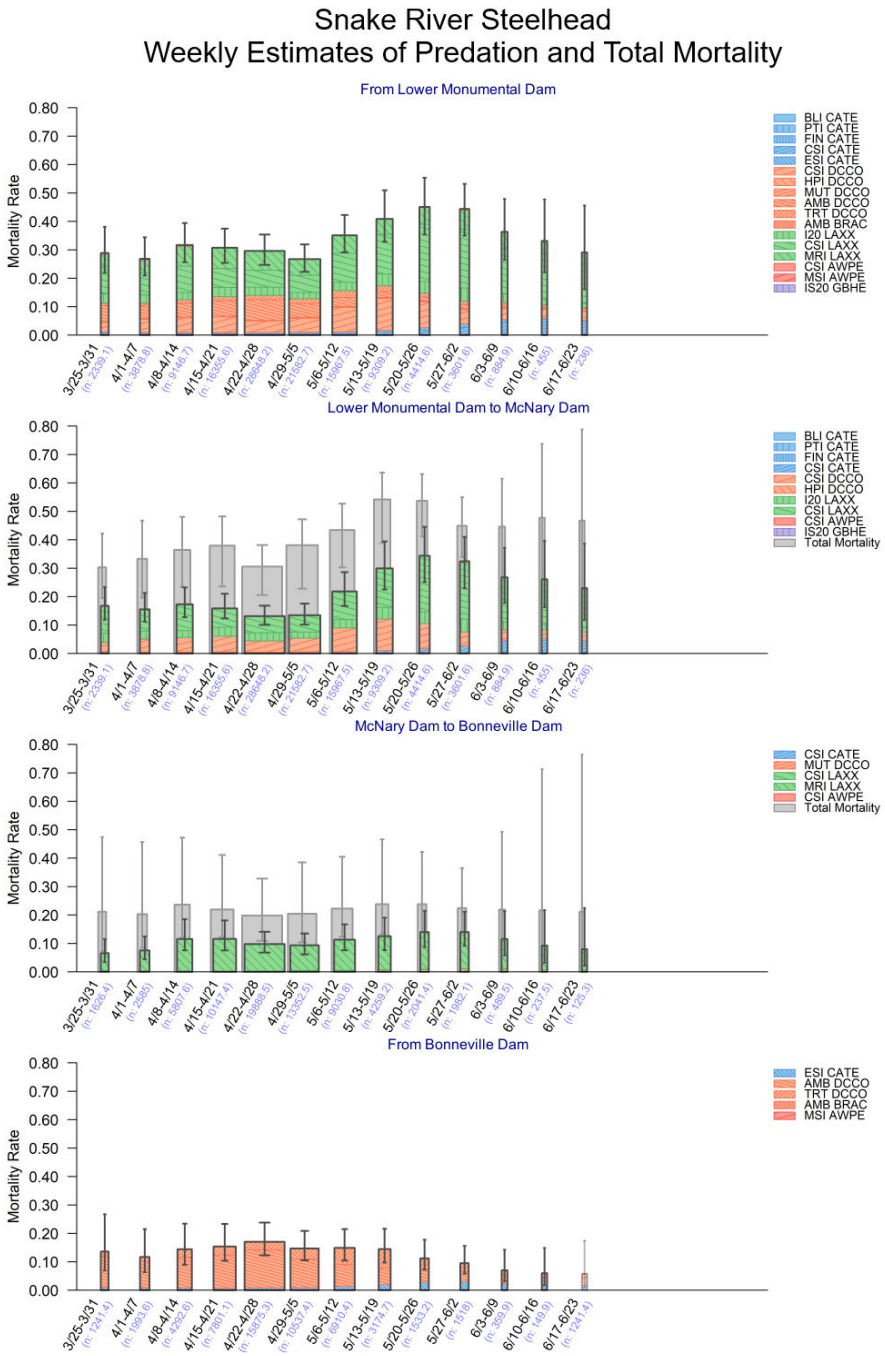


Figure D2. Estimated weekly total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Snake River steelhead during passage from Lower Monumental Dam to the Pacific Ocean (top), Lower Monumental Dam to McNary Dam, McNary Dam to Bonneville Dam, and following passage at Bonneville Dam. Bar widths are proportional to estimated sample abundance at Lower Monumental Dam and only weeks with > 99 PIT-tagged smolts were included. Bars are transparent for reach/weeks combinations for which the JMS model estimated less than < 100 surviving smolts entered the reach downstream of Lower Monumental Dam. Estimates are proportions with error bars representing 95% credible intervals. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), and American white pelicans (AWPE; see Map 1 for colony locations).

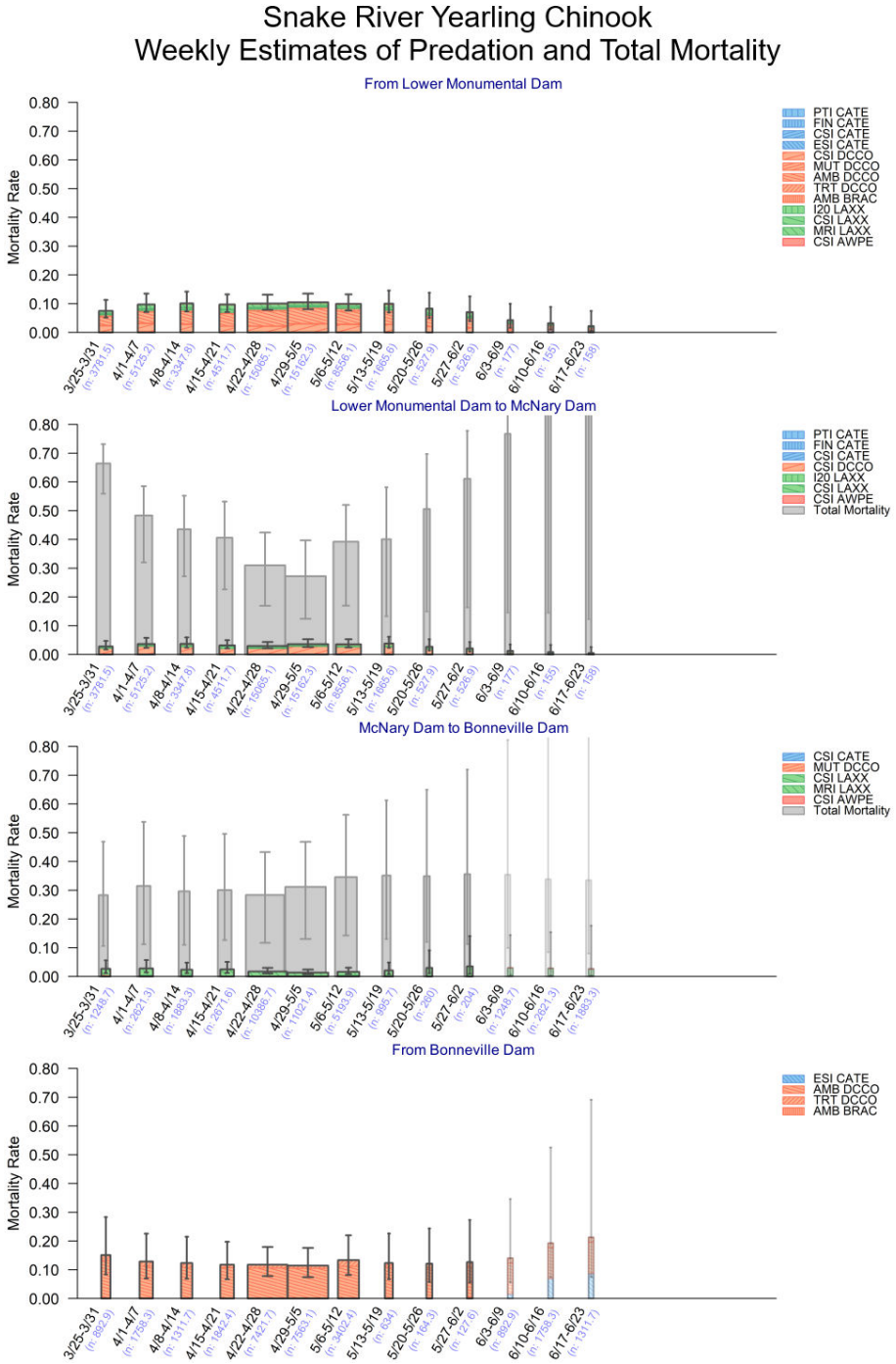


Figure D3. Estimated weekly total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Snake River yearling Chinook during passage from Lower Monumental Dam to the Pacific Ocean (top), Lower Monumental Dam to McNary Dam, McNary Dam to Bonneville Dam, and following passage at Bonneville Dam. Bar widths are proportional to estimated sample abundance at Lower Monumental Dam and only weeks with > 99 PIT-tagged smolts were included. Bars are transparent for reach/weeks combinations for which the JMS model estimated less than < 100 surviving smolts entered the reach downstream of Lower Monumental Dam. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), and American white pelicans (AWPE; see Map 1 for colony locations).

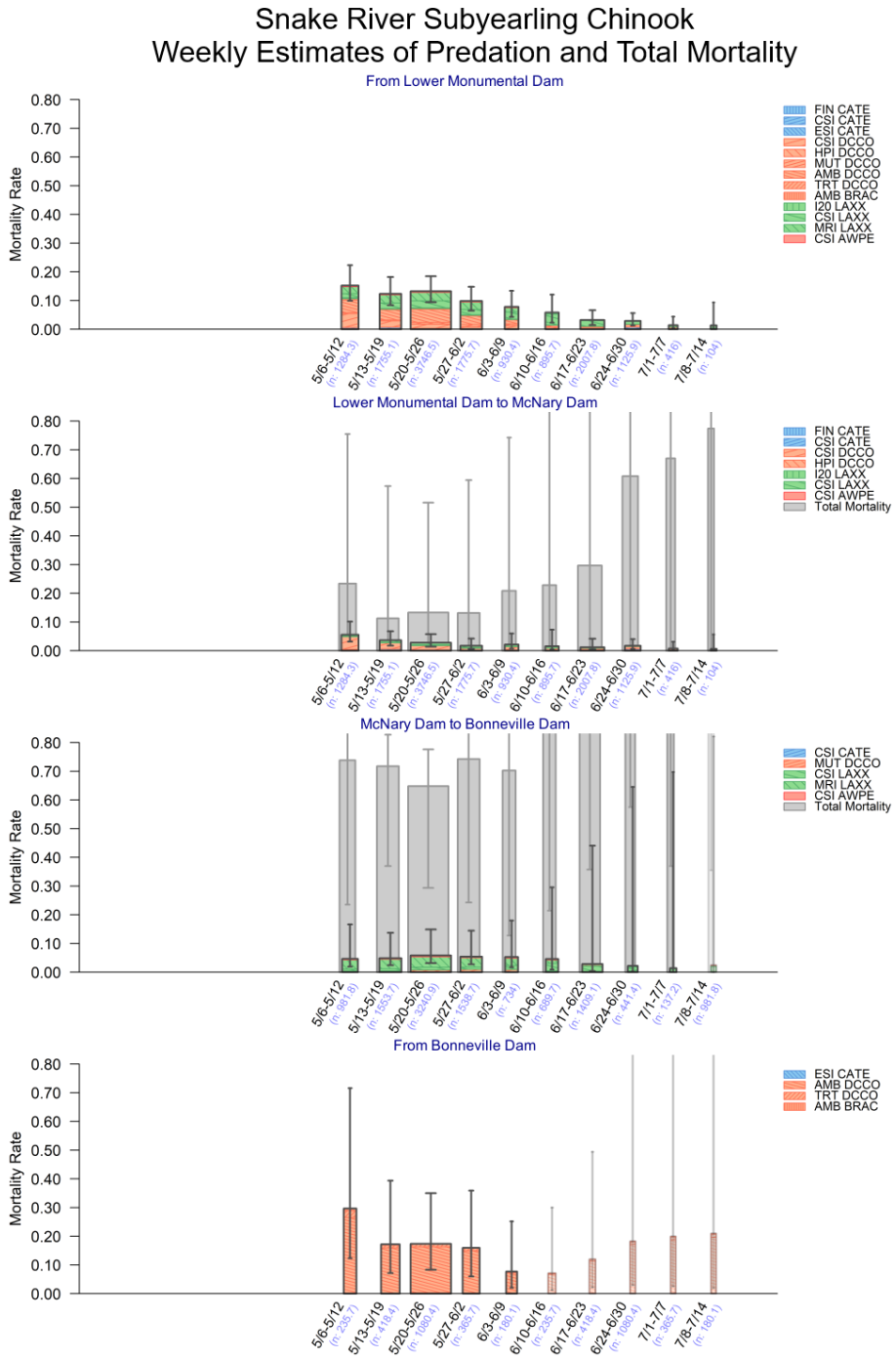


Figure D4. Estimated weekly total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Snake River subyearling Chinook during passage from Lower Monumental Dam to the Pacific Ocean (top), Lower Monumental Dam to McNary Dam, McNary Dam to Bonneville Dam, and following passage at Bonneville Dam. Bar widths are proportional to estimated sample abundance at Lower Monumental Dam and only weeks with > 99 PIT-tagged smolts were included. Bars are transparent for reach/weeks combinations for which the JMS model estimated less than < 100 surviving smolts entered the reach downstream of Lower Monumental Dam. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), and American white pelicans (AWPE; see Map 1 for colony locations).

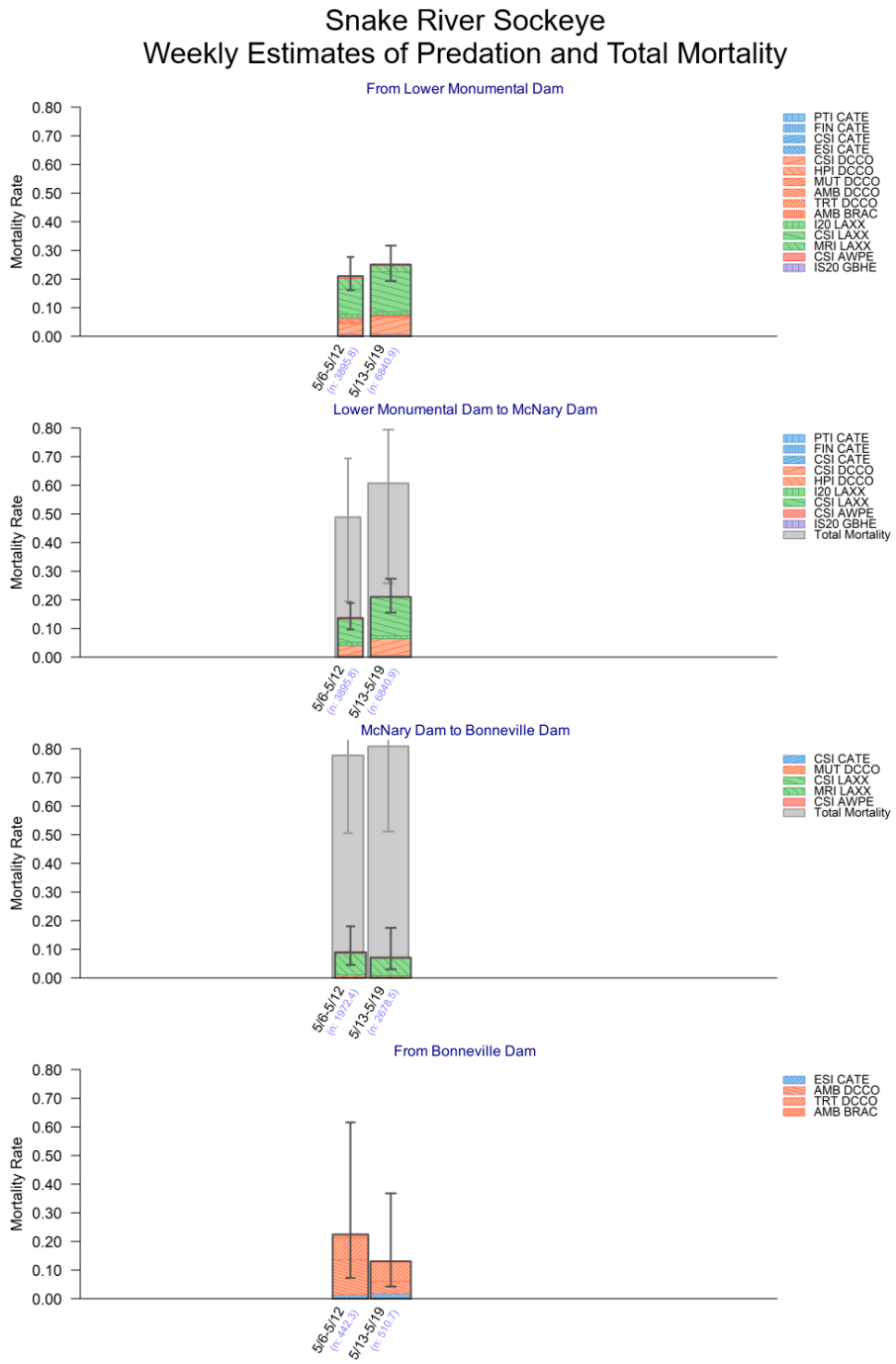


Figure D5. Estimated weekly total mortality (grey bars) and mortality associated with avian predation/consumption (colored bars) for Snake River sockeye during passage from Lower Monumental Dam to the Pacific Ocean (top), Lower Monumental Dam to McNary Dam, McNary Dam to Bonneville Dam, and following passage at Bonneville Dam. Bar widths are proportional to estimated sample abundance at Lower Monumental Dam and only weeks with > 99 PIT-tagged smolts were included. Bars are transparent for reach/weeks combinations for which the JMS model estimated less than < 100 surviving smolts entered the reach downstream of Lower Monumental Dam. Avian predator species include Caspian terns (CATE), double-crested cormorants (DCCO), California and ring-billed gulls (LAXX), and American white pelicans (AWPE; see Map 1 for colony locations).

APPENDIX E: DOUBLE-CRESTED CORMORANT COUNTS ON NAVIGATION AIDS DURING 2022–2025

Table E1. Location of navigation aids for double-crested cormorant colonies in the Columbia River estuary. Location names are from Roby et al. 2021, Lawonn 2023, and the associated United States Coast Guard (U.S.C.G.) number (No.) 2024.

Colony Complex	U.S.C.G	Name in Lawonn 2023	Name in Roby et al. 2021	Latitude	Longitude
Estuary Navigation Aids RKM 0-22	-	Jetty A Tower	Jetty A Channel Marker	46.265954	-124.037809
	9945	Sand Island Range Front Light	Sand Island Channel Marker #1	46.26582	-123.9929
	9950	Sand Island Range Rear Light	Sand Island Channel Marker #2	46.2673	-123.98103
Estuary Navigation Aids RKM 22-53	10185	Harrington Point Channel 52 Light	Estuary Channel Marker #1	46.2342	-123.71417
	10186	Tongue Point Channel Range Front Light	Estuary Channel Marker #2	46.23294	-123.71352
	10187	Tongue Point Channel Range Rear Light	Estuary Channel Marker #3	46.23515	-123.70581
	10175	Harrington Point Range Front Light	Estuary Channel Marker #4	46.25591	-123.677
	10180	Harrington Point Range Rear Light	Estuary Channel Marker #5	46.26176	-123.6656
	10215	Miller Sands Dike Light 5	Estuary Channel Marker #6	46.25652	-123.66856
	10230	Miller Sands Dike Light 11	Estuary Channel Marker #7	46.26118	-123.64197
	10195	Miller Sands Range Front Light	Estuary Channel Marker #8	46.26241	-123.63666
	10235	Pillar Rock Lower Range Front Light	Estuary Channel Marker #9	46.25275	-123.54342
	10240	Pillar Rock Lower Range Rear Light	Estuary Channel Marker #10	46.25173	-123.52938
	10275	Pillar Rock Upper Range Front Light	Estuary Channel Marker #11	46.26073	-123.51556
	10280	Piller Rock Upper Range Rear Light	Estuary Channel Marker #12	46.2617	-123.50288
Estuary Navigation Aids RKM 127-234	10950	Martin Island Bybee Ledge Channel Range Front	Not Reported	45.9571	-122.80797
	10955	Martin Island Bybee Ledge Channel Range Rear	Not Reported	45.95585	-122.80626
	11570	Vancouver to Bonneville Gary Island Light A Front	Not Reported	45.55180	-122.33982
	-	Unknown Marker (upstream of 11570)	Not Reported	45.54889	-122.33784
	11630	Fashion Reef Lower Range Front Light	Not Reported	45.58511	-122.12714
	11635	Fashion Reef Lower Range Rear Light	Not Reported	45.58629	-122.1193
	11715	Warrendale Lower Range Front Light	Not Reported	45.51379	-122.03487
	11720	Warrendale Lower Range Rear Light	Not Reported	45.61359	-122.03763

Table E2. Number of double-crested cormorant breeding pairs at known colonies on navigation aids downstream of Bonneville Dam during 2022–2025. NA denotes that counts were not available.

Colony	2022	2023	2024	2025
Jetty A Tower	21	NA	37	33
Sand Island Range Front Light	23	NA	20	16
Sand Island Range Rear Light	22	NA	15	0
Harrington Point Channel 52 Light	0	0	0	0
Tongue Point Channel Range Front Light	7	8	5	5
Tongue Point Channel Range Rear Light	10	0	0	0
Harrington Point Range Front Light	41	47	53	36
Harrington Point Range Rear Light	7	1	0	0
Miller Sands Dike Light 5	46	65	47	26
Miller Sands Dike Light 11	0	0	0	0
Miller Sands Range Front Light	10	12	11	10
Pillar Rock Lower Range Front Light	0	30	45	27
Pillar Rock Lower Range Rear Light	19	21	0	23
Pillar Rock Upper Range Front Light	44	50	34	16
Piller Rock Upper Range Rear Light	36	78	70	80
Martin Island Bybee Ledge Channel Range Front	NA	15	18	10
Martin Island Bybee Ledge Channel Range Rear Light	NA	50	49	31
Vancouver to Bonneville Gary Island Light A Front	NA	NA	0	7
Unknown Marker (upstream Gary Island Light A)	NA	10	0	0
Fashion Reef Lower Range Front Light	NA	15	9	10
Fashion Reef Lower Range Rear Light	NA	32	28	35
Warrendale Lower Range Front Light	NA	NA	NA	7
Warrendale Lower Range Rear Light	NA	10	20	28
Total	286	444	461	400

References

Lawonn, M. J. 2023. Summary of Double-crested Cormorant Monitoring in the Columbia River Estuary, 2020 and 2021. Science Bulletin 2023-02. Oregon Department of Fish and Wildlife, Salem, Oregon.

Roby D.D, A.F. Evans, and K. Collis (editors). 2021. Avian Predation on Salmonids in the Columbia River Basin: A Synopsis of Ecology and Management. A synthesis report submitted to the U.S. Army Corps of Engineers, Walla Walla, Washington; the Bonneville Power Administration, Portland, Oregon; the Grant County Public Utility District/Priest Rapids Coordinating Committee, Ephrata, Washington; and the Oregon Department of Fish and Wildlife, Salem, Oregon. 788 pp.

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